

# Chapter 5: Investigation Methodology

## Questions, Hypotheses, Timelines, and Relationships

### Companion Workbook to the Instructional Video

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#### How to Use This Workbook

This is the written companion to the instructional video on investigation methodology. It works as a follow-along guide, a standalone textbook chapter, and a practice manual.

**Where this chapter sits in the series.** Chapter 1 established what intelligence is. Chapter 2 established what OSINT is and why verification is central. Chapter 3 covered the source categories. Chapter 4 covered the collection methodology as a process. **This chapter is about the thinking.** Chapter 4 told you how to run the machine; Chapter 5 tells you how to reason inside it.

The distinction matters. A student can execute Chapter 4's nine phases flawlessly and still reach a wrong conclusion, because process discipline does not by itself produce good reasoning. The techniques in this chapter (question formulation, hypothesis testing, timeline construction, link analysis, assumption elimination, and relationship discovery) are what happens inside the requirements, exploitation, and analysis phases when they are done well.

**Labs 2 through 4 are chained** on a single fictional case, and Lab 5 uses a separate provided investigation. All evidence packets are in the workbook, so no external collection is required for the labs in this chapter.

#### Time estimate:

| Component                   | Estimated time        |
|-----------------------------|-----------------------|
| Video                       | 60 to 75 minutes      |
| Reading material            | 5 to 6 hours          |
| Knowledge assessment        | 1 hour                |
| Critical thinking exercises | 2 to 2.5 hours        |
| Labs 1 through 5            | 11 to 16 hours        |
| Capstone assignment         | 6 to 9 hours          |
| <b>Total</b>                | <b>26 to 35 hours</b> |

**Materials:** a word processor, a spreadsheet application, and paper or a whiteboard for link diagrams. Diagramming software is optional; hand-drawn charts are acceptable and often better for thinking. Internet access is needed only for the capstone's optional extension.

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## The Rules Still Apply

**Rule 1: No private individuals.** The named people in this chapter's exercises are fictional characters in fictional scenarios. Do not apply these techniques to a real private person as part of this coursework.

**Rule 2: Publicly available means genuinely public.**

**Rule 3: Document as you go.**

**Rule 4: No dark web access.**

**Rule 5: Automation is constrained.**

This chapter adds one more, because it is the chapter where you will build charts that connect people to one another:

**Rule 6: A relationship is a claim, and every claim needs a basis.** Every line you draw between two entities is an assertion about the world, and it must carry the evidence that supports it and the strength of that support. An unsourced line on a chart is the most persuasive form of unsupported assumption in the entire discipline, because diagrams look like conclusions even when they are guesses. Section 1.5.6 covers this in detail.

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## Learning Objectives

**Knowledge objectives (you can explain it):**

1. Explain what investigation methodology adds to collection methodology, and what fails without it.
2. Distinguish questions, assumptions, hypotheses, and conclusions, and explain how each converts into the next.
3. Distinguish facts, evidence, and inferences, and explain why the distinction governs how a finding may be worded.
4. Describe what makes an investigative question strong or weak, and name the six question types and what each is for.
5. Explain the method of multiple working hypotheses and why single-hypothesis investigation fails predictably.
6. Explain the difference between chronological and event-based timeline analysis and when each is appropriate.
7. Describe the components of a link chart and what each node and edge type represents.
8. Explain the difference between direct and indirect relationships, and what each supports and does not support.
9. Describe the major link analysis techniques and their applications.

10. Explain how unsupported assumptions enter an investigation and how each is surfaced.
11. Describe the mechanisms by which hidden relationships are discovered.

**Application objectives (you can do it):**

1. Convert a broad topic into prioritized, answerable investigative questions.
2. Generate at least three genuine competing hypotheses for a scenario, including one you consider unlikely, stated fairly.
3. Score evidence against hypotheses by disconfirmation and identify the discriminating items.
4. Construct a timeline with precision markers, source attribution, and inconsistency flags.
5. Detect and characterize timeline inconsistencies, and derive investigative steps from them.
6. Build a link chart with typed nodes, sourced edges, and strength ratings.
7. Identify brokers, clusters, and structural gaps in a relationship map.
8. Surface unsupported assumptions in your own and others' work, and rewrite conclusions using only verified evidence.
9. Discover indirect relationships through shared attributes, temporal correlation, and structural inference.

**Analysis objectives (you can judge it):**

1. Diagnose which reasoning step failed when an investigation reaches a wrong conclusion.
  2. Recognize the cognitive biases that operate at each stage and apply the specific control for each.
  3. Judge whether a conclusion is supported by the evidence actually cited, or by an unstated assumption.
  4. Assess whether a relationship chart shows what the evidence supports or what the analyst believes.
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## **Part 1: Reading Material**

### **1.1 Introduction to Investigation Methodology**

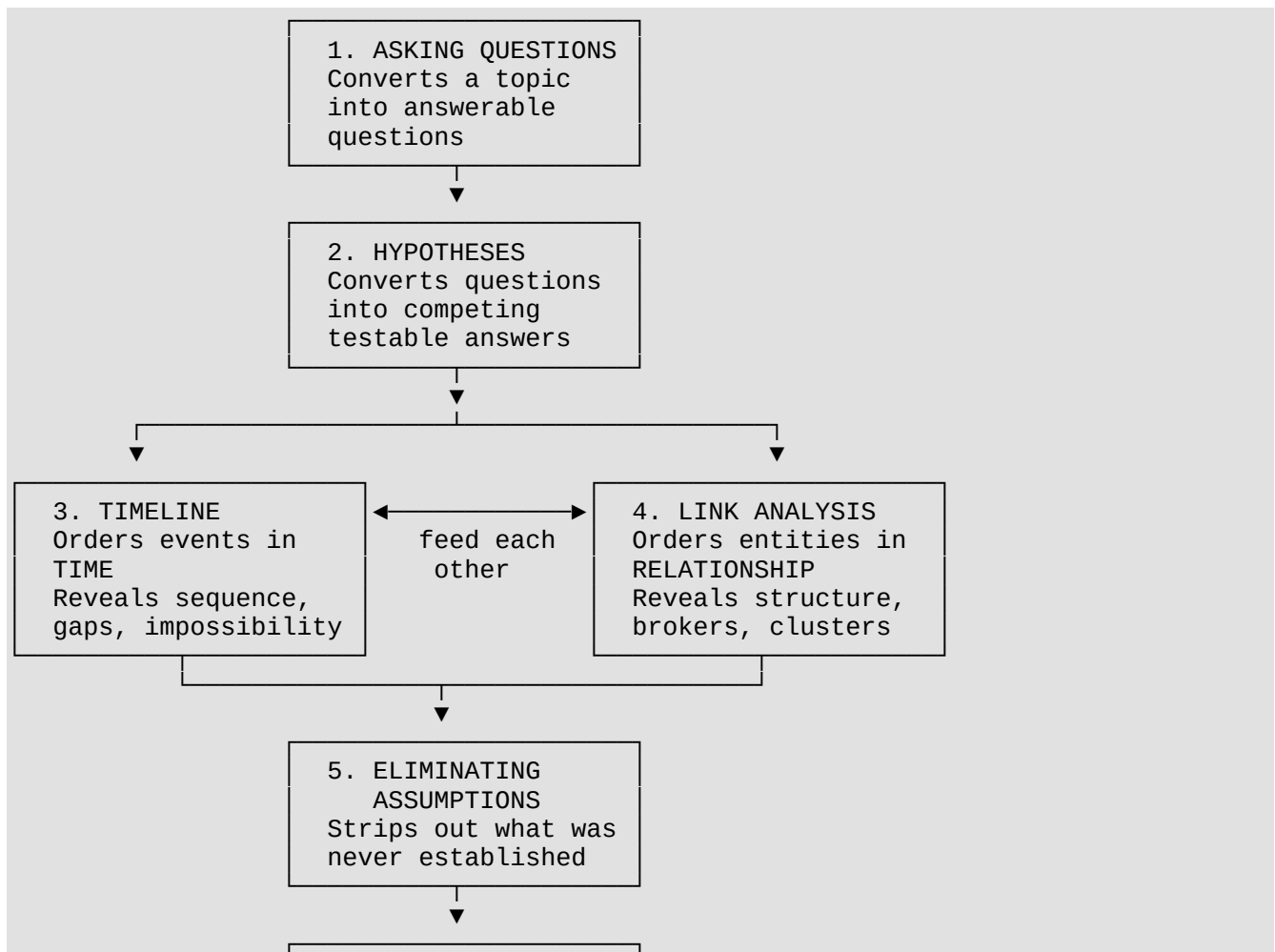
#### ***1.1.1 What Investigation Methodology Adds***

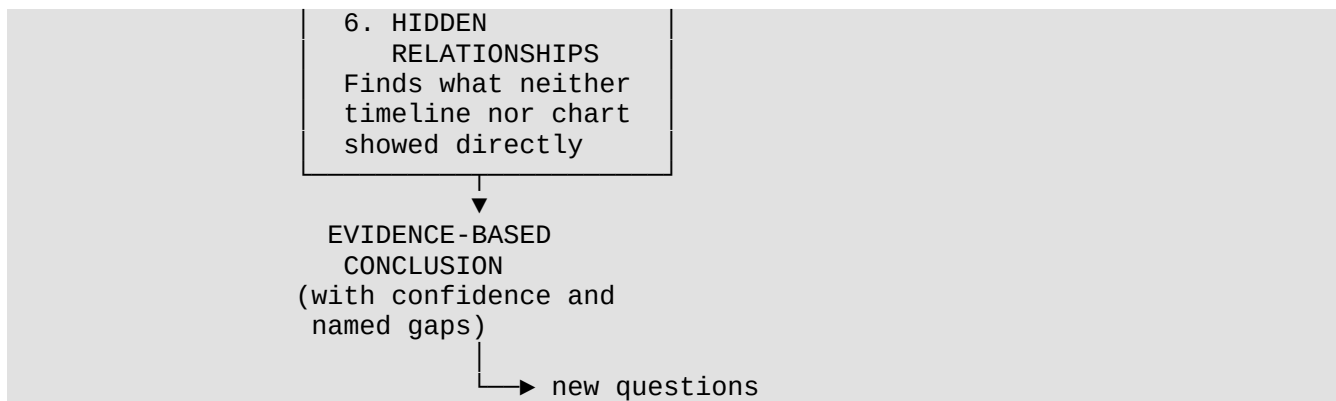
Chapter 4's collection methodology is a *process*: a sequence of phases with inputs, outputs, and gates. Investigation methodology is a set of *reasoning techniques* used inside that process. They are complementary and neither substitutes for the other.

|                    | Collection methodology (Ch.4)                        | Investigation methodology (this chapter)                                   |
|--------------------|------------------------------------------------------|----------------------------------------------------------------------------|
| Governs            | The sequence of work                                 | The quality of reasoning                                                   |
| Answers            | What do I do next?                                   | What should I believe, and why?                                            |
| Failure without it | Duplication, drift, unauditable work, legal exposure | Confident wrong conclusions produced by disciplined process                |
| Key artifacts      | Requirement record, collection plan, log, product    | Question set, hypothesis matrix, timeline, link chart, assumption register |
| Skill type         | Discipline                                           | Judgment                                                                   |

**The failure mode this chapter prevents.** An investigator with excellent process and poor reasoning produces work that is fully documented, entirely traceable, and wrong. Every source is logged, every date normalized, every finding archived, and the conclusion rests on an assumption nobody wrote down. That work is more dangerous than sloppy work, because its apparatus makes it persuasive.

### 1.1.2 The Six Techniques and How They Interlock





**Read the diagram for its two most important features.**

First, **timeline and link analysis feed each other continuously**. A timeline entry with an unexplained actor generates a link chart question. A link chart edge with no date generates a timeline question. Investigators who build one and then the other, sequentially and once, miss most of what the pair can produce.

Second, **assumption elimination comes after the analytical construction, not before**. You cannot strip assumptions from a picture you have not yet built. The order is build, then audit, then rebuild.

### **1.1.3 Facts, Evidence, and Inference**

The vocabulary that governs how everything in this chapter may be written.

**A fact** is a state of the world. It is true whether or not anyone knows it. "The warehouse fire began at 02:14."

**Evidence** is an item that bears on whether a proposed fact is true. It is always evidence *for or against something*, from a source, with a reliability. "The utility's meter log records a load loss at 02:14, item 12, rated A1."

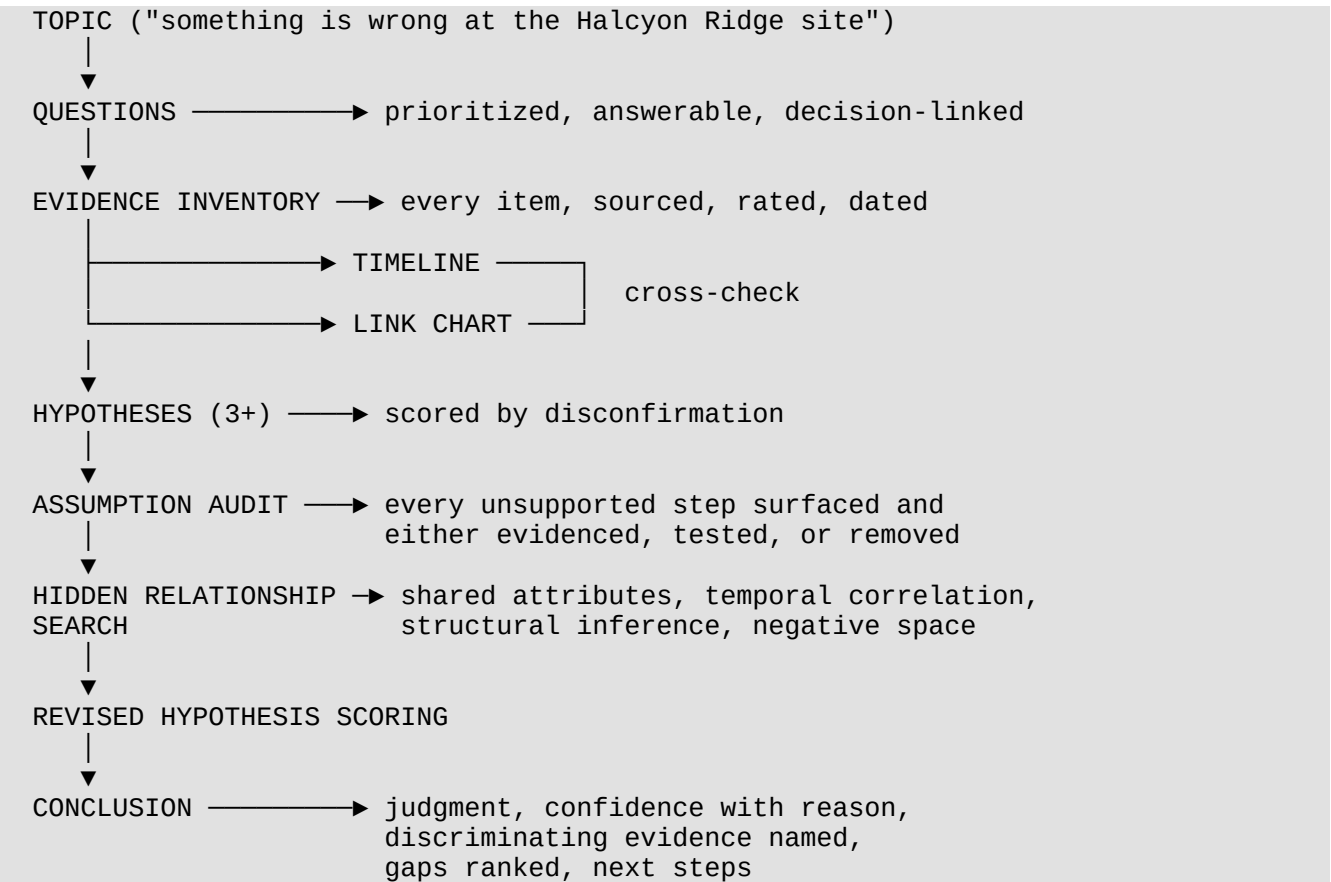
**An inference** is a step of reasoning from evidence to a conclusion, which necessarily adds something the evidence does not itself state. "The load loss at 02:14 indicates the fire had reached the service panel by that time."

|                   | <b>Fact</b>              | <b>Evidence</b>              | <b>Inference</b>                |
|-------------------|--------------------------|------------------------------|---------------------------------|
| Exists            | Independently of us      | As an item, from a source    | Only in an analyst's reasoning  |
| Can be wrong?     | No, by definition        | Yes: the source can be wrong | Yes: the reasoning can be wrong |
| Carries a source? | No                       | Always                       | Cites the evidence it rests on  |
| Carries a rating? | No                       | Reliability and credibility  | Confidence                      |
| How to write it   | Rarely asserted directly | "Item 12 records..."         | "We assess that..."             |

|              | Fact                 | Evidence                                  | Inference                 |
|--------------|----------------------|-------------------------------------------|---------------------------|
| Failure mode | Assuming we have one | Treating an item as the fact it describes | Presenting it as evidence |

**The practical rule for this entire chapter:** you almost never possess a fact. You possess evidence, and you make inferences from it. Products that say "the fire began at 02:14" without qualification are asserting a fact from an inference. Products that say "the utility meter log records a load loss at 02:14 (item 12, A1); we assess the fire reached the service panel by that time" are doing it correctly.

### 1.1.4 The Investigative Workflow



The loop back from hidden relationships to hypothesis scoring is the step students skip. A newly discovered indirect relationship changes the scoring of every hypothesis, and the matrix must be re-run rather than mentally adjusted.

### 1.1.5 The Four Failure Signatures

When an investigation reaches a wrong conclusion, the break is almost always in one of four places. Learning to recognize the signature is how you diagnose your own work and review someone else's.

| Signature                      | What you see                | Where the break is              |
|--------------------------------|-----------------------------|---------------------------------|
| <b>The premature narrative</b> | A clean story that explains | Hypothesis generation: only one |

| Signature                          | What you see                                                                                                   | Where the break is                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|                                    | everything, formed early, with contradicting items unexplained rather than addressed                           | was ever entertained                                              |
| <b>The load-bearing assumption</b> | A conclusion that survives only if an unstated premise holds; the premise appears nowhere in the evidence list | Assumption elimination: never performed                           |
| <b>The confident chart</b>         | A relationship diagram with lines nobody can source                                                            | Link analysis: edges asserted rather than evidenced               |
| <b>The smoothed timeline</b>       | An orderly sequence with no gaps and no conflicts, from messy sources                                          | Timeline construction: precision and inconsistency were flattened |

Note that none of these is a collection failure. All four are reasoning failures, and all four can occur in an investigation with perfect Chapter 4 documentation.

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## 1.2 Asking Questions

### 1.2.1 Why the Question Is the Investigation

The most consequential decision in an investigation is made before any evidence is examined: what are we asking? A well-formed question makes an investigation tractable. A badly formed one guarantees that no amount of evidence will feel sufficient, because there was never a condition under which the question could be answered.

Chapter 4 covered requirements from the consumer's side: what decision does this support. This section covers the investigator's side: how to turn a topic into a set of questions that can actually be worked.

### 1.2.2 From Topic to Question

A **topic** is a subject area. A **question** is something that can be answered. The conversion is a skill.

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TOPIC: "Problems at the Halcyon Ridge warehouse"
  |
  | Step 1: What is the concern, stated as a proposition?
  |
  ▼
CONCERN: "The fire may not have been accidental."
  |
  | Step 2: What would have to be true or false?
  |
  ▼
QUESTIONS:
Q1 When did the fire begin, and where within the structure?
Q2 What ignition sources were present at that location?
Q3 Who had physical access to that location in the
   preceding 12 hours?
Q4 What changed at the site in the 90 days before the fire?
Q5 Who benefits financially from the loss, and by how much?

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Q6 Are there prior events involving the same parties, locations, or patterns?

Step 3: Which of these can be answered, by what?

PRIORITIZED, SOURCED QUESTION SET

**The critical move is Step 2.** Ask what would have to be true for the concern to be justified, and what would have to be true for it to be unjustified. Questions generated from both directions produce a balanced question set; questions generated only from the concern produce an investigation that can only confirm.

### 1.2.3 The Six Question Types

Every investigation needs all six. Missing types produce characteristic blind spots.

| Type               | Form                                                | What it establishes                        | Blind spot if missing                                            |
|--------------------|-----------------------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| <b>Descriptive</b> | What, who, where, which                             | The basic entities and facts in play       | You investigate a situation you have not described               |
| <b>Temporal</b>    | When, in what order, how long                       | Sequence, duration, opportunity            | You cannot distinguish cause from coincidence                    |
| <b>Relational</b>  | Who is connected to whom, how                       | Structure, access, influence               | You see actors as isolated and miss the network                  |
| <b>Causal</b>      | Why, how did this produce that                      | Mechanism                                  | You have correlation and call it explanation                     |
| <b>Comparative</b> | Is this normal, how does it compare to the baseline | Whether anything anomalous occurred at all | You investigate ordinary variation as though it were significant |
| <b>Negative</b>    | What is absent that should be present               | Gaps, concealment, missing records         | You miss the most informative evidence in many cases             |

**The negative question deserves emphasis.** "Which records should exist and do not?" is frequently the highest-yield question in an investigation. A permit that was never filed, an inspection that never occurred, a log with a gap, a person who did not appear where their role required them: absence is evidence, and it is invisible to anyone who never asks for it.

### 1.2.4 Strong and Weak Questions

| Property   | Strong question                               | Weak question             |
|------------|-----------------------------------------------|---------------------------|
| Answerable | A specific state of the world could settle it | No answer would settle it |

| <b>Property</b>  | <b>Strong question</b>                | <b>Weak question</b>              |
|------------------|---------------------------------------|-----------------------------------|
| Bounded          | Names entities, a period, a location  | Open-ended                        |
| Neutral          | Does not presuppose the answer        | Loaded, presupposes wrongdoing    |
| Single-barrelled | Asks one thing                        | Asks three things joined by "and" |
| Falsifiable      | Something could show the answer is no | Cannot be shown false             |
| Sourced          | You can name what would answer it     | No source could answer it         |
| Consequential    | The answer changes what happens next  | Answering it changes nothing      |

### Worked rewrites:

| <b>Weak</b>                                                                      | <b>What is wrong</b>                              | <b>Strong</b>                                                                                                                                               |
|----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Is the company corrupt?"                                                        | Unbounded, loaded, unfalsifiable, no source       | "Do the county's award records show any contract to [entity] where the awarded value exceeded the posted estimate by more than 20%, in the past 36 months?" |
| "Why did the fire happen and who is responsible and what did the landlord know?" | Three questions; the third presupposes            | Split into three; rewrite the third as "What written notices, if any, did the landlord receive concerning the electrical system before the fire?"           |
| "Was the suspect at the scene?"                                                  | Presupposes a suspect; identity assumed           | "Which individuals' presence at the site between 22:00 and 03:00 is evidenced by access logs, and which access events are unattributed?"                    |
| "What is the competitor planning?"                                               | Unanswerable from open sources; asks about intent | "Which of the observable steps that necessarily precede a regional launch have occurred in the past 12 months?"                                             |
| "Is this normal?"                                                                | Unbounded, no baseline defined                    | "How does the volume of X in period Y compare with the same metric in the preceding four comparable periods?"                                               |

### 1.2.5 Prioritizing Questions

You will always have more questions than capacity. Rank on three dimensions and record the ranking:

| Dimension                   | Test                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Discriminating power</b> | Would the answer distinguish between your competing hypotheses? A question whose every possible answer is consistent with all hypotheses has low value regardless of how interesting it is |
| <b>Answerability</b>        | Is there a source that would settle it, at acceptable cost and within the deadline?                                                                                                        |
| <b>Consequence</b>          | Does the answer change what happens next: a decision, a course of action, another question?                                                                                                |

**The highest-value question in any investigation is the one that is both discriminating and answerable.** Students consistently over-invest in interesting questions with low discriminating power, which is how investigations become large and inconclusive.

### 1.2.6 Question Sets Evolve

A question set is a living document. Three movements are normal and each should be recorded:

**Answered.** Record the answer, the evidence, and the confidence. Then ask the follow-on question the answer generates.

**Refined.** The original question was too broad; it splits into three narrower ones. Record the split so the lineage is traceable.

**Retired.** The question turned out not to discriminate, or the decision it supported changed. Record the retirement and the reason. Silent abandonment is how investigations lose track of what they set out to learn.

### 1.2.7 Questions Investigators Fail to Ask

A checklist worth running against every investigation, because these are the recurring omissions:

- What would this look like if the innocent explanation were true?
- What is the base rate? How often does this happen anyway?
- Who else had the same opportunity, and why are they not in my analysis?
- What records should exist and do not?
- Who is not in my chart, and why?
- What did I learn from the source that most wants me to believe a particular thing?
- What would the subject say if asked, and does my evidence rule it out?
- If I am wrong, what is the most likely reason?

## 1.3 Developing Hypotheses

### 1.3.1 What a Hypothesis Is

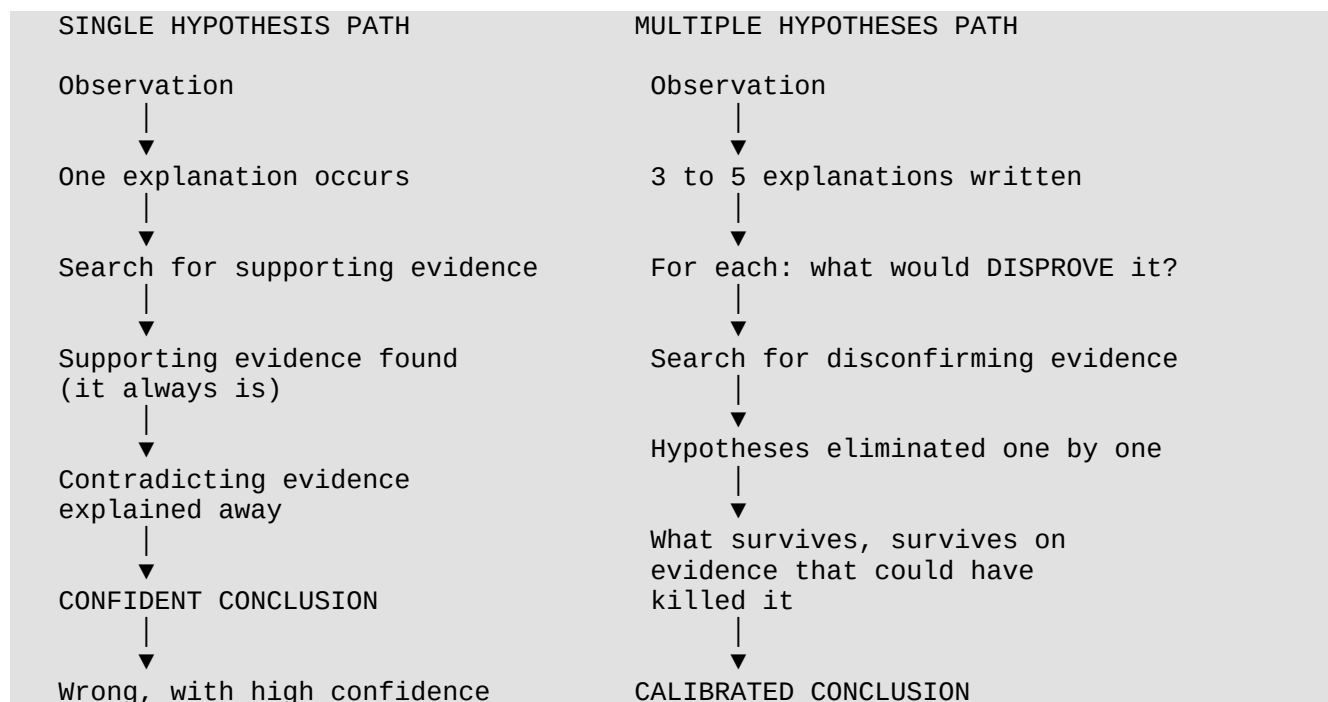
A **hypothesis** is a proposed explanation, stated so that evidence could support or undermine it. It is not a belief, not a theory of the case, and not a suspicion. It is a candidate answer to an investigative question, written in a form that can be tested.

|             | Suspicion                    | Hypothesis                                         |
|-------------|------------------------------|----------------------------------------------------|
| Stated      | Vaguely, often unstated      | Explicitly, in writing                             |
| Testable    | Not necessarily              | By construction                                    |
| Number held | One                          | Three or more, simultaneously                      |
| Attachment  | The investigator believes it | The investigator holds all of them at arm's length |
| Disproof    | Feels like a personal loss   | Is the point of the exercise                       |

### 1.3.2 The Method of Multiple Working Hypotheses

The central discipline of this chapter, and it predates modern intelligence work by more than a century. The idea is simple and psychologically difficult: **hold several explanations at once, deliberately, so that no single one accumulates the affection that turns an investigator into an advocate.**

The mechanism by which single-hypothesis investigation fails is well documented and entirely predictable:



**Why "supporting evidence found (it always is)" is the crucial line.** In any moderately complex situation, evidence consistent with almost any plausible explanation exists. Finding it proves nothing. This is why the search must be for evidence that would *disprove*, and why hypotheses are eliminated rather than selected.

### 1.3.3 Generating a Good Hypothesis Set

A hypothesis set should be **exhaustive enough** to include the true explanation and **mutually exclusive enough** that evidence can distinguish among the members.

**Generation techniques:**

| Technique                       | How                                                        | Guards against                                             |
|---------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| <b>The null hypothesis</b>      | "Nothing unusual occurred; this is ordinary variation"     | Investigating noise as though it were signal               |
| <b>The innocent explanation</b> | "The behavior occurred and has a benign cause"             | Presumption of wrongdoing                                  |
| <b>The obvious explanation</b>  | The one that first occurred to you, stated plainly         | Pretending you have no prior                               |
| <b>Role reversal</b>            | "What if the party I assume is the victim is the actor?"   | Fixed role assignment                                      |
| <b>Third-party</b>              | "Someone not currently in my analysis did this"            | Closed suspect pools                                       |
| <b>Compound</b>                 | "Two unrelated things happened close together"             | Forcing one explanation onto two events                    |
| <b>Deception</b>                | "Someone arranged the evidence to produce this appearance" | Taking a convenient evidence pattern at face value         |
| <b>Incompetence</b>             | "This is error, not design"                                | Attributing to intent what is explained by process failure |

**A minimum viable set for most investigations:** the null, the innocent explanation, the obvious explanation, and at least one of third-party, compound, or deception. Four hypotheses. If you cannot write four, you have not understood the situation well enough to investigate it.

### 1.3.4 Writing Hypotheses Properly

| Property      | Requirement                                              |
|---------------|----------------------------------------------------------|
| Declarative   | A statement, not a question                              |
| Specific      | Names actors, mechanism, and rough timing where possible |
| Falsifiable   | You can state what observation would defeat it           |
| Fairly stated | The one you consider least likely is written as its      |

| Property        | Requirement                                                                        |
|-----------------|------------------------------------------------------------------------------------|
|                 | own proponent would write it                                                       |
| Non-overlapping | Two hypotheses that would be supported by all the same evidence are one hypothesis |

**Worked example, from the Halcyon Ridge scenario used in Labs 2 through 4:**

| #  | Hypothesis                                                                                                                                            | What would disprove it                                                                                                                |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| H1 | The fire began accidentally from an electrical fault in the north bay service panel, which had documented deficiencies                                | Origin determined to be elsewhere; panel found intact; ignition sequence inconsistent with electrical fault                           |
| H2 | The fire was set deliberately by or on behalf of the tenant to realize insurance proceeds on inventory                                                | Tenant's coverage insufficient or unfavorable relative to the loss; inventory verified as present and valuable; no access opportunity |
| H3 | The fire was set deliberately by a third party unconnected to the tenant, for reasons unrelated to insurance                                          | Access controls exclude outside entry; no third party with motive or opportunity identified after directed search                     |
| H4 | The fire began accidentally from a non-electrical source (hot work, materials storage, or equipment) and the electrical deficiencies are coincidental | Origin located at the panel; no hot work or incompatible storage evidenced; timing inconsistent with those sources                    |

Note that H1 and H4 are both accidental but are separated, because different evidence discriminates between them and conflating them would hide that.

### **1.3.5 Scoring Evidence Against Hypotheses**

The matrix is the core artifact. Rows are evidence items and derived findings; columns are hypotheses.

| Item                                             | H1 Electrical | H2 Insurance | H3 Third party | H4 Other accidental | Discriminating ? |
|--------------------------------------------------|---------------|--------------|----------------|---------------------|------------------|
| E1 Panel deficiency notice issued 4 months prior | ++            | +            | 0              | 0                   | Weakly           |
| E2 Inventory insured at replacement cost, policy | 0             | ++           | 0              | 0                   | Yes              |

| Item                                                       | H1 Electrical | H2 Insurance | H3 Third party | H4 Other accidental | Discriminating ? |
|------------------------------------------------------------|---------------|--------------|----------------|---------------------|------------------|
| increased 6 weeks prior                                    |               |              |                |                     |                  |
| E3 Access log shows no badge entries after 19:40           | 0             | --           | --             | 0                   | Yes              |
| E4 Rear door contact sensor offline since prior week       | 0             | +            | ++             | 0                   | Yes              |
| E5 Weather: no lightning within 40 km                      | -             | 0            | 0              | -                   | Weakly           |
| E6 Fire origin assessed as north bay by responding officer | ++            | +            | +              | -                   | No               |

**Notation:** ++ strongly consistent, + consistent, 0 neutral, - inconsistent, -- strongly inconsistent.

### Three rules for scoring:

**Score consistency, not support.** The question for each cell is "if this hypothesis were true, would I expect to see this item?" not "does this item make me believe this hypothesis?"

**Weight the inconsistencies.** A single strong inconsistency does more work than five consistencies. In the matrix above, E3 does more to shape the conclusion than E1 and E6 combined.

**Ignore the row totals.** Counting plus signs is not analysis. E6 is consistent with three hypotheses and therefore contributes almost nothing, while E4 discriminates sharply. The discriminating column, not the arithmetic, drives the result.

### 1.3.6 Eliminating and Surviving

A hypothesis is eliminated when evidence is inconsistent with it in a way that cannot be reasonably explained. Two disciplines:

**Elimination must be recorded with its basis.** "H3 eliminated on E3 and E7: all perimeter access points evidenced as secured, with the rear sensor gap addressed by E9." A hypothesis that quietly stops appearing in your notes was not eliminated; it was forgotten.

**Surviving is not the same as proven.** When one hypothesis survives, the correct statement is that it is the best-supported explanation given the current evidence, at a stated confidence, with the gaps that

could change it. The survivors of an elimination process inherit the limits of that process, and if collection was incomplete, so is the elimination.

### ***1.3.7 When Hypotheses Should Be Revisited***

- New evidence arrives that is strongly inconsistent with the surviving hypothesis.
  - A hidden relationship is discovered (Section 1.7), which typically re-scores several rows at once.
  - An assumption audit (Section 1.6) removes a premise that a hypothesis rested on.
  - A timeline inconsistency is resolved, changing the sequence.
  - Someone outside the investigation proposes an explanation you did not generate. This is common, and the correct response is to add it to the matrix rather than to argue with it.
- 

## **1.4 Timeline Creation**

### ***1.4.1 What Timelines Do***

A timeline orders events in time. That sounds trivial and it is the single most productive analytic artifact in investigative work, because time constrains possibility. Four things a timeline establishes that nothing else does as cheaply:

**Sequence.** What happened before what. Causation requires the cause to precede the effect, so sequence errors destroy causal reasoning.

**Opportunity.** Who could have done what, given where they were and when.

**Impossibility.** The strongest analytic finding available. If the timeline shows an actor 300 km away 40 minutes before an event, a whole branch of hypotheses closes.

**Anomaly.** Clusters, gaps, and unusual timing. An action taken at 03:40 that is normally taken during business hours is a finding.

### ***1.4.2 Chronological and Event-Based Analysis***

Two ways to organize the same material, for different purposes.

**Chronological analysis** places every event on a single time axis, regardless of actor or subject. It answers "what was happening at this moment" and is how you find coincidence, convergence, and impossibility.

**Event-based analysis** groups events into episodes and traces each episode's internal structure. It answers "how did this particular thing unfold" and is how you find process, pattern, and repetition.

|                 | <b>Chronological</b> | <b>Event-based</b> |
|-----------------|----------------------|--------------------|
| Organizing axis | Absolute time        | The episode        |

|                  | <b>Chronological</b>                                                                                  | <b>Event-based</b>                                                 |
|------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Answers          | What was happening simultaneously?                                                                    | How did this episode develop?                                      |
| Reveals          | Convergence, coincidence, impossibility, gaps                                                         | Pattern across repeated episodes, internal sequence                |
| Best for         | Single-incident investigations, alibi and opportunity work                                            | Recurring conduct, campaigns, series offenses, fraud schemes       |
| Weakness         | Long or repeated conduct becomes an unreadable list                                                   | Loses cross-episode simultaneity                                   |
| Typical artifact | One master timeline, sorted by timestamp                                                              | One timeline per episode, plus a comparison matrix across episodes |
| Combined use     | Build both. Chronological for the incident; event-based for each prior similar incident, then compare |                                                                    |

**In practice, serious investigations build both**, and the comparison between them is productive: an event-based view reveals that a scheme has five stages, and the chronological view then reveals that stage three of episode B overlaps with stage one of episode C, which tells you the operation was running in parallel.

### **1.4.3 Timeline Entry Structure**

Every row carries these fields. The precision and source columns are what separate an analytic timeline from a narrative.

| <b>Field</b>           | <b>Content</b>                                       | <b>Why</b>                                                    |
|------------------------|------------------------------------------------------|---------------------------------------------------------------|
| <b>Timestamp</b>       | Normalized, with time zone                           | Chapter 4's processing standard                               |
| <b>Precision</b>       | Exact, minute, hour, day, month, or bracketed window | A bracketed change rendered as a specific date is a quiet lie |
| <b>Event</b>           | One sentence, stated as observed                     | Not as interpreted                                            |
| <b>Actors</b>          | Canonical entity names                               | Feeds the link chart                                          |
| <b>Location</b>        | Normalized                                           | Feeds opportunity analysis                                    |
| <b>Source item IDs</b> | Every item supporting this row                       | Traceability                                                  |
| <b>Categories</b>      | How many distinct source categories                  | Chapter 3's corroboration measure                             |
| <b>Type</b>            | Observed event, reported event, or inferred event    | Critical, see below                                           |
| <b>Confidence</b>      | In the event having occurred as described            | Separate from source reliability                              |

| Field | Content                              | Why                                |
|-------|--------------------------------------|------------------------------------|
| Flags | Conflict, gap, anomaly, unattributed | Drives the next investigative step |

The type field is the one investigators most often omit and most need. Three kinds of entry sit on a typical timeline:

- **Observed:** a sensor, log, record, or camera captured it. Highest weight.
- **Reported:** a person says it happened. Weight depends on the person and their basis.
- **Inferred:** nothing recorded it, but the analyst concludes it must have occurred. "The vehicle must have departed by 21:50 to arrive at 22:20."

Inferred entries are legitimate and necessary, and they must be visibly marked, because a timeline that mixes them silently presents the analyst's reasoning as though it were the record.

#### 1.4.4 A Worked Timeline Fragment

| TIME (UTC-6) | PREC  | TYPE | EVENT                                                                   | SRC | CAT  | CONF |
|--------------|-------|------|-------------------------------------------------------------------------|-----|------|------|
| FLAGS        |       |      |                                                                         |     |      |      |
| D-46         | month | OBS  | Panel deficiency notice issued to tenant by county electrical inspector | E1  | Gov  | High |
| D-42         | day   | OBS  | Contents coverage increased from \$1.2M to \$3.4M, effective D-40       | E2  | Corp | High |
| D-7          | day   | OBS  | Rear door contact sensor logged as                                      | E4  | Int  | High |
| ANOMALY      |       |      | offline; no repair ticket opened                                        |     |      |      |
| D-1 17:55    | min   | OBS  | Last badge-in of the day (staff ID 0412), north entrance                | E3  | Int  | High |
| D-1 19:40    | min   | OBS  | Last badge-out of the day (same ID)                                     | E3  | Int  | High |
| D-1 ~20:00   | hour  | INF  | Site unoccupied from approximately                                      | --  | --   | Mod  |
| INFERRED     |       |      | this time (no further access events)                                    |     |      |      |
| D-1 22:10    | min   | OBS  | Vehicle passes municipal camera                                         | E7  | Gov  | Mod  |
| PARTIAL      |       |      | 2 blocks north, direction of site;                                      |     |      |      |
| PLATE        |       |      | partial plate                                                           |     |      |      |
| D 02:14      | exact | OBS  | Utility meter records load loss at the service address                  | E12 | Gov  | High |
| D 02:19      | exact | OBS  | First 911 call, passing motorist                                        | E13 | Gov  | High |
| D 02:41      | min   | OBS  | Fire crews report heavy involvement, north bay                          | E14 | Gov  | High |
| D ~02:00     | hour  | INF  | Ignition occurred within                                                | --  | --   | Mod  |
| INFERRED     |       |      | approximately 20 minutes before                                         |     |      |      |
| CONFLICT     |       |      | the load loss                                                           |     |      |      |
| WITH E15     |       |      |                                                                         |     |      |      |
| D+3          | day   | REP  | Tenant states inventory was at                                          | E15 | Rep  | Low  |

|          |     |     |                                   |     |      |      |
|----------|-----|-----|-----------------------------------|-----|------|------|
| CONFLICT |     |     | full seasonal level               |     |      |      |
| WITH E16 |     |     |                                   |     |      |      |
| D+9      | day | OBS | Freight records show last inbound | E16 | Corp | High |
| CONFLICT |     |     | delivery 5 weeks prior; outbound  |     |      |      |
| WITH E15 |     |     | volume unchanged                  |     |      |      |

**Read what this fragment does.** It shows a sequence, marks two inferred entries as inferred, flags an anomaly seven days before the incident, records a conflict between a reported statement and a documentary record, and notes where a source is only partial. Every one of those flags is an investigative step waiting to be taken.

### 1.4.5 Finding Inconsistencies

Inconsistencies are the most valuable output of a timeline. Five types, each with a different meaning:

| Type                 | What it looks like                                                                                                    | What it usually means                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Contradiction</b> | Two sources place the same event at different times, or one says it happened and one says it did not                  | One source is wrong, or they describe different events, or someone is misrepresenting       |
| <b>Impossibility</b> | The sequence requires something physically impossible: travel time, simultaneous presence, an effect before its cause | A timestamp is wrong, an identity is wrong, or an assumption in the reconstruction is wrong |
| <b>Gap</b>           | A period with no evidence where evidence should exist                                                                 | Records missing, collection incomplete, or activity deliberately unlogged                   |
| <b>Anomaly</b>       | An event that occurred, but at an unusual time, frequency, or order                                                   | Genuinely significant, or a mundane explanation not yet found                               |
| <b>Convergence</b>   | Multiple independent threads intersect in a narrow window                                                             | Frequently the most important finding a timeline produces                                   |

**Handling discipline.** Do not resolve an inconsistency by choosing the source you prefer. Record it, state both versions, and derive the investigative step that would settle it. An investigation whose timeline shows no conflicts, built from messy real-world sources, has been smoothed, and the smoothing removed exactly the information the timeline existed to produce.

### 1.4.6 Deriving Next Steps From a Timeline

Each flag maps to an action:

| Flag                | Investigative step                                                                                                     |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Gap                 | Identify which record type would cover the period; determine whether it exists, was never created, or was not retained |
| Contradiction       | Identify the more reliable production process; seek a third, independent category                                      |
| Impossibility       | Re-verify the timestamps and identity attribution in both entries before concluding anything about the actors          |
| Anomaly             | Establish the baseline: is this actually unusual for this actor, site, or period?                                      |
| Convergence         | Test whether the intersection is coincidence: how often would these threads coincide by chance?                        |
| Inferred entry      | Identify what observation would convert it to observed                                                                 |
| Partial attribution | Determine what additional identifier would complete it, and whether that identifier is obtainable                      |

### 1.4.7 Timeline Failure Modes

- **Smoothing.** Removing precision and conflict to make the narrative readable. The most common and most damaging.
- **Publication-date contamination.** Placing an event at the date it was reported rather than when it occurred (Chapter 4, Section 1.5.3).
- **Time zone collapse.** Mixing zones, producing false sequence.
- **Unmarked inference.** Presenting reconstruction as record.
- **Selection.** Including only events that fit the leading hypothesis, which produces a timeline that proves whatever it was built to prove.
- **False precision.** "14 March" when the evidence supports "between 2 March and 19 April."

## 1.5 Link Analysis

### 1.5.1 What Link Analysis Does

Link analysis orders entities in relationship space, as a timeline orders events in time. It answers structural questions: who is connected to whom, through what, how tightly, and where the structure is vulnerable or informative.

Four things a link chart establishes:

**Structure.** Who occupies which position: centre, periphery, bridge.

**Access.** Who could reach what, through whom.

**Concentration.** Which small set of nodes carries a disproportionate share of the connections, and is therefore the highest-value investigative target.

**Absence.** Who or what is missing from a structure that should contain them. As with negative questions, absence is frequently the finding.

### **1.5.2 Anatomy of a Link Chart**

**Nodes** are entities. Type them, because untyped charts become unreadable and because type constrains what relationships are possible.

| <b>Node type</b> | <b>Examples</b>                                         | <b>Typical symbol convention</b> |
|------------------|---------------------------------------------------------|----------------------------------|
| Person           | Named individuals in their role                         | Circle                           |
| Organization     | Companies, agencies, associations                       | Rectangle                        |
| Location         | Addresses, parcels, facilities                          | Triangle                         |
| Event            | Meetings, transactions, incidents                       | Diamond                          |
| Asset            | Vehicles, equipment, accounts, property                 | Rounded rectangle                |
| Digital artifact | Domains, accounts, devices, certificates, phone numbers | Hexagon                          |
| Document         | Filings, contracts, permits, reports                    | Page shape                       |

**Edges** are relationships. Every edge carries four attributes, and Rule 6 requires all four:

| <b>Attribute</b> | <b>Content</b>                                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Type</b>      | Employment, ownership, kinship, communication, co-location, transaction, control, membership, shared attribute |
| <b>Direction</b> | Directed (A pays B, A controls B) or undirected (A and B share an address)                                     |
| <b>Basis</b>     | The specific evidence supporting it, by item ID                                                                |
| <b>Strength</b>  | Confirmed, probable, possible, or hypothesized                                                                 |

**Strength ratings, defined:**

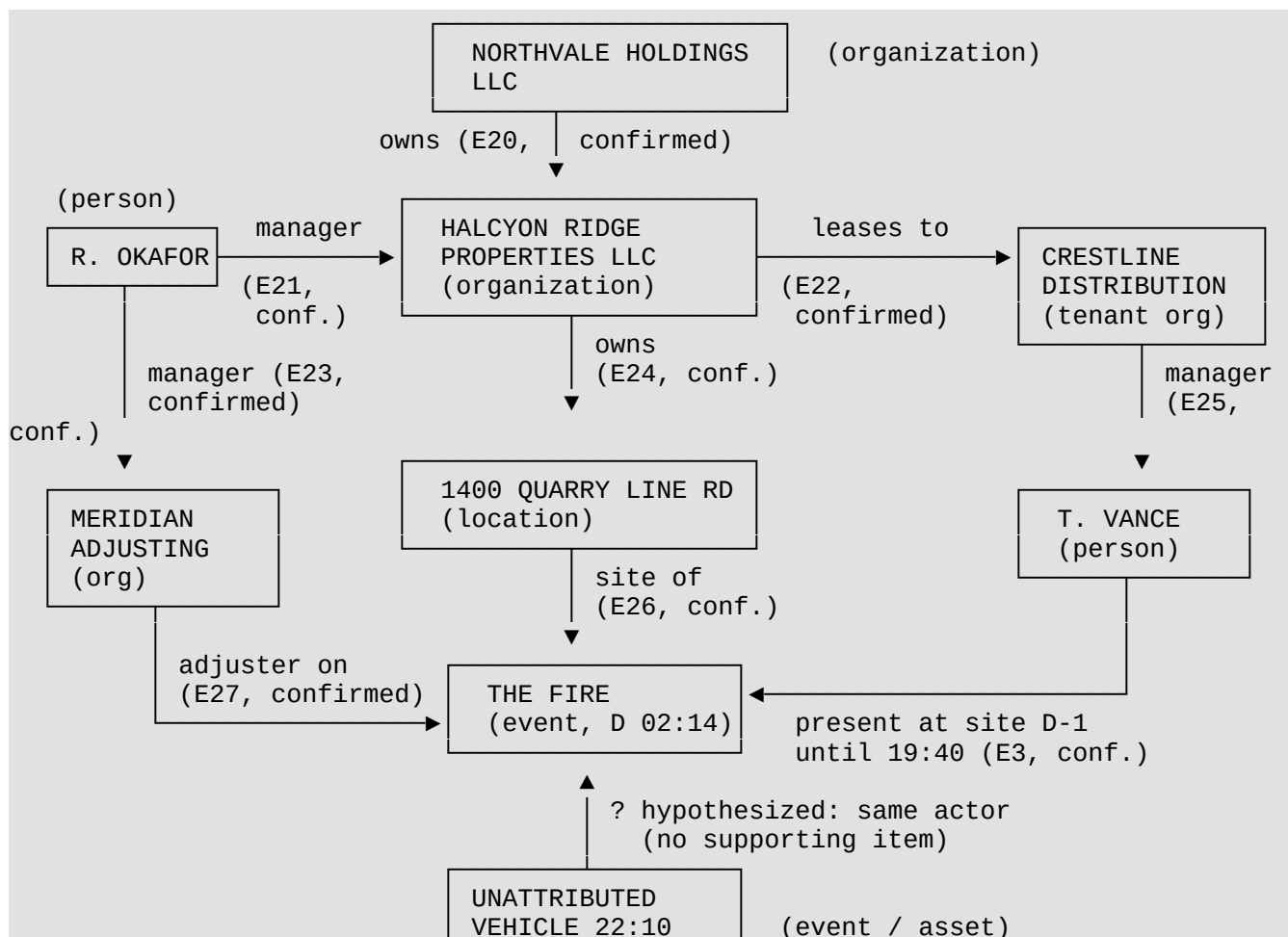
| <b>Rating</b> | <b>Meaning</b>         | <b>Rendering convention</b> |
|---------------|------------------------|-----------------------------|
| Confirmed     | Documented in a record | Solid line                  |

| Rating       | Meaning                                                                            | Rendering convention                                               |
|--------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|              | produced under obligation, or corroborated across two independent categories       |                                                                    |
| Probable     | Supported by one credible source, consistent with other evidence, no contradiction | Dashed line                                                        |
| Possible     | Suggested by circumstantial or single weak evidence                                | Dotted line                                                        |
| Hypothesized | Proposed by the analyst; no supporting evidence yet                                | Dotted line with a question mark, and it must be visually distinct |

**The rendering convention matters more than it sounds.** A chart in which confirmed and hypothesized edges look identical will be read as though every line is confirmed, by every reader, every time, including by you three weeks later.

### 1.5.3 A Worked Link Chart

Rendered in text for the Halcyon Ridge scenario:



partial plate

SHARED ATTRIBUTE (indirect, derived):

R. OKAFOR appears as manager of BOTH the landlord entity and the adjusting firm engaged on the loss (E21 + E23). Neither document states a relationship between the two organizations. The relationship is INDIRECT, derived from a shared officer, and it is the single most investigatively significant feature of this chart.

**What to notice.** The chart's most important content is not any single edge. It is the indirect relationship at the bottom, which no document asserts and which emerges only when two separate filings are placed on the same diagram. Section 1.7 is about finding exactly this.

Also notice that the one hypothesized edge is visually distinct and labeled as having no supporting item. That edge is a question, not a finding.

### 1.5.4 Direct and Indirect Relationships

|                          | Direct relationship                                      | Indirect relationship                                                                   |
|--------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Definition               | A single edge between two nodes, asserted by evidence    | A connection through one or more intermediate nodes, or derived from a shared attribute |
| Evidence                 | A record states it                                       | No record states it; it is inferred from the structure                                  |
| Example                  | "A is the manager of B" per a registry filing            | "A and C are connected because both list B as their registered agent"                   |
| Strength                 | As strong as the record                                  | Depends entirely on how discriminating the intermediary is                              |
| What it supports         | The specific relationship asserted                       | Opportunity, access, possible coordination                                              |
| What it does NOT support | More than it says                                        | Any claim of intent, agreement, or coordination without further evidence                |
| Failure mode             | Over-reading a formal relationship as an operational one | Treating a shared attribute as a relationship when the attribute is common              |

**The discriminating-intermediary test.** An indirect relationship is only meaningful if the intermediate node is *rare*. Two companies sharing a registered agent that serves 40,000 entities tells you almost nothing. Two companies sharing an unusual office suite address, a specific individual, or an uncommon phone number tells you a great deal. Before drawing an indirect relationship, establish how many other entities share the same intermediary. If you cannot establish that, mark the edge as possible rather than probable and record the gap.

**The most common student error in this chapter** is drawing indirect relationships at the same visual weight as direct ones and then reasoning about the chart as though every line meant the same thing.

### 1.5.5 Link Analysis Techniques

| Technique                                | What it does                                                                                                         | When to use it                                                    | What it produces                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Association matrix</b>                | A grid of every entity against every other, marked confirmed, suspected, or none                                     | Early, before drawing anything; forces you to consider every pair | A complete pairwise assessment and a list of pairs you had not considered               |
| <b>Relationship (link) chart</b>         | The visual network with typed nodes and rated edges                                                                  | Once the matrix is populated                                      | Structure, clusters, isolated nodes                                                     |
| <b>Activity matrix</b>                   | Entities against events, activities, or locations                                                                    | When you care about who participated in what                      | Participation patterns and co-participation                                             |
| <b>Centrality analysis</b>               | Identifies which nodes are most connected (degree), most between others (betweenness), or closest to all (closeness) | Once the chart is stable                                          | High-degree hubs and high-betweenness brokers                                           |
| <b>Broker / cut-point identification</b> | Finds nodes whose removal disconnects the network                                                                    | Disruption planning, or identifying the essential intermediary    | The node that matters most structurally, which is frequently not the most connected one |
| <b>Cluster detection</b>                 | Groups of nodes more connected to each other than to the rest                                                        | Large charts                                                      | Sub-organizations, cells, or unrelated components accidentally combined                 |
| <b>Path analysis</b>                     | The shortest and alternate routes between two nodes                                                                  | Access and influence questions                                    | How A could reach B, and through whom                                                   |
| <b>Temporal layering</b>                 | The chart rendered at several points in time                                                                         | Any investigation spanning months or years                        | When relationships formed and dissolved, which is often the finding                     |
| <b>Attribute overlay</b>                 | Shared attributes (addresses, phones, agents, devices) rendered as edges                                             | Hidden relationship search                                        | Indirect connections invisible in the direct chart                                      |
| <b>Negative space analysis</b>           | Examining expected edges that are absent                                                                             | Structural review                                                 | Missing intermediaries, deliberate separation, or gaps in collection                    |

**Degree versus betweenness deserves a sentence.** The most connected node is the most visible. The

node with the highest betweenness sits on the most paths between other nodes, and it is frequently the broker who makes the structure work. In many investigations the high-degree node is a well-known principal and the high-betweenness node is an administrator, an agent, or a facilitator nobody had noticed. Ask for both.

### **1.5.6 Rule 6 in Practice**

A chart is a persuasive object. Readers accept lines as facts. Three disciplines protect against that:

**Every edge has an item ID.** No exceptions. If you cannot cite an item, the edge is hypothesized and must be rendered as such.

**Maintain an edge register alongside the chart.** A table listing every edge, its type, direction, basis, strength, and the analyst who added it. The chart is the display; the register is the record.

**Review the chart for edges you cannot defend.** Before any product, walk every line and ask what item supports it. In practice, students find one to three indefensible edges in their own charts, every time. Removing them is the exercise working.

### **1.5.7 Link Analysis Failure Modes**

- **The confident chart.** Unsourced edges, uniform rendering, read as conclusion.
- **Guilt by association.** A connection is not participation. That two entities are connected says nothing about whether either knew what the other did.
- **The common intermediary.** Indirect relationships built on non-discriminating shared attributes.
- **Scope explosion.** Adding every entity that appears anywhere until the chart is unreadable and every node connects to every other. Charts should be bounded by the question.
- **Static charts for dynamic structures.** A chart with no temporal layer implies all relationships existed simultaneously, which is frequently false and occasionally decisive.
- **Missing node types.** Charts of only people, when the informative connections run through addresses, agents, and digital artifacts.

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## **1.6 Eliminating Assumptions**

### **1.6.1 Assumptions Are Not Optional**

Every investigation rests on assumptions, and it must. You assume the timestamps are approximately right, the records describe what they claim to describe, the entity names refer to the entities you think, and the world behaved normally except where evidence says otherwise. Without some assumptions you could not reason at all.

The problem is never that assumptions exist. It is that **unexamined assumptions are invisible, and invisible assumptions carry weight without earning it.**

The goal of this section is not to eliminate assumptions. It is to convert every hidden assumption into either an evidenced finding, an explicit and tested assumption, or a removed premise.

### 1.6.2 Where Assumptions Enter

| Entry point                  | Typical assumption                                                           | How it damages                                    |
|------------------------------|------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Question formulation</b>  | The concern is justified; something wrong occurred                           | Every subsequent question searches for wrongdoing |
| <b>Entity resolution</b>     | Same name means same entity                                                  | Invented relationships, or missed ones            |
| <b>Source acceptance</b>     | The record describes what its title suggests                                 | Findings built on a misread field                 |
| <b>Coverage</b>              | If it happened, a record would exist                                         | Absence read as evidence of non-occurrence        |
| <b>Timeline construction</b> | Reported times are accurate; the sequence implies causation                  | False causal chains                               |
| <b>Link analysis</b>         | Connection implies coordination; formal role implies operational involvement | Guilt by association                              |
| <b>Behavioral reasoning</b>  | People act rationally, and in the way I would act                            | Mirror imaging (Chapter 1)                        |
| <b>Baseline</b>              | This is unusual                                                              | Ordinary variation investigated as anomaly        |
| <b>Continuity</b>            | Roles, ownership, and addresses that are current were current then           | Anachronistic structures                          |
| <b>Completeness</b>          | I have most of the relevant evidence                                         | Confidence unwarranted by coverage                |

### 1.6.3 The Assumption Register

The artifact that makes this technique operational. One row per assumption, maintained throughout the investigation.

| Field                      | Content                                                                |
|----------------------------|------------------------------------------------------------------------|
| ID                         | For reference in the product                                           |
| Assumption, stated plainly | "The badge log records all entries to the building"                    |
| Where it entered           | Which analytic step relies on it                                       |
| What rests on it           | Which findings or hypotheses fail if it is wrong                       |
| Type                       | Coverage, identity, source, temporal, behavioral, baseline, continuity |
| Status                     | Untested, tested and supported, tested and refuted,                    |

| Field                | Content                                       |
|----------------------|-----------------------------------------------|
|                      | untestable                                    |
| Test performed       | What was done                                 |
| Result               | What was found                                |
| Consequence if wrong | Stated explicitly                             |
| Residual             | If untestable, the confidence penalty applied |

### 1.6.4 Surfacing Hidden Assumptions

Five techniques, in increasing order of difficulty:

- 1. The "therefore" audit.** Walk your written findings and circle every "therefore," "which means," "indicating," "shows that," and "confirms." Each is a joint between evidence and conclusion, and an assumption lives in every joint. Write it down.
- 2. The negation test.** Take each finding and ask: what would have to be true for this to be wrong? If the answer is "the source is lying" you have a source assumption. If it is "the record system covers this" you have a coverage assumption.
- 3. The stranger test.** Hand your conclusion, with only the evidence list, to someone with no context. Ask what they would conclude. The difference between their conclusion and yours is composed of your assumptions.
- 4. The reverse construction.** Start from your conclusion and work backward, item by item, asking at each step what would need to be true to get from the previous step to this one. This surfaces chained assumptions that individually look reasonable.
- 5. The steel-man reconstruction.** Write the strongest case for the hypothesis you rejected. The evidence you find yourself reaching for to defeat it will reveal which of your premises were doing the work.

### 1.6.5 Cognitive Biases and Their Specific Controls

Assumptions are frequently the residue of bias. Each bias has a control that works, and generic exhortations to "be objective" are not among them.

| Bias                     | How it operates in an investigation                                                                          | Control                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Confirmation bias</b> | Evidence consistent with the leading hypothesis is noticed and weighted; contrary evidence is explained away | Score matrices by consistency rather than support; require the disconfirmation question per hypothesis |
| <b>Anchoring</b>         | The first hypothesis or the first evidence item sets the frame                                               | Generate all hypotheses before scoring any evidence; use Chapter 4's collection order                  |

| <b>Bias</b>                 | <b>How it operates in an investigation</b>                                 | <b>Control</b>                                                                        |
|-----------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Availability</b>         | Vivid or recent evidence outweighs mundane evidence                        | Weight by production process and discriminating power, not by salience                |
| <b>Mirror imaging</b>       | Assuming the subject reasons as you do                                     | Explicitly write the subject's incentives from their position                         |
| <b>Narrative fallacy</b>    | The coherent story is preferred because it is coherent                     | Require every unexplained item to be listed and addressed, not omitted                |
| <b>Base rate neglect</b>    | An unusual event is treated as significant without asking how common it is | Establish the baseline before calling anything anomalous                              |
| <b>Guilt by association</b> | Connection is read as participation                                        | Rule 6: edge type and strength; participation requires its own evidence               |
| <b>Hindsight bias</b>       | Once the outcome is known, prior indicators look obvious                   | Reconstruct what was knowable at the time; count the indicators that led nowhere      |
| <b>Sunk cost</b>            | A hypothesis is retained because effort was invested in it                 | Elimination is recorded with its basis, so abandoning is a documented act, not a loss |
| <b>Authority bias</b>       | An official or senior source is accepted without evaluation                | Rate every source on production process, including official ones                      |
| <b>Attribution error</b>    | Behavior attributed to character rather than circumstance                  | Include the incompetence and process-failure hypotheses in every set                  |

### ***1.6.6 Rewriting Conclusions on Verified Evidence Only***

The output of an assumption audit is a rewritten conclusion. The procedure:

- 1. List every claim in the conclusion**, one per line.
- 2. For each, cite the specific evidence items.** Not "the investigation found" but item IDs.
- 3. Any claim with no cited item is an assumption.** Either find the evidence, mark it explicitly as an assumption with its consequence, or delete the claim.
- 4. Any claim citing an item that does not actually state it is an inference.** Rewrite it as an assessment with a confidence level.
- 5. Recalculate confidence** for the whole conclusion, because removing load-bearing assumptions almost always lowers it.

6. **Record what changed**, so the revision is visible.

**What a rewritten conclusion typically looks like.** Shorter, more heavily qualified, and more useful. Lab 5 is built entirely on this procedure, and students routinely find that a confident three-paragraph conclusion reduces to one paragraph of supported findings plus a list of open questions. That reduction is the correct outcome, not a failure.

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## 1.7 Finding Hidden Relationships

### 1.7.1 What "Hidden" Means

A hidden relationship is a connection that no single item states and that no party disclosed. It is not hidden because someone concealed it, necessarily. Most hidden relationships are hidden because the information that reveals them sits in two different records that nobody had placed side by side.

Three categories:

| Category         | Mechanism                                                                   | Example                                                                 |
|------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Latent</b>    | The information exists in the collected material but has not been assembled | Two filings, six months apart, listing the same officer                 |
| <b>Derived</b>   | The relationship exists only as an inference from a pattern                 | Two entities whose activity always occurs within 48 hours of each other |
| <b>Concealed</b> | Someone acted to prevent the relationship from being visible                | Ownership routed through an intermediary jurisdiction                   |

Most of what investigators find is latent. That is encouraging: the highest-yield technique in this section is systematic assembly, not cleverness.

### 1.7.2 The Five Discovery Mechanisms

**Mechanism 1: Shared attribute analysis.** The workhorse. Take every attribute you hold for every entity and look for repeats.

| Attribute class | Examples                                                                           | Discriminating power                                          |
|-----------------|------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Physical        | Address, suite number, parcel                                                      | High if specific; low for large office buildings              |
| Personnel       | Officers, managers, registered agents, signatories                                 | High for named individuals; low for commercial agent services |
| Digital         | Domains, name servers, hosting, certificates, analytics identifiers, phone numbers | Usually high                                                  |

| Attribute class | Examples                                                                | Discriminating power            |
|-----------------|-------------------------------------------------------------------------|---------------------------------|
| Financial       | Bank references in filings, lien holders, insurers, bonding companies   | Moderate to high                |
| Temporal        | Registration dates, filing dates, incorporation dates in tight clusters | Moderate; strong when clustered |
| Documentary     | Identical language, shared templates, uncorrected names in copied text  | Very high when present          |
| Operational     | Same vendors, same auditors, same counsel, same equipment               | Moderate                        |

**The procedure:** build an attribute table with entities as rows and attribute classes as columns, then sort by each column in turn. Repeats jump out. This is a mechanical exercise that finds things brilliant reading does not.

**Mechanism 2: Temporal correlation.** Events that consistently occur close together are related, or share a cause, or one causes the other. Look for:

- Actions by different entities within a narrow window, repeatedly.
- One entity's activity ceasing as another's begins.
- Registration or formation clustering.
- Lead and lag patterns: A always precedes B by roughly the same interval.

**The chance test.** Before treating a temporal correlation as meaningful, estimate how often the pattern would occur by chance given the base rates. Two events within a week in a dataset where both types occur monthly is unremarkable. Five paired occurrences at a consistent five-week lag is not.

**Mechanism 3: Structural inference.** The chart's shape reveals relationships its edges do not state.

- **Missing intermediary.** Two clusters that clearly interact but share no visible node. Something connects them and you have not found it. This is one of the most productive findings available.
- **Improbable isolation.** A node that should be connected given its role and is not, which suggests either a missing edge or deliberate separation.
- **Structural equivalence.** Two nodes with identical connection patterns to the same third parties, which often indicates they are the same actor, or interchangeable roles in a system.
- **Broker position.** A node sitting between clusters, which is frequently the actor that makes an arrangement function.

**Mechanism 4: Negative space.** What is absent that should be present.

- A party whose role in a transaction is required, and who appears nowhere.

- A record that every comparable transaction generated, missing for this one.
- A person present at every similar event, absent from this one.
- A gap in a log that is otherwise continuous.

**Mechanism 5: Cross-domain assembly.** The relationship becomes visible only when material from different Chapter 3 categories is placed on one artifact. A corporate filing plus a court exhibit plus a permit application produce a picture none of them contains. This is why cross-category collection matters analytically and not just for corroboration.

### 1.7.3 Testing a Discovered Relationship

Discovery is not confirmation. Every candidate relationship goes through four tests before it is drawn as anything stronger than possible:

| Test                           | Question                                                                                                                                             |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Discrimination</b>          | How many other entities share this intermediary or attribute?                                                                                        |
| <b>Alternative explanation</b> | What benign or structural reason would produce this pattern? (Shared industry, shared jurisdiction, common service provider, regulatory requirement) |
| <b>Independence</b>            | Do the two items evidencing it derive from a common origin?                                                                                          |
| <b>Temporal validity</b>       | Did the relationship exist at the time that matters, or only later?                                                                                  |

**The alternative explanation test catches the most errors.** Two companies using the same accounting firm in a small city where three firms serve the market is not a finding. The same two companies using the same accounting firm in a metropolitan market with four hundred firms is worth a second look. Both cases look identical on a chart.

### 1.7.4 Wording a Hidden Relationship Finding

Because these findings are inferential and frequently concern people, wording discipline is not optional.

| Do write                                                                                                      | Do not write                                          |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| "Registry filings list the same individual as manager of both entities (E21, E23)"                            | "The two companies are secretly the same"             |
| "The two entities share a registered agent address; we have not established how many other entities share it" | "The entities are connected through a shared address" |
| "We assess it is likely the two operations are coordinated, based on X, Y, Z; alternative                     | "The operations are coordinated"                      |

| Do write                                                                                | Do not write                     |
|-----------------------------------------------------------------------------------------|----------------------------------|
| explanations considered were A and B"                                                   |                                  |
| "No document in the collected material discloses a relationship between these entities" | "The relationship was concealed" |

That last row is the one that gets investigators into trouble. **Non-disclosure is not concealment.** Concealment requires evidence of an act intended to hide, and an inference of intent needs its own basis. The professional formulation states what the record shows and does not show, and stops there.

### 1.7.5 What Hidden Relationships Do to the Analysis

When you find one, three things happen and all three must actually be done:

1. **Re-score the hypothesis matrix.** A new relationship typically changes several rows at once, and mentally adjusting is not sufficient.
2. **Revisit the timeline.** When did this relationship form, and does its timing relative to the events change the picture?
3. **Re-audit assumptions.** A discovered relationship often invalidates an assumption of independence that other findings rested on.

## 1.8 Putting It Together

### 1.8.1 The Integrated Workflow

| PHASE                | PRIMARY ARTIFACT                               | FEEDS                                        |
|----------------------|------------------------------------------------|----------------------------------------------|
| Questions            | Question set, prioritized                      | → Hypotheses, collection plan                |
| Hypotheses           | Hypothesis list with disproof conditions       | → Evidence scoring, question refinement      |
| Evidence             | Inventory: sourced, rated, dated, typed        | → Timeline, chart                            |
| Timeline             | Master timeline with precision and flags       | → Opportunity, sequence, inconsistency steps |
| Link analysis        | Chart + edge register                          | → Structure, brokers, access                 |
| Scoring              | Hypothesis matrix                              | → Elimination, confidence                    |
| Assumptions          | Assumption register                            | → Rewritten findings, confidence adjustment  |
| Hidden relationships | Attribute table, structural findings           | → Re-scoring, new questions                  |
| Conclusion           | Product with confidence, gaps, recommendations | → Feedback, next investigation               |

### 1.8.2 Iteration Is Not Failure

New investigators expect to move through this once. Practitioners expect three or four passes, each

shorter than the last. A finding in the hidden relationship stage sends you back to hypothesis scoring; a timeline inconsistency sends you back to questions; an assumption audit sends you back to the chart.

**A useful stopping heuristic:** stop when a full pass produces no change to the hypothesis ranking, no new discriminating evidence, and no new question with high discriminating power. That is analytic saturation, and it is the analogue of Chapter 4's collection saturation.

### 1.8.3 Documentation Set for an Investigation

At the end, these artifacts should exist and be internally consistent:

| Artifact                      | Purpose                                                     |
|-------------------------------|-------------------------------------------------------------|
| Question set with status      | What we asked, answered, refined, retired                   |
| Evidence inventory            | Every item, sourced, rated, typed                           |
| Master timeline               | Sequence with precision, types, and flags                   |
| Link chart plus edge register | Structure with every edge sourced                           |
| Hypothesis matrix             | Scoring with disproof questions answered                    |
| Assumption register           | Every assumption, tested or acknowledged                    |
| Attribute table               | The shared-attribute search that produced indirect findings |
| Findings document             | Claims, each with cited items and confidence                |
| Gap list                      | Ranked, each with the source that would resolve it          |
| Decision log                  | Significant investigative choices and why                   |

**The consistency test.** Every claim in the findings document traces to items in the inventory, which appear on the timeline or the chart, which are scored in the matrix, which survived the assumption register. If a claim exists that cannot be traced through that chain, it is an assumption wearing a conclusion's clothes.

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## Part 2: Comparison Tables

### 2.1 Questions vs. Assumptions vs. Hypotheses vs. Conclusions

| Dimension  | Question                                       | Assumption                                   | Hypothesis                                            | Conclusion                       |
|------------|------------------------------------------------|----------------------------------------------|-------------------------------------------------------|----------------------------------|
| Definition | A request for information that can be answered | A premise accepted without being established | A proposed explanation stated so evidence can test it | A judgment reached after testing |
| Form       | Interrogative                                  | Declarative, usually unstated                | Declarative, explicitly stated                        | Declarative, with confidence     |

| <b>Dimension</b>    | <b>Question</b>                                      | <b>Assumption</b>                                          | <b>Hypothesis</b>                              | <b>Conclusion</b>                                  |
|---------------------|------------------------------------------------------|------------------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| Stated openly?      | Yes                                                  | Usually not, which is the problem                          | Always, in writing                             | Always                                             |
| Number held         | Many, prioritized                                    | Many, mostly invisible                                     | Three or more, simultaneously                  | One, with alternatives noted                       |
| Evidence            | Points toward what to collect                        | None by definition                                         | Scored against it                              | Cited for every claim                              |
| Carries confidence? | No                                                   | No; carries a consequence if wrong                         | No; carries a disproof condition               | Yes, with a stated reason                          |
| How it fails        | Unanswerable, loaded, or non-discriminating          | Invisible and load-bearing                                 | Only one entertained, or straw-manned          | Overstated, or resting on an unexamined assumption |
| Correct handling    | Prioritize by discriminating power and answerability | Surface, register, test or acknowledge                     | Generate broadly, eliminate by disconfirmation | Cite items, state confidence, name gaps            |
| Becomes             | A hypothesis set                                     | A tested finding, an explicit caveat, or a deleted premise | A surviving explanation or an eliminated one   | A new question                                     |
| Artifact            | Question set                                         | Assumption register                                        | Hypothesis matrix                              | Findings document                                  |

## 2.2 Facts vs. Evidence vs. Inference

| <b>Dimension</b>     | <b>Fact</b>                | <b>Evidence</b>                                    | <b>Inference</b>                                       |
|----------------------|----------------------------|----------------------------------------------------|--------------------------------------------------------|
| What it is           | A state of the world       | An item bearing on whether a proposed fact is true | A reasoning step from evidence to a claim              |
| Exists               | Independently of observers | As a record, from a source                         | Only in the analyst's reasoning                        |
| Can be wrong         | No, by definition          | Yes: sources err, mislead, or are misread          | Yes: reasoning can be invalid or rest on a bad premise |
| Carries a source     | No                         | Always, with an item ID                            | Cites the items it rests on                            |
| Carries a rating     | No                         | Reliability (A-F), credibility (1-6)               | Confidence, with a reason                              |
| Analyst possesses it | Almost never directly      | Yes, this is what you                              | Yes, this is what you                                  |

| Dimension       | Fact                                                              | Evidence                                                     | Inference                                             |
|-----------------|-------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|
|                 |                                                                   | actually have                                                | actually produce                                      |
| Correct wording | Rarely asserted without qualification                             | "Item 12 records that..."                                    | "We assess that... [confidence], based on..."         |
| Failure mode    | Believing you have one                                            | Treating the item as the thing it describes                  | Presenting it as evidence, or omitting the confidence |
| In a product    | Appears only where corroborated across categories and uncontested | Key facts section                                            | Assessment section                                    |
| Example         | The fire began at 02:14                                           | The utility meter log records a load loss at 02:14 (E12, A1) | The fire had reached the service panel by 02:14       |

## 2.3 Strong vs. Weak Investigative Questions

| Property             | Strong                                        | Weak                                           | Diagnostic test                                                |
|----------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------------------------|
| Answerability        | A specific state of the world would settle it | No answer would settle it                      | Can you name the answer that would close it?                   |
| Boundedness          | Names entities, period, location              | Open-ended                                     | Could the answer be a library?                                 |
| Neutrality           | Does not presuppose                           | Loaded; presupposes wrongdoing or an actor     | Does the question survive if the innocent explanation is true? |
| Single-barrelled     | One question                                  | Several joined by "and"                        | Count the question marks it deserves                           |
| Falsifiability       | Something could show the answer is no         | Nothing could                                  | What observation would defeat it?                              |
| Sourcing             | You can name what would answer it             | No source could                                | Which system holds the answer?                                 |
| Discriminating power | The answer distinguishes among hypotheses     | Every answer is consistent with all hypotheses | Score the possible answers against your matrix                 |
| Consequence          | The answer changes what happens next          | Answering changes nothing                      | What do you do differently under each answer?                  |
| Example              | "Which access events between 19:40 and        | "Was anyone suspicious in the building?"       |                                                                |

| Property | Strong                                           | Weak | Diagnostic test |
|----------|--------------------------------------------------|------|-----------------|
|          | 02:14 are recorded, and which are unattributed?" |      |                 |

## 2.4 Single Hypothesis vs. Multiple Working Hypotheses

| Dimension                                 | Single hypothesis                                                                      | Multiple working hypotheses                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Number entertained                        | One, usually the first that occurred                                                   | Three to five, generated deliberately                                             |
| Search behavior                           | For supporting evidence                                                                | For disconfirming evidence, per hypothesis                                        |
| Evidence handling                         | Contrary items explained away                                                          | Contrary items are the point; they eliminate                                      |
| Analyst's relationship to the explanation | Ownership; disproof feels like loss                                                    | Detachment; disproof is progress                                                  |
| Effect of new evidence                    | Absorbed into the existing story                                                       | Re-scores the whole matrix                                                        |
| Typical confidence                        | High, early                                                                            | Moderate, and calibrated                                                          |
| Speed                                     | Fast to a conclusion                                                                   | Slower to a conclusion, faster to a correct one                                   |
| Documentation                             | A narrative                                                                            | A matrix with disproof conditions                                                 |
| Vulnerability to deception                | Very high: a deceiver only has to satisfy one story                                    | Lower: the deception must satisfy several tests                                   |
| Failure signature                         | The premature narrative (Section 1.1.5)                                                | Over-generation: hypotheses so numerous or overlapping that nothing discriminates |
| When single is acceptable                 | Never as a method; a single explanation may survive, but only after others were tested |                                                                                   |

## 2.5 Chronological vs. Event-Based Analysis

| Dimension       | Chronological                                  | Event-based                                  |
|-----------------|------------------------------------------------|----------------------------------------------|
| Organizing axis | Absolute time                                  | The episode or incident                      |
| Core question   | What was happening at this moment?             | How did this episode unfold?                 |
| Reveals         | Simultaneity, convergence, impossibility, gaps | Internal structure, repeated pattern, stages |
| Scale           | Single incident or a bounded                   | Recurring conduct across months              |

| <b>Dimension</b> | <b>Chronological</b>                                                                | <b>Event-based</b>                                                                |
|------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                  | period                                                                              | or years                                                                          |
| Best for         | Alibi and opportunity, incident reconstruction, disaster sequencing                 | Fraud schemes, series offences, campaigns, recurring outages                      |
| Artifact         | One master timeline sorted by timestamp                                             | One timeline per episode, plus a cross-episode comparison matrix                  |
| Weakness         | Long or repeated conduct becomes an unreadable list                                 | Loses cross-episode simultaneity                                                  |
| Detects          | That two threads intersected at 02:10                                               | That every episode has the same five stages, and stage three is always outsourced |
| Feeds            | Opportunity and access analysis                                                     | Modus operandi and predictive indicators                                          |
| Combined use     | Build both: chronological for the incident, event-based across priors, then compare |                                                                                   |

## 2.6 Direct vs. Indirect Relationships

| <b>Dimension</b> | <b>Direct</b>                                                           | <b>Indirect</b>                                                                |
|------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Definition       | One edge, asserted by evidence                                          | A path through intermediates, or derived from a shared attribute               |
| Evidence         | A record states the relationship                                        | No record states it; it is inferred                                            |
| Typical basis    | Registry filing, contract, employment record, communication record      | Shared officer, agent, address, domain infrastructure, timing pattern          |
| Strength driver  | The production process of the record                                    | How discriminating the intermediary is                                         |
| Rendering        | Solid line if confirmed                                                 | Distinct rendering; never the same weight as direct                            |
| Supports         | Exactly what it states                                                  | Opportunity, access, possible coordination                                     |
| Does not support | More than the record says; a formal role is not operational involvement | Intent, agreement, coordination, or knowledge, absent further evidence         |
| Required test    | Is the record what it appears to be?                                    | How many others share the intermediary? What benign explanation produces this? |
| Failure mode     | Over-reading formality as                                               | Common-intermediary error:                                                     |

| Dimension | Direct                                 | Indirect                                                                               |
|-----------|----------------------------------------|----------------------------------------------------------------------------------------|
|           | operational reality                    | treating a shared commercial agent as a relationship                                   |
| Example   | "A is manager of B" per a filing       | "A and C both list agent B; agent B serves 12 entities, all in one office suite"       |
| Wording   | "Filings list A as manager of B (E21)" | "A and C share a registered agent; we have not established how many other entities do" |

## 2.7 Link Analysis Techniques and Their Applications

| Technique                                | Input               | Output                                             | Best application                                                       | Limitation                                                                |
|------------------------------------------|---------------------|----------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Association matrix</b>                | Entity list         | Pairwise assessment of every possible relationship | Early stage; forces consideration of pairs you would not have drawn    | Grows quadratically; unusable beyond roughly 40 entities                  |
| <b>Link chart</b>                        | Populated matrix    | Visual network with typed nodes and rated edges    | Communicating structure; spotting clusters and isolates                | Persuasive beyond its evidence if edges are unsourced                     |
| <b>Activity matrix</b>                   | Entities and events | Who participated in what                           | Co-participation patterns; identifying who is present at every episode | Says nothing about the nature of participation                            |
| <b>Degree centrality</b>                 | Chart               | Most-connected nodes                               | Identifying principals and hubs                                        | The most connected node is often the most visible, not the most important |
| <b>Betweenness centrality</b>            | Chart               | Nodes on the most paths between others             | Finding brokers and facilitators, frequently overlooked administrators | Sensitive to missing edges; a gap in collection distorts it badly         |
| <b>Cut-point / broker identification</b> | Chart               | Nodes whose removal disconnects the network        | Disruption planning; identifying the essential intermediary            | Assumes the chart is complete                                             |
| <b>Cluster detection</b>                 | Chart               | Groups more connected                              | Large charts; separating sub-                                          | Cluster boundaries are sensitive to                                       |

| Technique                      | Input                        | Output                                     | Best application                                            | Limitation                                                      |
|--------------------------------|------------------------------|--------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------|
|                                |                              | internally than externally                 | organizations or unrelated components                       | how edges were rated                                            |
| <b>Path analysis</b>           | Chart plus two nodes         | Shortest and alternate routes between them | Access and influence questions; "could A have reached B?"   | A path is opportunity, never proof of use                       |
| <b>Temporal layering</b>       | Chart plus dated edges       | The network at several points in time      | Any multi-year structure; showing when relationships formed | Requires dated edges, which many records do not provide         |
| <b>Attribute overlay</b>       | Attribute table              | Indirect edges from shared attributes      | Hidden relationship discovery, Section 1.7                  | Requires the discrimination test or it manufactures connections |
| <b>Negative space analysis</b> | Chart plus role expectations | Expected-but-absent edges and nodes        | Structural review; detecting missing intermediaries         | Requires knowing what the structure should look like            |

## 2.8 Common Investigative Mistakes and How to Avoid Them

| Mistake                                     | What it looks like                      | Where it originates   | Downstream damage                                           | Prevention                                                                |
|---------------------------------------------|-----------------------------------------|-----------------------|-------------------------------------------------------------|---------------------------------------------------------------------------|
| Investigating a topic instead of a question | Broad research, no stopping point       | Question formulation  | Effort with no conclusion                                   | Convert topic to concern to questions; prioritize by discriminating power |
| Loaded question                             | "Why did they falsify the report?"      | Question formulation  | Only confirming evidence is sought                          | Neutrality test; write the innocent-explanation version                   |
| Single hypothesis                           | A clean narrative formed early          | Hypothesis generation | Confident wrong conclusion; high vulnerability to deception | Minimum four hypotheses including the null and innocent                   |
| Straw-man alternative                       | Alternatives listed but written to fail | Hypothesis generation | The appearance of rigor without the substance               | Write the least-likely hypothesis as its proponent would                  |

| <b>Mistake</b>                     | <b>What it looks like</b>                                | <b>Where it originates</b> | <b>Downstream damage</b>                       | <b>Prevention</b>                                                      |
|------------------------------------|----------------------------------------------------------|----------------------------|------------------------------------------------|------------------------------------------------------------------------|
| Scoring for support                | Counting plus signs                                      | Hypothesis scoring         | Non-discriminating evidence drives the result  | Score consistency; weight inconsistencies; identify discriminators     |
| Smoothed timeline                  | Orderly sequence, no conflicts, from messy sources       | Timeline construction      | The most informative content is deleted        | Precision column; flag every conflict, gap, and anomaly                |
| Unmarked inference                 | Reconstruction presented as record                       | Timeline construction      | The analyst's reasoning is read as evidence    | Type field: observed, reported, inferred                               |
| Publication-date contamination     | Events dated when reported                               | Timeline construction      | Wrong sequence, therefore wrong causation      | Dual dates (Ch.4)                                                      |
| Unsourced edge                     | Lines on a chart with no item ID                         | Link analysis              | Diagram read as conclusion                     | Rule 6; edge register; pre-product edge walk                           |
| Guilt by association               | Connection read as participation                         | Link analysis              | Wrong actor centred; defamation exposure       | Edge type and strength; participation requires its own evidence        |
| Common-intermediary error          | Indirect edges from non-discriminating shared attributes | Hidden relationships       | Manufactured networks                          | Discrimination test: how many others share it?                         |
| Non-disclosure read as concealment | "The relationship was hidden"                            | Hidden relationships       | Asserted intent with no basis                  | State what the record shows and does not show; stop there              |
| Load-bearing assumption            | Conclusion survives only if an unstated premise holds    | Assumption handling        | The entire conclusion collapses under review   | Assumption register; therefore-audit; negation test                    |
| Coverage assumption                | "If it happened, a record would exist"                   | Assumption handling        | Absence misread as non-occurrence              | Establish whether the record type exists, is required, and is retained |
| Base rate neglect                  | Anomaly declared without a baseline                      | Reasoning                  | Ordinary variation investigated as significant | Establish the baseline before calling anything unusual                 |

| Mistake                  | What it looks like                       | Where it originates | Downstream damage                   | Prevention                                                      |
|--------------------------|------------------------------------------|---------------------|-------------------------------------|-----------------------------------------------------------------|
| Scope explosion          | Chart or timeline containing everything  | Any stage           | Unreadable artifacts; no conclusion | Bound artifacts by the question                                 |
| Stopping at confirmation | Search ends when the theory is supported | Any stage           | Premature closure                   | Stopping condition defined in advance; analytic saturation test |

## Part 3: Case Studies

Each case walks the six techniques and shows the decisions made. Cases marked *historical* describe real documented events in general terms; cases marked *composite* are built from documented professional practice and are not real organizations or people.

### 3.1 Cybersecurity: An Insider Data Exfiltration Question

(*composite*)

**Situation.** A mid-sized engineering firm discovers that a competitor's bid matched its own pricing structure with improbable precision on two consecutive tenders. Leadership suspects an insider. The security lead must determine what actually happened, and the firm has 14 people with access to the pricing model.

| Technique         | How it was applied                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Questions</b>  | The initial framing, "who leaked the pricing," was rejected as loaded and presupposing.<br>Reformulated into: Q1 what specific information would be required to produce the competitor's bid structure; Q2 which systems hold that information and what do their access logs show; Q3 what is the base rate of bid similarity in this market; Q4 what routes other than an insider could produce this outcome; Q5 which records that should exist do not |
| <b>Hypotheses</b> | H1 insider exfiltration; H2 the competitor derived the structure independently from public tender history and market rates; H3 information passed through a shared third party (a supplier, a consultant, or the tendering authority itself); H4 a compromise of the firm's systems by an external                                                                                                                                                       |

| Technique                   | How it was applied                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | actor; H5 coincidence within normal market convergence                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Timeline</b>             | Chronological timeline of both tender cycles: model changes, internal approvals, document exports, external sharing, and the competitor's submission dates. Two inferred entries were marked as such                                                                                                                                                                                                                                             |
| <b>Link analysis</b>        | Nodes for the 14 people with access, the tendering authority, two shared suppliers, a consultant engaged by both firms in a prior year, plus documents and systems. Edges typed and rated                                                                                                                                                                                                                                                        |
| <b>Assumptions</b>          | The register surfaced four: that the access log captured all reads (it did not; the model was also emailed as an attachment, unlogged by the DLP rule in force); that only the 14 had access (a departed employee's account remained active); that the competitor's bid required the internal model (Q3 had never been asked); and that similarity implied copying                                                                               |
| <b>Hidden relationships</b> | Attribute analysis on external parties found that the consultant engaged by the firm three years earlier had, per public filings, since become a director of a supplier that served both firms. Neither relationship was disclosed and neither was individually improper                                                                                                                                                                         |
| <b>Conclusion</b>           | H3 and H5 could not be eliminated; H1 could not be established. Base rate work under Q3 found that in this market, bid structures converge more than leadership assumed. The assessment reported that the evidence did not support an insider explanation at any useful confidence, identified the unlogged email path and the active departed account as genuine control failures, and recommended closing both regardless of the leak question |

## Lessons.

**The reformulated question changed the outcome.** "Who leaked" would have produced a search for the most plausible-looking insider and, given 14 candidates and normal workplace evidence, would have found one.

**The base rate question was decisive and nearly went unasked.** Q3 is the sort of question that feels like a diversion and is frequently where the answer lives.

**Two real findings emerged from a false premise.** The control failures were genuine and would not have been found without the investigation, which is a common and underappreciated outcome: the investigation was right to run and its original hypothesis was wrong.

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## 3.2 Criminal Investigation: The Halcyon Ridge Warehouse Fire

*(composite; this is the scenario used in Labs 2 through 4)*

**Situation.** A fire destroys a leased warehouse. The tenant claims a substantial inventory loss. The county fire investigator's origin-and-cause work is inconclusive between electrical fault and undetermined. An insurance special investigations unit and a county investigator both need to know what happened.

| Technique     | How it was applied                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Questions     | Descriptive: what was in the building and what was the structure's condition. Temporal: when did ignition occur and what is the last evidenced human presence. Relational: who holds interests in the property, the tenant, and the claim. Causal: what ignition mechanisms are consistent with the observed burn pattern. Comparative: how does this loss compare to the tenant's prior claims and to comparable losses. Negative: which records that should exist do not |
| Hypotheses    | The four in Section 1.3.4: electrical, insurance-motivated arson, third-party arson, other accidental cause                                                                                                                                                                                                                                                                                                                                                                |
| Timeline      | The fragment in Section 1.4.4. Key features: a coverage increase 42 days prior, a disabled sensor 7 days prior with no repair ticket, the last badge-out at 19:40, an unattributed vehicle at 22:10, and a conflict between the tenant's inventory statement and the freight records                                                                                                                                                                                       |
| Link analysis | The chart in Section 1.5.3. The landlord entity and the adjusting firm engaged on the loss share a manager, which no document in either file disclosed                                                                                                                                                                                                                                                                                                                     |
| Assumptions   | Register surfaced: that the badge log captured all entry (the rear door sensor gap defeats it); that the disabled sensor was a malfunction (untested); that the inventory was present (contradicted by freight records); that the tenant is the only beneficiary (the shared-manager finding defeats it); and that the origin determination was reliable (it was expressly inconclusive)                                                                                   |

| Technique            | How it was applied                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hidden relationships | The shared manager between the landlord and the adjuster, found by attribute analysis across two registry filings. The discrimination test was applied: the individual is a specific named person, not a commercial agent, so the intermediary is highly discriminating                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Conclusion           | H2 was not established, and neither was H1. The assessment reported: origin inconclusive per the authority of record; the freight records materially contradict the claimed inventory, which is a claims issue independent of cause; the sensor gap creates an unlogged access route that defeats the opportunity analysis in all directions; and the landlord-adjuster relationship is a conflict requiring disclosure and an independent adjuster, stated as a structural finding without any allegation of wrongdoing. Recommended steps: independent origin review, sensor forensic examination, inventory reconstruction from freight and sales records, and disclosure of the shared officer to the carrier |

### Lessons.

**The most important finding was not about the fire.** The inventory contradiction and the adjuster conflict were both discovered by techniques aimed at the cause question, and both mattered more than the cause question to the decision at hand.

**The sensor gap defeated the opportunity analysis in every direction.** Students frequently treat an access-control gap as supporting the insider hypothesis. It does the opposite: it removes the analytic value of the access log entirely, which weakens the case against every candidate equally.

**Nothing here supports an allegation.** The correct product states structural findings and recommends steps. An investigator who wrote "the tenant burned the building for the insurance" from this evidence would be wrong, unprofessional, and exposed.

## 3.3 Fraud Detection: A Recurring Vendor Overbilling Pattern

*(composite)*

**Situation.** A hospital system's internal audit flags that one facilities vendor's invoices have grown 34% over two years while service volume has not.

| <b>Technique</b>            | <b>How it was applied</b>                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Questions</b>            | Comparative first, deliberately: is 34% unusual against peer vendors and against input cost inflation? Then descriptive (what exactly is being billed), temporal (when did the divergence begin and what changed then), relational (who approves these invoices and who selected this vendor), and negative (which supporting documentation that policy requires is absent) |
| <b>Hypotheses</b>           | H1 billing fraud by the vendor; H2 legitimate scope growth not reflected in the volume metric; H3 internal collusion between an approver and the vendor; H4 a control failure allowing duplicate or unsupported billing without anyone intending fraud; H5 a measurement artifact in the volume metric                                                                      |
| <b>Timeline</b>             | Event-based rather than chronological, because the conduct was recurring. Each invoice cycle treated as an episode; the internal structure of each mapped: issue, receipt, approval, payment. The comparison across 26 episodes was the productive artifact                                                                                                                 |
| <b>Link analysis</b>        | Activity matrix of approvers against invoice episodes revealed that 21 of 26 anomalous invoices were approved by one of two approvers, out of eleven with authority                                                                                                                                                                                                         |
| <b>Assumptions</b>          | Register surfaced that "approval implies review" was untested; the approval workflow permitted bulk approval, so an approver's name on 21 invoices might reflect a queue assignment rather than a decision                                                                                                                                                                  |
| <b>Hidden relationships</b> | Attribute analysis across vendor registry filings, the hospital's conflict-of-interest disclosures, and property records found no relationship between either approver and the vendor. This negative finding was reported explicitly and materially changed the assessment                                                                                                  |
| <b>Conclusion</b>           | H3 was substantially weakened by the negative relationship finding and by the bulk-approval mechanism. H4 became best-supported: the pattern was consistent with a control failure in which unsupported line items passed through bulk approval. H1 could not be eliminated and remained open pending the vendor's supporting documentation. Recommendations targeted the   |

| Technique | How it was applied       |
|-----------|--------------------------|
|           | control before the actor |

### Lessons.

**The comparative question was asked first, on purpose.** Establishing whether anything anomalous occurred before investigating the anomaly prevents the most common fraud-investigation error.

**A negative relationship finding is a finding.** Reporting that no connection was found, with the categories searched and the coverage boundary, changed the conclusion and protected two employees from an unsupported inference.

**Event-based analysis was correct here** because the conduct was recurring. A chronological timeline of 26 invoice cycles would have been an unreadable list.

## 3.4 Investigative Journalism: A Municipal Contract Series

*(historical pattern, described in general terms)*

**Situation.** Local newsrooms routinely investigate whether municipal contracts have been awarded improperly. The pattern below reflects published methodology from that work.

| Technique            | How it runs                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Questions</b>     | Narrow and answerable rather than thematic: not "is the city corrupt" but "in the past 60 months, which contracts were awarded without competitive bidding, to whom, and under which exemption?"                                                |
| <b>Hypotheses</b>    | Routine use of a lawful emergency or sole-source exemption; a systemic procurement weakness producing the pattern without intent; a relationship between decision makers and recipients; and coincidence arising from a small local vendor pool |
| <b>Timeline</b>      | Chronological across award dates, plus event-based per award: solicitation, exemption invoked, approval, amendment history. Amendment history is frequently where the story is, because the awarded value and the final value diverge           |
| <b>Link analysis</b> | Nodes for officials, vendors, entities, addresses, and campaign or disclosure filings. Edges from public filings only. Charts are built for internal use and only their evidenced portions are published                                        |
| <b>Assumptions</b>   | Standard newsroom audit: does the disclosure                                                                                                                                                                                                    |

| Technique                   | How it runs                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | regime actually require the relationship to be reported (a coverage assumption), and does a formal role imply involvement in the specific decision                                                                                                                                                                                               |
| <b>Hidden relationships</b> | Attribute analysis across corporate registries, campaign finance filings, property records, and professional filings. The discrimination test is applied rigorously, because a small vendor pool produces coincidental overlap that is not a story                                                                                               |
| <b>Conclusion</b>           | Published findings state what records show, quantify the pattern, distinguish lawful exemptions from questionable ones, and present the response of every named party before publication. Where a relationship exists but no impropriety is established, the finding is reported as a disclosure and conflict question rather than as wrongdoing |

### Lessons.

**The small-pool base rate is the discipline this domain teaches.** In a town with six qualified contractors, the same names recur for reasons that have nothing to do with impropriety. Establishing the base rate is what separates a story from an accusation.

**Amendment history is the underused artifact.** Award value is public and often unremarkable; the sequence of amendments frequently carries the finding.

**Right of reply is a methodological step, not just an ethical one.** A subject's response regularly surfaces a benign explanation the investigator had not generated, which is a hypothesis-generation mechanism.

## 3.5 Competitive Intelligence: A Supplier Capacity Question

*(composite)*

**Situation.** A manufacturer must decide whether to single-source a critical component from a supplier that claims it has doubled capacity at its primary plant.

| Technique        | How it was applied                                                                                                                                                                                                                                    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Questions</b> | Descriptive (what does doubled capacity mean in this process), negative (which permits, filings, or hiring would necessarily accompany a doubling and are they present), comparative (what did the last comparable expansion by this supplier require |

| <b>Technique</b>            | <b>How it was applied</b>                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | and produce), temporal (when would each observable step have had to occur), relational (who else does this supplier serve and would they absorb the new capacity)                                                                                                                                                                                                    |
| <b>Hypotheses</b>           | H1 capacity genuinely doubled; H2 capacity increased materially but less than claimed; H3 nominal capacity increased through a second shift rather than plant expansion, which has different resilience implications; H4 the claim is forward-looking and describes planned rather than realized capacity                                                            |
| <b>Timeline</b>             | Chronological across permits, imagery captures, hiring postings, and public statements over 30 months. The productive finding was ordering: the capacity claim preceded the building permit by five months                                                                                                                                                           |
| <b>Link analysis</b>        | Modest chart: the supplier, its parent, its two largest disclosed customers, the plant location, and the equipment vendor named in a trade publication                                                                                                                                                                                                               |
| <b>Assumptions</b>          | Register surfaced a coverage assumption (that a permit would be required for the works implied, which had to be checked against that jurisdiction's thresholds) and a continuity assumption (that the parent structure at the time of the claim matched the current structure; it did not, an acquisition had intervened)                                            |
| <b>Hidden relationships</b> | Temporal correlation found that the supplier's hiring surge aligned not with plant expansion but with the acquisition of a second facility 90 km away, which the capacity claim did not mention                                                                                                                                                                      |
| <b>Conclusion</b>           | H3 and a variant of H1 both survived, and the distinction mattered enormously to the decision: capacity across two sites is a different resilience proposition from capacity at one. Assessed at moderate confidence, with the recommendation that the client seek site-level capacity representations contractually, since open sources could not resolve the split |

## **Lessons.**

**The ordering finding did the work.** A claim preceding the permit that would enable it is a discriminating observation available only from a timeline.

**The conclusion refined the question rather than answering it.** "Doubled, but across two sites" was more useful to the client than a yes or a no, and identifying that the open-source ceiling had been reached was a legitimate and valuable output.

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### 3.6 Missing Person Investigation: A Fictional Vulnerable Adult Case

*(composite; all persons are fictional)*

**Situation.** An adult with a documented health condition has not been seen for four days. Family reported them missing on day two. This case is included because missing person work is where structured methodology most directly affects a person's safety, and where premature narrative most often causes harm.

| Technique     | How it was applied                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Questions     | Temporal: what is the last independently evidenced sighting, as distinct from the last reported one. Descriptive: what was the person carrying, wearing, and dependent on. Relational: who forms the person's routine contact set. Negative: which routine activities did not occur, and when did each stop. Comparative: has this person been out of contact before, and for how long |
| Hypotheses    | H1 voluntary absence; H2 medical incident during ordinary travel; H3 accident in an unfrequented location; H4 harm by another; H5 the person is where they usually are and the reporting chain is wrong. All five were carried simultaneously, and resource allocation was matched to each hypothesis's search implications rather than to the most emotionally salient                |
| Timeline      | Chronological, with the type field doing critical work: the last <i>reported</i> sighting was day zero evening, but the last <i>independently evidenced</i> one was day zero afternoon. That five-hour discrepancy relocated the search area                                                                                                                                           |
| Link analysis | Routine contact map: household, workplace, care providers, transport, and regular locations, with the last confirmed contact date on every edge. Negative space analysis identified a weekly appointment that had been missed with no cancellation, which had not been reported by anyone                                                                                              |
| Assumptions   | Register surfaced: that the reported last sighting was accurate (it was second-hand); that the                                                                                                                                                                                                                                                                                         |

| Technique                   | How it was applied                                                                                                                                                                                                                                                                   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | person's routine was as described (family described the routine as of a year prior); that a device would generate location signals (the person's device had been left behind, which was itself evidence); and that absence from home implied travel                                  |
| <b>Hidden relationships</b> | Not a covert relationship but an unnoticed one: the missed appointment provider had assumed the family had cancelled, and the family had assumed the appointment had occurred. Neither party knew the other's assumption. Cross-domain assembly of two routine records surfaced it   |
| <b>Conclusion</b>           | The evidenced timeline shortened the unaccounted window by five hours and shifted the search geography. The missed-appointment finding narrowed the window further. H2 and H3 were prioritized for resource allocation on the basis of the geography, without eliminating the others |

#### Lessons.

**The distinction between reported and evidenced sightings is the single most valuable technique in this domain.** Reported last sightings are routinely wrong, not through deception but through ordinary memory error, and a search area built on one can be materially misplaced.

**Negative space produced the second most valuable finding.** What did not happen (an appointment, uncanceled) told investigators more than most of what did.

**Carrying all five hypotheses protected the person.** Prematurely settling on voluntary absence, which is common and often correct, would have de-prioritized exactly the searches that the geography supported.

### 3.7 National Security: Assessing an Ambiguous Facility

*(historical pattern, described in general terms)*

**Situation.** Analysts must assess the purpose of a facility under construction in a country of concern, where the state has offered a civilian explanation and no access is available.

| Technique        | How it runs                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Questions</b> | Descriptive (dimensions, layout, power and water provision, security perimeter), comparative (how does this compare to known facilities of each candidate type), negative (which features that a |

| Technique            | How it runs                                                                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | facility of type X requires are absent), temporal (construction sequence and its rate), relational (which entities are associated with the construction and what else have they built)                                                                                                                                                                  |
| Hypotheses           | The stated civilian purpose; each plausible alternative purpose; a dual-use facility; and a decoy or deception construction. The stated purpose is included as a genuine hypothesis, not as a formality, and it is sometimes correct                                                                                                                    |
| Timeline             | Construction sequence from repeated imagery. The order in which elements appear is highly diagnostic: power provision, containment, effluent handling, and security perimeters are built in different sequences for different purposes                                                                                                                  |
| Link analysis        | Entities: contractors, equipment suppliers where identifiable from public procurement or shipping records, and the facility's institutional owner. Edges from documentary sources only                                                                                                                                                                  |
| Assumptions          | Register surfaces the central one in this domain: that the analyst's own design conventions apply to another country's engineering practice. That is mirror imaging, and it has produced documented historical errors. Also surfaced: that imagery coverage is representative, and that absence of a feature in available imagery means absence in fact |
| Hidden relationships | Contractor overlap with previously assessed facilities of a known type, found by attribute analysis across procurement and shipping records                                                                                                                                                                                                             |
| Conclusion           | Assessments in this domain are explicitly probabilistic, differentiate confidence by element (high confidence on dimensions, lower on purpose), name the discriminating features that would resolve the question, and state what collection would provide them                                                                                          |

## Lessons.

**Construction sequence is the timeline technique that carries this domain.** A static image shows what exists; the sequence shows what was prioritized, and priority reveals purpose.

**Mirror imaging is the assumption that has caused documented historical error here.** Assuming another country's engineers would build as yours do is a behavioral assumption of exactly the kind the register exists to catch.

**The stated explanation is a real hypothesis.** Treating an adversary's public account as automatically false is itself an assumption, and it has been wrong.

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### 3.8 Cross-Case Comparison

| Case                 | Question that mattered most                                 | Decisive technique                      | Most common error avoided                        |
|----------------------|-------------------------------------------------------------|-----------------------------------------|--------------------------------------------------|
| Insider exfiltration | The base rate question (Q3)                                 | Assumption register                     | Searching for a plausible insider among 14       |
| Warehouse fire       | The negative question (which records are missing)           | Hidden relationship via shared officer  | Alleging arson from motive plus opportunity      |
| Vendor overbilling   | The comparative question, asked first                       | Event-based analysis across 26 episodes | Treating an approver's name count as involvement |
| Municipal contracts  | Which awards bypassed competition, and under what exemption | Amendment-history timeline              | Small-pool coincidence read as a pattern         |
| Supplier capacity    | Which observable steps must precede a doubling              | Timeline ordering (claim before permit) | Accepting or rejecting the claim wholesale       |
| Missing person       | Last evidenced versus last reported sighting                | Negative space analysis                 | Premature settlement on voluntary absence        |
| Ambiguous facility   | Which required features are absent                          | Construction-sequence timeline          | Mirror imaging on engineering practice           |

Two patterns run through all seven. **The most productive question in six of the seven cases was a comparative or negative question**, not a descriptive one; asking what is normal and what is missing outperforms asking what happened. And **in every case the decisive technique produced a finding the investigator was not looking for**, which is the practical argument for running the full method rather than the parts that seem relevant to the leading theory.

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## Part 4: Chapter Summary

**Investigation methodology is the reasoning that runs inside the collection process.** Chapter 4's phases govern what you do next; this chapter governs what you should believe and why. An investigator with excellent process and poor reasoning produces fully documented, entirely traceable, wrong conclusions, and that work is more dangerous than sloppy work because its apparatus makes it persuasive.

**You almost never possess a fact.** You possess evidence, from sources, with reliability ratings, and you

make inferences from it. A fact is a state of the world; evidence is an item bearing on whether a proposed fact is true; an inference is a reasoning step that necessarily adds something the evidence does not state. Products that assert facts where they have inferences are the most common wording failure in investigative work.

**The question is the investigation.** Convert a topic into a concern, then into questions, by asking what would have to be true for the concern to be justified and what would have to be true for it to be unjustified. Generate all six question types: descriptive, temporal, relational, causal, comparative, and negative. The comparative question ("is this even unusual?") and the negative question ("which records should exist and do not?") are the highest-yield and the most frequently omitted. Prioritize by discriminating power, answerability, and consequence, and remember that a question whose every possible answer is consistent with all your hypotheses has almost no value however interesting it is.

**Hold multiple working hypotheses, deliberately.** Single-hypothesis investigation fails predictably, because evidence consistent with almost any plausible explanation exists and finding it proves nothing. Generate at least four, including the null, the innocent explanation, the obvious one, and at least one of third-party, compound, deception, or incompetence. Write the least likely as its own proponent would write it. Score evidence by consistency rather than support, weight inconsistencies over consistencies, ignore row totals, and eliminate hypotheses on recorded bases. What survives is the best-supported explanation given current evidence, not a proven one.

**Timelines constrain possibility.** They establish sequence, opportunity, impossibility, and anomaly. Every entry carries a precision marker, a type (observed, reported, or inferred), source item IDs, and flags. Chronological analysis answers what was happening simultaneously; event-based analysis answers how a recurring episode unfolds; serious investigations build both and compare them. The five inconsistency types (contradiction, impossibility, gap, anomaly, convergence) are the timeline's most valuable output, and each maps to a specific investigative step. A timeline built from messy sources that shows no conflicts has been smoothed, and the smoothing deleted exactly what the timeline existed to produce.

**Link analysis orders entities in relationship space.** Type your nodes, and give every edge a type, a direction, a basis by item ID, and a strength rating rendered so that confirmed and hypothesized edges never look alike. Direct relationships are asserted by records; indirect relationships are derived through intermediaries and are only meaningful when the intermediary is discriminating. The techniques (association matrix, activity matrix, centrality, brokers and cut-points, clusters, paths, temporal layering, attribute overlay, negative space) each answer a different question, and the highest-betweenness node is frequently a facilitator nobody had noticed. Rule 6 governs everything here: a chart is persuasive beyond its evidence, and an unsourced line is the most convincing form of unsupported assumption in the discipline.

**Assumptions cannot be eliminated, only surfaced.** They enter at question formulation, entity resolution, source acceptance, coverage, timeline construction, link analysis, behavioral reasoning, baselines, continuity, and completeness. Maintain an assumption register. Surface hidden ones with the therefore-audit, the negation test, the stranger test, reverse construction, and steel-manning. Every cognitive bias has a specific control, and generic instructions to be objective are not among them. The

output is a rewritten conclusion in which every claim cites items, unsupported claims are removed or explicitly marked, and confidence is recalculated, which almost always lowers it.

**Hidden relationships are usually latent rather than concealed.** Most become visible through systematic assembly, not cleverness. The five mechanisms are shared attribute analysis, temporal correlation, structural inference, negative space, and cross-domain assembly. Every candidate relationship passes four tests before being drawn: discrimination, alternative explanation, independence, and temporal validity. Word the finding as what the record shows and does not show, and never as concealment, because non-disclosure is not concealment and an inference of intent needs its own basis. When you find one, re-score the hypothesis matrix, revisit the timeline, and re-audit the assumptions, all three, actually.

**The techniques interlock and iterate.** Timeline and link analysis feed each other continuously. Assumption elimination comes after construction, because you cannot audit a picture you have not built. Hidden relationship discovery sends you back to hypothesis scoring. Expect three or four passes, and stop at analytic saturation: a full pass that produces no change in hypothesis ranking, no new discriminating evidence, and no new high-value question.

**Four failure signatures identify a broken investigation:** the premature narrative, the load-bearing assumption, the confident chart, and the smoothed timeline. None is a collection failure. All four occur in investigations with perfect documentation, and recognizing them is how you review your own work and someone else's.

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## Part 5: Key Terms

Grouped by theme. Terms carried from earlier chapters are marked and restated briefly.

### 5.1 Foundations

**Investigation methodology.** The set of reasoning techniques applied within a collection process: question formulation, hypothesis testing, timeline construction, link analysis, assumption elimination, and relationship discovery.

**Fact.** A state of the world, true independently of observation. Rarely possessed directly by an investigator.

**Evidence.** An item bearing on whether a proposed fact is true, always from a source, always carrying reliability and credibility ratings.

**Inference.** A reasoning step from evidence to a claim, adding something the evidence does not itself state; carries a confidence level.

**Assessment (Ch.1).** A stated inference with its confidence level and basis.

**Confidence level (Ch.1).** How solid the basis for a judgment is; distinct from probability.

**Discriminating evidence (Ch.4).** An item that distinguishes between competing hypotheses, as

opposed to one consistent with all of them.

**Analytic saturation.** The point at which a full analytic pass produces no change in hypothesis ranking, no new discriminating evidence, and no new high-value question.

**Failure signature.** A recognizable pattern indicating which reasoning step broke: the premature narrative, the load-bearing assumption, the confident chart, or the smoothed timeline.

## 5.2 Questions

**Topic.** A subject area; not answerable as stated.

**Concern.** A topic converted into a proposition that could be true or false.

**Investigative question.** A specific request for information that a state of the world could settle.

**Descriptive question.** What, who, where, which. Establishes entities and basic facts.

**Temporal question.** When, in what order, how long. Establishes sequence, duration, and opportunity.

**Relational question.** Who is connected to whom and how. Establishes structure and access.

**Causal question.** Why, and how did this produce that. Establishes mechanism.

**Comparative question.** Is this normal; how does it compare to a baseline. Establishes whether anything anomalous occurred.

**Negative question.** What is absent that should be present. Establishes gaps, missing records, and concealment indicators.

**Loaded question.** One that presupposes its answer or an actor's culpability.

**Single-barrelled.** Asking exactly one thing; the opposite of compound questions joined by "and."

**Discriminating power (of a question).** Whether the answer would distinguish among competing hypotheses.

**Base rate.** How often something occurs anyway, independent of the case at hand. Required before calling anything anomalous.

**Question set.** The living document recording questions, their status (open, answered, refined, retired), and their lineage.

## 5.3 Hypotheses

**Hypothesis.** A proposed explanation stated so that evidence could support or undermine it.

**Method of multiple working hypotheses.** Holding several explanations simultaneously and deliberately, so that no single one accumulates undue attachment.

**Null hypothesis.** "Nothing unusual occurred; this is ordinary variation."

**Innocent explanation hypothesis.** "The behavior occurred and has a benign cause."

**Third-party hypothesis.** "Someone not currently in the analysis did this."

**Compound hypothesis.** "Two unrelated things happened close together."

**Deception hypothesis.** "Someone arranged the evidence to produce this appearance."

**Incompetence hypothesis.** "This is error or process failure, not design."

**Disproof condition.** The observation that would defeat a given hypothesis, written when the hypothesis is written.

**Disconfirmation scoring.** Evaluating evidence by asking whether each hypothesis would predict it, and searching for items that would defeat rather than support.

**Consistency (in scoring).** Whether an item is what you would expect if the hypothesis were true; distinct from whether the item makes you believe the hypothesis.

**Hypothesis matrix (ACH matrix) (Ch.1).** The grid of evidence items against hypotheses, scored for consistency.

**Elimination.** Removing a hypothesis on the basis of recorded inconsistent evidence.

**Straw-man alternative.** An alternative hypothesis written so as to fail, producing the appearance of rigor without the substance.

**Steel-manning.** Writing the strongest good-faith version of an explanation you find least plausible.

## 5.4 Timelines

**Timeline.** An ordered record of events with timestamps, precision, types, sources, and flags.

**Chronological analysis.** Events organized on a single absolute time axis, revealing simultaneity, convergence, and impossibility.

**Event-based analysis.** Events grouped into episodes, each traced internally, revealing pattern and stages across repeated conduct.

**Precision (Ch.4).** Whether a timestamp is exact, minute, hour, day, month, or a bracketed window.

**Entry type.** Whether an entry is observed (captured by a record or sensor), reported (asserted by a person), or inferred (reconstructed by the analyst).

**Inferred entry.** A timeline row that no record captured, marked as such so the analyst's reasoning is not read as record.

**Opportunity.** Whether an actor could have done something, given evidenced location and time.

**Impossibility.** A sequence requiring something physically impossible; the strongest analytic finding a timeline produces.

**Contradiction.** Two sources placing the same event differently, or disagreeing on whether it occurred.

**Gap.** A period with no evidence where evidence should exist.

**Anomaly.** An event occurring at an unusual time, frequency, or order relative to a baseline.

**Convergence.** Multiple independent threads intersecting in a narrow window.

**Smoothing.** Removing precision markers and conflicts to produce a readable narrative; the most damaging timeline failure.

**Publication-date contamination (Ch.4).** Dating an event to when it was reported rather than when it occurred.

## 5.5 Link Analysis

**Link analysis.** Ordering entities in relationship space to reveal structure, access, concentration, and absence.

**Node.** An entity in a chart: person, organization, location, event, asset, digital artifact, or document.

**Edge.** A relationship, carrying a type, a direction, a basis (item ID), and a strength rating.

**Edge register.** The tabular record of every edge and its attributes, maintained alongside the visual chart.

**Strength rating.** Confirmed, probable, possible, or hypothesized, rendered so that they are visually distinguishable.

**Direct relationship.** A single edge asserted by evidence.

**Indirect relationship.** A connection through intermediates or derived from a shared attribute; no record states it.

**Discriminating intermediary.** A shared attribute rare enough that sharing it is meaningful. A commercial registered agent serving thousands of entities is not one.

**Association matrix.** A pairwise grid of every entity against every other, marked confirmed, suspected, or none.

**Activity matrix.** Entities against events, activities, or locations, revealing participation and co-participation.

**Degree centrality.** How many connections a node has; identifies hubs.

**Betweenness centrality.** How often a node sits on paths between other nodes; identifies brokers.

**Broker.** A node connecting otherwise separated parts of a network; frequently a facilitator rather than a principal.

**Cut-point.** A node whose removal disconnects the network.

**Cluster.** A group of nodes more connected to each other than to the rest of the network.

**Path analysis.** Identifying the routes between two nodes; establishes possible access, never its use.

**Temporal layering.** Rendering the chart at several points in time to show when relationships formed and dissolved.

**Attribute overlay.** Rendering shared attributes as edges to expose indirect relationships.

**Negative space analysis.** Examining expected relationships or nodes that are absent.

**Guilt by association.** Reading a connection as participation; a connection is not evidence of knowledge, agreement, or involvement.

**Scope explosion.** Adding entities until the chart is unreadable and uninformative.

## 5.6 Assumptions and Bias

**Assumption.** A premise accepted without being established.

**Load-bearing assumption.** One on which a conclusion depends, such that the conclusion fails if it is wrong.

**Assumption register.** The record of every assumption, its type, what rests on it, its test status, and the consequence if wrong.

**Coverage assumption.** "If it happened, a record would exist." The assumption OSINT analysts most frequently miss.

**Identity assumption.** "The same name means the same entity."

**Continuity assumption.** "The current structure, ownership, or role was the structure then."

**Completeness assumption.** "I have most of the relevant evidence."

**Behavioral assumption.** A premise about how people act, including mirror imaging.

**Therefore-audit.** Circling every connective in written findings and naming the assumption living in each joint.

**Negation test.** Asking what would have to be true for a finding to be wrong.

**Stranger test.** Giving the evidence list without context to an uninvolved person and comparing their conclusion to yours.

**Reverse construction.** Working backward from a conclusion, step by step, naming what must be true at each link.

**Confirmation bias (Ch.1).** Noticing and weighting evidence that supports an existing belief.

**Anchoring (Ch.1).** Excessive influence of the first hypothesis or first evidence encountered.

**Availability bias.** Weighting vivid or recent evidence over mundane evidence.

**Mirror imaging (Ch.1).** Assuming a subject reasons and values as the analyst does.

**Narrative fallacy.** Preferring an explanation because it is coherent.

**Base rate neglect.** Declaring an event significant without establishing how common it is.

**Hindsight bias (Ch.2).** Finding prior indicators obvious once the outcome is known.

**Sunk cost.** Retaining a hypothesis because effort was invested in it.

**Authority bias.** Accepting official or senior sources without evaluating their production process.

**Attribution error.** Attributing behavior to character rather than to circumstance or process.

## 5.7 Hidden Relationships

**Hidden relationship.** A connection that no single item states and no party disclosed.

**Latent relationship.** Present in collected material but not yet assembled; the most common type.

**Derived relationship.** Existing only as an inference from a pattern.

**Concealed relationship.** One that someone acted to keep invisible; requires evidence of the concealing act.

**Shared attribute analysis.** Systematically comparing entity attributes to find repeats.

**Temporal correlation.** Recurring proximity in time between events involving different entities.

**Lead-lag pattern.** One entity's activity consistently preceding another's by a stable interval.

**Structural inference.** Deriving a relationship from the shape of a network, including missing intermediaries, improbable isolation, and structural equivalence.

**Structural equivalence.** Two nodes with identical connection patterns, suggesting a shared identity or interchangeable role.

**Missing intermediary.** Two clusters that interact but share no visible connecting node.

**Cross-domain assembly.** Placing material from different source categories on one artifact so that a relationship none of them contains becomes visible.

**Discrimination test.** Establishing how many other entities share an intermediary before treating it as meaningful.

**Alternative explanation test.** Identifying benign or structural reasons that would produce the observed pattern.

**Non-disclosure versus concealment.** The distinction between a relationship that was not stated and one that was actively hidden; asserting the second requires evidence of the act.

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## Part 6: Knowledge Assessment

### 6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Explain what investigation methodology adds to collection methodology, and describe the failure mode it prevents.

2. Distinguish facts, evidence, and inference, and give the correct wording for each in a product.
3. Describe the conversion from topic to concern to questions, and explain why questions should be generated from both directions.
4. Name the six question types and state the blind spot produced by omitting each.
5. Explain why the negative question is frequently the highest-yield question in an investigation.
6. State four properties of a strong investigative question and rewrite one weak question of your own devising.
7. Explain the three dimensions on which questions are prioritized, and why an interesting question with low discriminating power should be deferred.
8. Explain the method of multiple working hypotheses and describe the mechanism by which single-hypothesis investigation fails.
9. Name six hypothesis generation techniques and state what each guards against.
10. Explain why row totals in a hypothesis matrix should be ignored, and what should be examined instead.
11. Explain the difference between eliminating a hypothesis and a hypothesis surviving, and how each should be worded.
12. Describe the four things a timeline establishes that nothing else establishes as cheaply.
13. Explain the entry type field and why unmarked inference on a timeline is a serious error.
14. Distinguish chronological from event-based analysis and state when each is appropriate.
15. Name the five timeline inconsistency types and give the investigative step that each implies.
16. Describe the four attributes every edge in a link chart must carry, and explain Rule 6.
17. Distinguish direct from indirect relationships and explain the discriminating-intermediary test.
18. Explain the difference between degree and betweenness centrality and why the latter often matters more.
19. Explain why assumptions cannot be eliminated, and describe what an assumption register converts them into.
20. Name the five discovery mechanisms for hidden relationships and explain why most hidden relationships are latent rather than concealed.

## **6.2 Multiple Choice (15 questions)**

1. An investigator with excellent process discipline and poor reasoning most characteristically produces: a) Incomplete work b) Fully documented, traceable, wrong conclusions c) Work that cannot be audited d) Legally exposed collection
2. "The utility meter log records a load loss at 02:14" is: a) A fact b) Evidence c) An inference d) A

hypothesis

3. "The fire had reached the service panel by 02:14" is: a) A fact b) Evidence c) An inference d) A question
4. Which is a negative question? a) Who was present at the site? b) When did the fire begin? c) Which inspection records that policy requires are absent? d) Why did the panel fail?
5. The primary purpose of the comparative question is to establish: a) Who is responsible b) Whether anything anomalous occurred at all c) The sequence of events d) The relationships among actors
6. A minimum viable hypothesis set should include the null, the innocent explanation, the obvious explanation, and: a) The most serious allegation b) At least one of third-party, compound, deception, or incompetence c) The explanation the consumer prefers d) Nothing further
7. In a hypothesis matrix, an item consistent with every hypothesis: a) Strongly supports the leading hypothesis b) Should be removed from the evidence base c) Has little discriminating power and contributes almost nothing d) Should be weighted most heavily
8. A timeline entry that no record captured but the analyst concludes must have occurred should be: a) Omitted b) Recorded and marked as inferred c) Recorded as observed, since the reasoning is sound d) Recorded without a type designation
9. Event-based analysis is most appropriate for: a) A single-incident reconstruction b) An alibi and opportunity question c) Recurring conduct such as a fraud scheme across many cycles d) Establishing physical impossibility
10. The strongest analytic finding a timeline can produce is: a) A convergence b) An anomaly c) An impossibility d) A gap
11. Under Rule 6, an edge on a link chart with no supporting item must be: a) Removed silently b) Rendered identically to other edges c) Rendered as hypothesized and visually distinct, and recorded as such d) Supported by the analyst's professional judgment
12. Two entities share a registered agent that serves 40,000 companies. This indirect relationship is: a) Confirmed b) Probable c) Of little value, because the intermediary is not discriminating d) Evidence of coordination
13. The node most likely to be an overlooked facilitator is the one with the highest: a) Degree centrality b) Betweenness centrality c) Number of documents attached d) Closeness to the analyst's leading hypothesis
14. "If it happened, a record would exist" is an example of: a) An identity assumption b) A coverage assumption c) A continuity assumption d) A behavioral assumption
15. An investigator writes: "The relationship was concealed." The evidence is that no filing disclosed it. This is: a) Correct, since non-disclosure is concealment b) Incorrect, because concealment asserts an intentional act requiring its own evidence c) Correct if the relationship is material d) Correct if the entities are related

## 6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. Investigators usually possess facts and reason from them directly.
  2. Evidence consistent with a hypothesis is proof that the hypothesis is correct.
  3. A hypothesis set should include the explanation the investigator considers least likely, stated fairly.
  4. A timeline built from messy real-world sources that shows no conflicts has probably been smoothed.
  5. A connection between two entities on a link chart is evidence that both participated in the same conduct.
  6. Absence of a record is always evidence that the underlying event did not occur.
  7. Assumption elimination should be performed before the timeline and link chart are built.
  8. Most hidden relationships are latent rather than deliberately concealed.
  9. The most connected node in a network is always the most investigatively important.
  10. Rewriting a conclusion using only verified evidence usually lowers the stated confidence.
- 

## Part 7: Critical Thinking Exercises

Each group targets one technique. Produce the artifact the exercise names.

### Developing investigative questions

**7.1** A city council member says: "Something is wrong with how the transit authority awards its maintenance contracts." Convert this topic into a concern, then into eight questions covering all six types. Mark which two you would work first and why.

**7.2** Rewrite each of these as a strong question, and name the property that made the original weak: (a) "Was the CFO involved?"; (b) "Why did they hide the second facility?"; (c) "Is the vendor legitimate and are the invoices real and who approved them?"; (d) "Is this amount of overtime suspicious?"

**7.3** You have twelve questions and capacity for four. Two are highly interesting and non-discriminating; two are dull and highly discriminating; four are answerable at low cost; four require records you cannot obtain in time. Produce your prioritized set with the reasoning, and write the sentence you would send the consumer about the four you are dropping.

### Generating multiple hypotheses

**7.4** A regional bank's fraud team observes that eleven customer accounts, opened within one month at three branches, each received a wire transfer of just under the reporting threshold and then closed

within 60 days. Write five hypotheses, including the null and the innocent explanation, each with its disproof condition.

**7.5** Take your least likely hypothesis from 7.4 and write it as its strongest proponent would, in at least five sentences. Then state which evidence you would need to defeat it.

**7.6** An investigator tells you their leading hypothesis is supported by 27 of 34 evidence items. Explain in writing why that statistic is close to meaningless, and describe the analysis you would run instead.

## **Building chronological timelines**

**7.7** From the following fragments, build a timeline with precision markers and entry types, and identify every inconsistency: a witness says they saw a delivery "sometime Tuesday afternoon"; the delivery company's record shows a scan at 09:12 Tuesday; the receiving dock's log has no entry for Tuesday; the site camera has a 40-minute gap beginning 09:00 Tuesday; an invoice is dated Wednesday; an email attaching the invoice was sent Tuesday at 16:20.

**7.8** For each inconsistency you identified in 7.7, state its type and the specific investigative step that would resolve it.

**7.9** You are investigating conduct that recurred across nineteen months. Explain whether you would build a chronological timeline, an event-based analysis, or both, and describe what the comparison between them would be designed to reveal.

## **Creating relationship maps**

**7.10** Build a link chart, in text or on paper, from these facts, using typed nodes and rated edges: filings list M. Adeyemi as director of Company A and of Company B; Company A and Company C share a suite address; Company B leases a facility from Company D; a court exhibit lists Company C and Company D as co-defendants; an invoice from Company A names a bank account also referenced in a Company C filing. State which edges are direct and which are indirect, and mark the strength of each.

**7.11** In your chart from 7.10, identify the node with the highest degree and the node with the highest betweenness. Explain which you would investigate first and why.

**7.12** One edge in 7.10 rests on a shared suite address. Write the discrimination test you would perform, state what you would conclude if the suite houses 400 registered entities, and state what you would conclude if it houses three.

## **Identifying unsupported assumptions**

**7.13** A findings paragraph reads: "Access logs show the last entry at 19:40. Therefore no one was in the building after that time. The fire must therefore have started from an unattended source." Perform a therefore-audit, name every assumption, and rewrite the paragraph using only what the evidence supports.

**7.14** An investigator concludes that a supplier has no capacity constraint because no complaint has been filed with the regulator. Name the assumption class, describe how you would test it, and state the confidence penalty if it proves untestable.

**7.15** Take any conclusion from your own recent work, coursework, or otherwise. Apply the negation test and the stranger test, and report what each surfaced.

## Discovering hidden relationships

**7.16** You have a table of 40 entities with columns for address, registered agent, officers, domain, name server, incorporation date, and bank reference. Describe the exact procedure you would follow to find indirect relationships, and state which two columns you would sort first and why.

**7.17** You observe that Entity X files a permit within eight days of Entity Y filing one, on six occasions over three years. Describe the chance test you would apply, and state what additional evidence would convert this from a temporal correlation into a supported relationship.

**7.18** Two clusters in your chart clearly interact (their members appear together at events) but share no visible connecting node. Explain what this structural feature suggests, and describe the three collection steps you would take next.

## Explaining how evidence supports or disproves theories

**7.19** Write a paragraph explaining, to a non-specialist stakeholder, why the discovery that an access control sensor was disabled weakens the case against every candidate rather than strengthening the case against one.

**7.20** For a hypothesis of your choice, write the three sentences that would defeat it, then describe what you would collect to determine whether any of the three is true.

**7.21** A stakeholder says: "You have not proven anything. Why should I act on this?" Write the response an analyst should give, addressing evidence, confidence, alternatives, gaps, and the decision the stakeholder actually faces.

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# Part 8: Hands-On Lab Exercises

## General Lab Guidance

**Labs 2 through 4 are chained** on one fictional case, the Halcyon Ridge warehouse fire. Lab 2's hypothesis matrix feeds Lab 3's timeline, which feeds Lab 4's chart, and Lab 4 sends you back to re-score Lab 2. Keep every artifact. Lab 1 and Lab 5 are standalone.

**All evidence in these labs is fictional.** Every person, company, and address is invented. Nothing here describes a real event.

**Rules 1 through 6 apply.** In particular, Rule 6: every edge you draw carries a basis and a strength.

**Rubric scale.** 0 to 4 per criterion: 4 professional, 3 competent, 2 developing, 1 attempted, 0 missing or fundamentally incorrect.

**Universal requirements, graded in every lab.** Every claim cites item IDs. Every judgment carries a confidence level with a stated reason. Every artifact distinguishes observation from interpretation.

Every conclusion names at least one ranked gap.

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## Lab 1: From Broad Topic to Prioritized Questions

**Time required:** 90 to 120 minutes **Format:** Individual, then small group comparison **Difficulty:** Foundational

### Objectives

1. Convert a broad topic into a stated concern and then into answerable questions.
2. Generate questions across all six types.
3. Distinguish strong from weak questions and repair the weak ones.
4. Prioritize by discriminating power, answerability, and consequence.

### Prerequisites

- Section 1.2 and comparison tables 2.1 and 2.3.
- No collection required; this is a formulation lab.

### The Topic

*Fictional briefing note, provided by your consumer.*

**To:** Analyst **From:** Chair, Riverbend Unified School District Board **Re:** Transportation department

Several board members have raised concerns about the transportation department. Over the past two years the department's budget has grown while the number of routes has stayed roughly the same. There have been complaints from parents about late buses in the north attendance area. A maintenance contractor was changed eighteen months ago without a board vote, which staff say was within the superintendent's authority. Two long-serving mechanics left within a month of each other last spring. The department has requested a \$2.1M capital allocation for vehicle replacement at the June meeting.

The board needs to know what is actually going on before the June vote. Please advise.

### Steps

**Step 1 (15 min).** Write the decision statement: what the board decides, when, and what a yes and a no would each mean. Then list every distinct concern embedded in the briefing note, separately. There are at least six.

**Step 2 (15 min).** Convert each concern into a proposition that could be true or false, on Worksheet 1A. A concern stated as "there have been complaints" is not a proposition; "service reliability in the north attendance area has declined relative to other areas" is.

**Step 3 (35 min).** Generate questions on Worksheet 1B. Minimum eighteen questions, with at least two of each of the six types. For each: the type, what would answer it, and which concern it serves.

**Step 4 (15 min).** Quality pass on Worksheet 1C. Apply the seven properties from table 2.3 to every question. Repair every question that fails, retaining the original so the repair is visible. At least three of your initial questions will fail something; if none do, you have been generous with yourself.

**Step 5 (20 min).** Prioritization on Worksheet 1D. Score every question on discriminating power, answerability, and consequence. Select the six you would work first. Then write the two sentences you would send the Chair explaining the selection and naming anything the board is asking that cannot be answered before June.

**Step 6 (15 min).** Write the null and innocent-explanation propositions for the two concerns you ranked highest, in preparation for hypothesis work. State what evidence would be consistent with each.

**Step 7 (5 min).** Self-check: is any of your questions loaded? Does any presuppose an actor? Would the six you selected still be worth asking if every concern in the briefing note had a benign explanation?

### ***Worksheet 1A: Concerns as Propositions***

| # | Concern as stated in the note | Restated as a proposition | Could be false how? |
|---|-------------------------------|---------------------------|---------------------|
|---|-------------------------------|---------------------------|---------------------|

### ***Worksheet 1B: Question Generation***

| Q# | Question | Type | What would answer it | Serves which concern |
|----|----------|------|----------------------|----------------------|
|----|----------|------|----------------------|----------------------|

### ***Worksheet 1C: Quality Pass***

| Q# | Answerable | Bounded | Neutral | Single | Falsifiable | Sourced | Consequential | Repair |
|----|------------|---------|---------|--------|-------------|---------|---------------|--------|
|----|------------|---------|---------|--------|-------------|---------|---------------|--------|

### ***Worksheet 1D: Prioritization***

| Q# | Discriminating power (H/M/L) | Answerability (H/M/L) | Consequence (H/M/L) | Rank | Reason |
|----|------------------------------|-----------------------|---------------------|------|--------|
|----|------------------------------|-----------------------|---------------------|------|--------|

### ***Expected Outcomes***

- At least six distinct concerns identified. Students who treat the note as one concern have merged budget growth, service reliability, procurement process, staff departures, and the capital request into a single undifferentiated worry, which is exactly the error the lab targets.
- Eighteen-plus questions with all six types represented. Comparative and negative questions are

the ones most often missing.

- The two highest-value questions in most strong submissions are comparative and negative: how does per-route cost compare to comparable districts and to prior years, and which required procurement documentation for the contractor change is absent.
- At least three repaired questions with the originals retained.
- A prioritization that reflects the June deadline rather than analytic interest.
- Null and innocent-explanation propositions that are genuine: budget growth explained by fuel, wage, and insurance cost increases; the contractor change lawful within delegated authority; mechanic departures explained by the regional labour market.

### ***Discussion Questions***

1. Which concern in the note was hardest to state as a falsifiable proposition, and why?
2. Which of your questions presupposed an actor before you repaired it?
3. Your two highest-ranked questions are probably comparative or negative. Why do these outperform descriptive questions here?
4. If every concern has a benign explanation, which of your six questions still needs answering before the June vote?

### ***Grading Rubric: Lab 1 (44 points)***

| <b>Criterion</b>                  | <b>Weight</b> | <b>4 points looks like</b>                                           |
|-----------------------------------|---------------|----------------------------------------------------------------------|
| Decision statement                | x1            | Decision, date, and what yes and no each mean                        |
| Concern separation                | x2            | Six-plus distinct concerns, each as a falsifiable proposition        |
| Question volume and type coverage | x3            | Eighteen-plus questions, all six types, two-plus each                |
| Question quality                  | x3            | Seven properties applied; three-plus repairs with originals retained |
| Neutrality                        | x2            | No loaded questions; no presupposed actors                           |
| Prioritization reasoning          | x2            | Three dimensions applied; deadline respected                         |
| Consumer sentence                 | x1            | Names what cannot be answered before June                            |
| Null and innocent propositions    | x1            | Genuine, with consistent evidence identified                         |

Total possible: 60 weighted. Passing: 41.

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## The Chained Case: The Halcyon Ridge Warehouse Fire

*Everything below is fictional.*

**Background.** At approximately 02:14 on 9 March, fire destroyed a single-storey warehouse at 1400 Quarry Line Road in Halcyon Ridge. The building was leased by Crestline Distribution, a regional wholesaler of seasonal goods. Crestline has submitted a claim for building contents totalling \$2.9M. The building itself is insured separately by the property owner. No injuries occurred.

**Your role.** You are an analyst supporting a joint review: the county fire investigator's office needs a cause determination, and the contents insurer's special investigations unit needs a claims assessment. Both consumers receive your work.

**The county fire investigator's origin-and-cause report** classifies the origin as the north bay and the cause as **undetermined**, expressly noting that fire damage precluded a definitive determination between an electrical fault at the service panel and other possible causes.

### ***Evidence Packet A (for Labs 2 and 3)***

| ID | Item                                                                                                                                                                                       | Source category        | Rel./Cred. |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|
| E1 | County electrical inspection notice, dated 22 January, citing "deteriorated conductor insulation and loose lugs" at the north bay service panel; corrective action required within 90 days | Government records     | A1         |
| E2 | Endorsement to Crestline's contents policy increasing coverage from \$1.2M to \$3.4M, executed 26 January, effective 28 January                                                            | Corporate/insurer file | A1         |
| E3 | Building access control log, 1 March to 9 March: 41 badge events, all attributed to five staff IDs. Last event 8 March 19:40, badge-out, ID 0412                                           | Internal record        | A2         |
| E4 | Alarm system service                                                                                                                                                                       | Internal record        | A2         |

| ID  | Item                                                                                                                                                                                | Source category    | Rel./Cred. |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------|
|     | report noting the rear door contact sensor reporting "fault, open circuit" since 2 March; no repair ticket found in the system                                                      |                    |            |
| E5  | National weather record for 8 to 9 March: no lightning within 40 km; wind 8 to 12 km/h; temperature 4 C                                                                             | Government records | A1         |
| E6  | County fire investigator's origin and cause report: origin north bay, cause undetermined                                                                                            | Government records | A1         |
| E7  | Municipal traffic camera still, 8 March 22:10, two blocks north of the site, showing a light commercial vehicle travelling south; plate partially legible, three characters         | Government records | B3         |
| E8  | Crestline's recorded statement to the insurer, 12 March: inventory was "at full seasonal level, approximately \$2.8M at cost"                                                       | Reported statement | C3         |
| E9  | Freight receiving records obtained from Crestline's logistics provider: last inbound delivery to the site 1 February; outbound shipments continued at normal volume through 7 March | Corporate records  | A2         |
| E10 | Crestline's prior-year claims history: two claims in six years, both weather-related, both under \$40,000                                                                           | Insurer file       | A1         |

| <b>ID</b> | <b>Item</b>                                                                                                                                                                              | <b>Source category</b>        | <b>Rel./Cred.</b> |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------|
| E11       | Utility interval meter data for the service address: load loss recorded 9 March 02:14:38                                                                                                 | Government/utility records    | A1                |
| E12       | 911 call record: first call 9 March 02:19 from a passing motorist reporting "heavy smoke from the roof"                                                                                  | Government records            | A1                |
| E13       | Fire crew arrival and progress log: first unit on scene 02:31; heavy involvement north bay; roof collapse north section 02:52                                                            | Government records            | A1                |
| E14       | Photographs of the north bay taken during overhaul, showing the service panel enclosure severely damaged and the panel interior not recoverable                                          | Government records            | A1                |
| E15       | Statement of Crestline's site supervisor (ID 0412): departed 8 March at approximately 19:40; states the building was secure and all equipment powered down except lighting and the alarm | Reported statement            | B3                |
| E16       | Invoice from an electrical contractor to Crestline dated 14 February for "panel assessment, north bay," \$850, marked paid                                                               | Corporate records             | A2                |
| E17       | No permit on file with the county for electrical remediation at 1400 Quarry Line Road at any time after 22 January                                                                       | Government records (negative) | A1                |

| <b>ID</b> | <b>Item</b>                                                                                                                                                                               | <b>Source category</b> | <b>Rel./Cred.</b> |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|
| E18       | Property lease between Halcyon Ridge Properties LLC and Crestline Distribution, commencing 3 years prior, with a clause assigning responsibility for electrical maintenance to the tenant | Corporate records      | A1                |
| E19       | Crestline's outbound sales records for January and February, showing revenue 31% below the same period the prior year                                                                     | Corporate records      | B2                |

***Evidence Packet B (added at Lab 3)***

| <b>ID</b> | <b>Item</b>                                                                                                                                             | <b>Source category</b>      | <b>Rel./Cred.</b> |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------|
| E20       | Fuel card transaction, Crestline company vehicle, 8 March 21:52, service station 6 km north of the site                                                 | Corporate records           | A2                |
| E21       | Second recorded statement from ID 0412, 21 March: states they "may have returned briefly that evening to collect a laptop" but does not recall the time | Reported statement          | C4                |
| E22       | Traffic camera still, 8 March 22:47, same camera, showing a light commercial vehicle travelling north; plate not legible                                | Government records          | B3                |
| E23       | Cell site records were not sought; the insurer has no authority to obtain them and no request was made to law enforcement                               | (negative / process record) | A1                |

| <b>ID</b> | <b>Item</b>                                                                                                                                                          | <b>Source category</b> | <b>Rel./Cred.</b> |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|
| E24       | Alarm system event log: no alarm signals of any type transmitted between 8 March 19:41 and 9 March 02:22, when a communication failure signal was recorded           | Internal record        | A2                |
| E25       | Alarm monitoring contract: the rear door contact is a monitored zone; a zone in fault condition suppresses signalling for that zone only                             | Corporate records      | A1                |
| E26       | Statement of a neighbouring business owner: reports seeing "lights on in the back" of the warehouse "late, maybe eleven or midnight" on a night "sometime that week" | Reported statement     | C4                |
| E27       | Weather record confirms full cloud cover and no moon visibility on the night of 8 to 9 March                                                                         | Government records     | A1                |
| E28       | Crestline's bank facility agreement: a covenant requiring minimum inventory value of \$2.0M, tested quarterly, next test 31 March                                    | Corporate records      | A2                |
| E29       | Email from Crestline's controller to the logistics provider, 3 March: "hold all inbound until further notice"                                                        | Corporate records      | A2                |
| E30       | Fire investigator's supplemental note, 18 March: burn pattern analysis "does not                                                                                     | Government records     | A2                |

| ID | Item                                                                                             | Source category | Rel./Cred. |
|----|--------------------------------------------------------------------------------------------------|-----------------|------------|
|    | exclude" an origin at floor level in the north bay approximately 4 metres from the service panel |                 |            |

***Evidence Packet C (added at Lab 4)***

| ID  | Item                                                                                                                                        | Source category   | Rel./Cred. |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| E31 | State registry: Halcyon Ridge Properties LLC, active, formed 7 years ago. Manager: R. Okafor. Registered agent: Statewide Agents Inc.       | Corporate filings | A1         |
| E32 | State registry: Northvale Holdings LLC, active. Sole member of Halcyon Ridge Properties LLC per its annual report. Manager: R. Okafor       | Corporate filings | A1         |
| E33 | State registry: Meridian Adjusting Services LLC, active, formed 4 years ago. Manager: R. Okafor. Registered agent: Statewide Agents Inc.    | Corporate filings | A1         |
| E34 | Insurer file: the independent adjuster assigned to the Crestline contents claim is Meridian Adjusting Services LLC                          | Insurer file      | A1         |
| E35 | State registry: Crestline Distribution Inc. Officers: T. Vance (president), L. Serrano (secretary). Registered agent: Statewide Agents Inc. | Corporate filings | A1         |

| <b>ID</b> | <b>Item</b>                                                                                                                                                       | <b>Source category</b> | <b>Rel./Cred.</b> |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|
| E36       | Statewide Agents Inc. is a commercial registered agent service listing approximately 31,000 entities in the state                                                 | Corporate filings      | A1                |
| E37       | County property records: 1400 Quarry Line Road owned by Halcyon Ridge Properties LLC; acquired 7 years ago                                                        | Government records     | A1                |
| E38       | County property records: three additional industrial parcels in the county owned by Northvale Holdings LLC                                                        | Government records     | A1                |
| E39       | Court records: two prior fire loss claims in the county involving properties owned by Northvale Holdings LLC, 5 and 3 years ago, both resolved without litigation | Insurer/industry file  | B2                |
| E40       | Electrical contractor invoice (E16) letterhead lists a business address identical to the registered address of Northvale Holdings LLC                             | Corporate records      | B2                |
| E41       | Professional profile for L. Serrano lists prior employment at Meridian Adjusting Services, ending 2 years ago                                                     | Public profile         | C3                |
| E42       | No disclosure of any relationship between Halcyon Ridge Properties LLC and Meridian Adjusting Services LLC appears in the insurer's file                          | (negative)             | A1                |

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## Lab 2: Competing Hypotheses

**Time required:** 2 to 2.5 hours **Format:** Individual **Difficulty:** Intermediate

### Objectives

1. Generate at least four genuine competing hypotheses with disproof conditions.
2. Score evidence by consistency rather than support.
3. Identify discriminating evidence and state what the judgment actually rests on.
4. Eliminate hypotheses on recorded bases and calibrate confidence.

### Prerequisites

- Section 1.3 and comparison tables 2.1, 2.2, and 2.4.
- Evidence Packet A.

### Steps

**Step 1 (15 min).** Before reading the packet a second time, write down the explanation that first occurred to you. Seal it (fold the page, or note it in a separate file). You will compare it against your conclusion in Step 8. This step is graded and its purpose is to make your prior visible rather than to embarrass you.

**Step 2 (30 min).** Generate hypotheses on Worksheet 2A. Minimum four, and you must include the null or an innocent explanation, plus at least one of third-party, compound, deception, or incompetence. For each: the hypothesis in declarative form, its disproof condition, and how a proponent of it would summarize the case in one sentence.

**Step 3 (45 min).** Score Packet A on Worksheet 2B. Every item against every hypothesis, using ++, +, 0, -, --. For each cell, ask "if this hypothesis were true, would I expect to see this item?" Record a one-line reason for every ++ and every --.

**Step 4 (15 min).** Disconfirmation pass on Worksheet 2C. For each hypothesis: what would disprove it, did you look for that in the packet, and what did you find? At least one honest "the packet does not address this" is expected.

**Step 5 (15 min).** Discriminating evidence on Worksheet 2D. Identify every item that distinguishes between two or more hypotheses, and every item consistent with all of them. Then write one sentence stating how many items your emerging judgment actually rests on.

**Step 6 (15 min).** Elimination on Worksheet 2E. For each hypothesis you eliminate: the basis, the item IDs, and whether the elimination is provisional pending a named gap.

**Step 7 (20 min).** Write the interim assessment, half a page: the best-supported hypothesis or

hypotheses, the confidence with its reason, the discriminating evidence named, the alternatives and why they were downgraded, and at least three ranked gaps.

**Step 8 (10 min).** Open your Step 1 note. Write a paragraph comparing your initial explanation to your assessment. If they match, describe specifically what evidence would have changed your mind and whether you looked for it.

### **Worksheet 2A: Hypotheses**

| H# | Hypothesis<br>(declarative) | Type | Disproof condition | One-sentence<br>proponent<br>summary |
|----|-----------------------------|------|--------------------|--------------------------------------|
|----|-----------------------------|------|--------------------|--------------------------------------|

### **Worksheet 2B: Scoring Matrix**

| Item | H1 | H2 | H3 | H4 | H5 | Reason for<br>any ++ or -- |
|------|----|----|----|----|----|----------------------------|
|------|----|----|----|----|----|----------------------------|

### **Worksheet 2C: Disconfirmation**

| H# | What would disprove it | Did I look? | What I found |
|----|------------------------|-------------|--------------|
|----|------------------------|-------------|--------------|

### **Worksheet 2D: Discrimination**

| Item | Distinguishes which hypotheses? | Or consistent with all? |
|------|---------------------------------|-------------------------|
|------|---------------------------------|-------------------------|

### **Worksheet 2E: Elimination**

| H# | Eliminated? | Basis (item IDs) | Provisional pending |
|----|-------------|------------------|---------------------|
|----|-------------|------------------|---------------------|

### **Expected Outcomes**

- Four-plus hypotheses covering at minimum: accidental electrical, accidental non-electrical, deliberate for insurance benefit, deliberate by a third party. Strong submissions add a compound hypothesis (an accidental fire at a site where an unrelated inventory misstatement exists) and recognize that the inventory question and the cause question are separable.
- **The separability insight is the lab's central lesson.** E8 conflicts with E9, and E19, E28, and E29 bear on financial condition. None of this establishes anything about ignition. A student whose matrix treats financial motive items as strongly consistent with the arson hypothesis and inconsistent with the accidental hypotheses has confused motive with causation.
- E6 (cause undetermined) should be scored as consistent with nearly all hypotheses and therefore

non-discriminating, despite feeling important.

- E4 (sensor in fault since 2 March) should be recognized as reducing the value of E3 for every hypothesis, not as supporting one. Students who score it as supporting third-party entry have missed that it equally undermines the exclusion of tenant re-entry.
- E17 (no remediation permit) is genuinely discriminating between the accidental-electrical hypothesis and the others, and it is a negative finding.
- Confidence should be low to moderate at this stage, and no hypothesis should be eliminated outright on Packet A alone. Students who reach a confident arson conclusion here have run ahead of the evidence, which the packet is designed to invite.
- The Step 8 comparison should show movement. Where it does not, the required paragraph on what would have changed your mind is graded strictly.

### ***Discussion Questions***

1. Which items did you initially score as strongly supporting a hypothesis and later reclassify as non-discriminating?
2. E8 and E9 conflict. What does the conflict establish, and what does it not establish?
3. How many items does your emerging judgment rest on? Was that number what you expected?
4. Did your Step 1 explanation survive? What does your answer tell you about the value of Step 1?

### ***Grading Rubric: Lab 2 (60 points)***

| <b>Criterion</b>               | <b>Weight</b> | <b>4 points looks like</b>                                     |
|--------------------------------|---------------|----------------------------------------------------------------|
| Prior recorded before analysis | x1            | Written and sealed at Step 1, honestly compared at Step 8      |
| Hypothesis set breadth         | x3            | Four-plus, required types present, no overlaps                 |
| Hypotheses fairly stated       | x2            | The least likely written as its proponent would                |
| Disproof conditions            | x2            | Specific and observable for every hypothesis                   |
| Scoring by consistency         | x3            | Cells answer the correct question; reasons given for ++ and -- |
| Motive and causation separated | x3            | Financial items not scored as evidence of ignition mechanism   |
| Disconfirmation pass           | x3            | Answered per hypothesis, including honest gaps                 |
| Discrimination identification  | x3            | Non-discriminating items correctly identified; count stated    |

| Criterion              | Weight | 4 points looks like                             |
|------------------------|--------|-------------------------------------------------|
| Elimination discipline | x2     | Recorded bases; provisional eliminations marked |
| Confidence calibration | x3     | Low to moderate, with the reason stated         |
| Ranked gaps            | x2     | Three-plus, each with what would resolve it     |

Total possible: 108 weighted. Passing: 74.

---

## Lab 3: Timeline Construction and Inconsistency Analysis

**Time required:** 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

### Objectives

1. Construct a timeline with precision markers, entry types, and source attribution.
2. Detect and classify inconsistencies.
3. Derive investigative steps from each flag.
4. Re-score the hypothesis matrix in light of the timeline.

### Prerequisites

- Section 1.4 and comparison table 2.5.
- Lab 2 complete. Evidence Packets A and B.

### Steps

**Step 1 (60 min).** Build the master timeline on Worksheet 3A from Packets A and B combined. Every dated or datable item becomes at least one row. Requirements:

- Precision marker on every row: exact, minute, hour, day, month, or bracketed window.
- Entry type on every row: observed, reported, or inferred.
- Source item IDs on every row.
- At least three inferred entries, each marked, each with the reasoning stated.
- Two views: a full chronological view, and a focused view of 8 March 17:00 to 9 March 03:00 at minute resolution.

**Step 2 (30 min).** Inconsistency analysis on Worksheet 3B. Identify every contradiction, impossibility, gap, anomaly, and convergence. Minimum eight flags. For each: the type, the items involved, and what the inconsistency does and does not establish.

**Step 3 (20 min).** Investigative steps on Worksheet 3C. For each flag, the specific step that would resolve it, the source category it would come from, and whether it is available to your consumers (note that E23 tells you something about what is available).

**Step 4 (20 min).** Opportunity analysis on Worksheet 3D. Given the evidenced timeline, state for each hypothesis from Lab 2 whether opportunity existed, was excluded, or is indeterminate, and why. Handle E4, E24, and E25 carefully.

**Step 5 (25 min).** Re-score. Return to your Lab 2 matrix and add the Packet B items. Record which cells changed and why on Worksheet 3E. Then restate your confidence and note whether it moved.

**Step 6 (15 min).** Write the timeline findings, half a page: what the sequence establishes, what it excludes, the two most significant inconsistencies, and the single investigative step you would take first.

### ***Worksheet 3A: Master Timeline***

| Date/<br>time | Precision | Type | Event<br>(as<br>observed<br>) | Actors | Location | Item IDs | Categories | Confidence | Flags |
|---------------|-----------|------|-------------------------------|--------|----------|----------|------------|------------|-------|
|               |           |      |                               |        |          |          |            |            |       |

### ***Worksheet 3B: Inconsistencies***

| Flag # | Type | Items involved | What it establishes | What it does NOT establish |
|--------|------|----------------|---------------------|----------------------------|
|        |      |                |                     |                            |

### ***Worksheet 3C: Derived Steps***

| Flag # | Investigative step | Source category | Available to our consumers? |
|--------|--------------------|-----------------|-----------------------------|
|        |                    |                 |                             |

### ***Worksheet 3D: Opportunity***

| Hypothesis | Opportunity existed /<br>excluded / indeterminate | Reasoning | Key items |
|------------|---------------------------------------------------|-----------|-----------|
|            |                                                   |           |           |

### ***Worksheet 3E: Re-scoring***

| Item | Hypothesis | Old score | New score | Why it changed |
|------|------------|-----------|-----------|----------------|
|      |            |           |           |                |

### ***Expected Outcomes***

- A timeline with genuine precision variation. E26 ("late, maybe eleven or midnight," "sometime

that week") must be recorded as a bracketed window at day precision or worse, not placed on 8 March. Students who place it on the night of the fire have committed false precision, and it is the most common error in this lab.

- At least eight flags. The expected set includes: the E8 and E9 contradiction; the E15 and E21 contradiction; the E3 and E4 combination creating a gap in access coverage; the anomaly of E4 with no repair ticket; the anomaly of E29 (inbound hold) five days before the fire; the convergence of E1, E2, E16, E17, E19, E28, and E29 in a six-week window; the E7 and E22 vehicle sightings bracketing a 37-minute absence; and the E20 fuel transaction at 21:52 placing a company vehicle 6 km north.
- **The E7, E20, E22 combination is the lab's centrepiece.** A fuel transaction at 21:52 six km north, a southbound vehicle at 22:10 two blocks north, and a northbound vehicle at 22:47 are consistent with a round trip to the site. They are also consistent with any other vehicle on a public road. The correct handling is to record the pattern, note that the plate is only partially legible on one and illegible on the other, mark the identification as unestablished, and derive the step (obtain the full plate, or obtain the vehicle's telematics if any) rather than asserting the trip occurred.
- E24 and E25 together must be read correctly: the alarm's silence between 19:41 and 02:22 does not establish that no one entered, because the rear zone was in fault. Students who cite alarm silence as evidence of no entry have made a coverage assumption.
- The opportunity analysis should conclude indeterminate for most hypotheses, and the reason should be the sensor fault. This is the correct and counterintuitive result.
- Confidence should remain low to moderate after Packet B. The additional items add texture and conflict, not resolution.

### ***Discussion Questions***

1. Where did you place E26, and what would placing it on 8 March have done to your analysis?
2. The alarm was silent. Explain in two sentences why that establishes less than it appears to.
3. The vehicle sightings are suggestive. Write the sentence you would use in a product, and the sentence a careless investigator would use.
4. Which flag would you resolve first, and what makes it the highest-value step?
5. Did your confidence move after Packet B? In which direction, and why is that direction defensible?

### ***Grading Rubric: Lab 3 (68 points)***

| Criterion             | Weight | 4 points looks like                                 |
|-----------------------|--------|-----------------------------------------------------|
| Timeline completeness | x2     | Every datable item represented; both views produced |

| Criterion                   | Weight | 4 points looks like                                                                   |
|-----------------------------|--------|---------------------------------------------------------------------------------------|
| Precision markers           | x3     | Genuine variation; E26 correctly bracketed                                            |
| Entry types                 | x3     | Observed, reported, inferred correctly assigned; three-plus inferred entries reasoned |
| Source attribution          | x2     | Item IDs on every row                                                                 |
| Inconsistency detection     | x3     | Eight-plus flags, correctly typed                                                     |
| "Does not establish" column | x3     | Rigorously completed; over-reading avoided                                            |
| Vehicle-sighting handling   | x3     | Pattern recorded, identification not asserted, step derived                           |
| Alarm and sensor reasoning  | x3     | Coverage assumption avoided; opportunity correctly indeterminate                      |
| Derived steps               | x2     | Specific, sourced, availability assessed                                              |
| Re-scoring record           | x2     | Changed cells identified with reasons                                                 |
| Confidence movement         | x2     | Stated, with a defensible direction                                                   |

Total possible: 112 weighted. Passing: 77.

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## Lab 4: Link Analysis and Hidden Relationships

**Time required:** 2.5 to 3 hours **Format:** Individual **Difficulty:** Advanced

### Objectives

1. Build a link chart with typed nodes and fully attributed edges.
2. Distinguish direct from indirect relationships and apply the discrimination test.
3. Identify brokers, clusters, and structural gaps.
4. Discover hidden relationships and re-score the hypothesis matrix.

### Prerequisites

- Sections 1.5 and 1.7, and comparison tables 2.6 and 2.7.
- Labs 2 and 3 complete. Evidence Packets A, B, and C.

## Steps

**Step 1 (20 min).** Entity inventory on Worksheet 4A. Every person, organization, location, event, asset, digital artifact, and document appearing anywhere in Packets A through C. Type each. Minimum eighteen nodes.

**Step 2 (25 min).** Association matrix on Worksheet 4B. Every entity against every other, marked confirmed, suspected, or none. Do this before drawing anything. The purpose is to force consideration of pairs you would not have drawn.

**Step 3 (45 min).** Build the chart on Worksheet 4C and the edge register on Worksheet 4D. Every edge carries type, direction, basis (item IDs), and strength. Render strengths distinguishably. Minimum twenty edges.

**Step 4 (25 min).** Shared attribute analysis on Worksheet 4E. Build the attribute table: entities as rows, attribute classes as columns (address, registered agent, officers, letterhead address, prior employment, insurer relationships). Sort by each column. Record every repeat.

**Step 5 (20 min).** Discrimination testing on Worksheet 4F. For every indirect relationship you found, apply the four tests from Section 1.7.3. Pay particular attention to E36.

**Step 6 (20 min).** Structural analysis on Worksheet 4G: highest degree, highest betweenness, any cut-point, any cluster, and any negative-space finding. Explain which node you would investigate first and why.

**Step 7 (25 min).** Re-score and write. Add the Packet C findings to your matrix, record what changed, and write the half-page relationship findings: the structural findings, each worded per Section 1.7.4, with confidence and the alternative explanations considered.

### Worksheet 4A: Entity Inventory

| Node ID | Entity | Type | First appears in |
|---------|--------|------|------------------|
|---------|--------|------|------------------|

### Worksheet 4B: Association Matrix

|  | N1 | N2 | N3 | ... |
|--|----|----|----|-----|
|--|----|----|----|-----|

### Worksheet 4C: Chart

Text diagram or drawing. Strengths must be visually distinguishable.

### Worksheet 4D: Edge Register

| Edge # | From | To | Type | Direction | Basis (item IDs) | Strength | Added by |
|--------|------|----|------|-----------|------------------|----------|----------|
|--------|------|----|------|-----------|------------------|----------|----------|

### **Worksheet 4E: Attribute Table**

| Entity | Address | Registered agent | Officers | Other attributes | Repeats found |
|--------|---------|------------------|----------|------------------|---------------|
|--------|---------|------------------|----------|------------------|---------------|

### **Worksheet 4F: Discrimination Testing**

| Indirect relationship | Discrimination | Alternative explanation | Independence | Temporal validity | Verdict |
|-----------------------|----------------|-------------------------|--------------|-------------------|---------|
|-----------------------|----------------|-------------------------|--------------|-------------------|---------|

### **Worksheet 4G: Structural Analysis**

| Measure | Node | Reasoning |
|---------|------|-----------|
|---------|------|-----------|

### **Expected Outcomes**

- **The central finding:** R. Okafor manages Halcyon Ridge Properties LLC (the landlord, E31), Northvale Holdings LLC (its parent, E32), and Meridian Adjusting Services LLC (E33), which is the independent adjuster assigned to the claim (E34). No document in the insurer's file discloses this (E42). This is a latent hidden relationship found by shared-officer analysis across three registry filings, and it is highly discriminating because the intermediary is a named individual.
- **The trap:** E36 establishes that Statewide Agents Inc. serves approximately 31,000 entities. Any student who draws a meaningful relationship between Crestline (E35) and the Okafor entities on the basis of the shared registered agent has failed the discrimination test, and this is graded heavily. The shared agent is worthless as an indicator.
- **The secondary finding:** E40, the electrical contractor's letterhead address matching Northvale Holdings' registered address, is an indirect relationship of moderate strength requiring the alternative-explanation test (shared commercial building, prior tenancy, a service arrangement). It should be flagged for verification, not asserted.
- **The third finding:** E41, L. Serrano's prior employment at Meridian, is a temporal-validity question. The employment ended two years ago, and its relevance depends on whether any relationship persisted. The correct handling is to record it as possible with the temporal caveat.
- **Structural:** R. Okafor is the highest-betweenness node and the cut-point connecting the property side to the adjusting side. Students should identify that removing that node disconnects the two clusters, which is precisely what makes it significant.
- **Negative space:** E42 is an expected-but-absent disclosure. Students should record it as a negative finding and word it as non-disclosure, not concealment.
- **Re-scoring:** the Packet C findings do not establish anything about the cause of the fire. They

establish a conflict of interest in the claims process. Students who score the Okafor finding as evidence supporting the arson hypothesis have made the guilt-by-association error, and this is the second heavily graded trap.

### **Discussion Questions**

1. Statewide Agents Inc. connects three entities in your chart. Why is that connection worthless, and what would make a shared agent meaningful?
2. R. Okafor has high betweenness. What does that establish about the fire? What does it establish about the claim?
3. Write the finding about the landlord and the adjuster in one sentence that a lawyer would approve, and one sentence that would create liability.
4. E42 is an absence. Explain why absence of disclosure is a finding and why "concealment" is not the word for it.
5. What did the association matrix make you consider that you would not have drawn?

### **Grading Rubric: Lab 4 (76 points)**

| <b>Criterion</b>                   | <b>Weight</b> | <b>4 points looks like</b>                                                  |
|------------------------------------|---------------|-----------------------------------------------------------------------------|
| Entity inventory and typing        | x2            | Eighteen-plus nodes, correctly typed                                        |
| Association matrix completed first | x2            | Full pairwise assessment before drawing                                     |
| Edge attribution                   | x3            | Every edge has type, direction, basis, strength                             |
| Strength rendering                 | x2            | Confirmed, probable, possible, hypothesized visually distinct               |
| Shared attribute analysis          | x3            | Systematic table; all repeats found                                         |
| Discrimination test on E36         | x4            | Registered agent correctly rejected as non-discriminating                   |
| Okafor finding identified          | x4            | Landlord, parent, and adjuster connection found and correctly characterized |
| Guilt-by-association avoided       | x4            | Okafor finding not scored as evidence of ignition cause                     |
| Secondary findings handled         | x2            | E40 and E41 flagged with the correct tests and caveats                      |
| Structural analysis                | x3            | Betweenness, cut-point, clusters, and negative space identified             |
| Wording discipline                 | x3            | Non-disclosure not called                                                   |

| Criterion         | Weight | 4 points looks like                   |
|-------------------|--------|---------------------------------------|
|                   |        | concealment; no intent asserted       |
| Re-scoring record | x2     | Changed cells identified with reasons |

Total possible: 136 weighted. Passing: 93.

---

## Lab 5: Auditing a Completed Investigation

**Time required:** 2.5 to 3 hours **Format:** Individual **Difficulty:** Advanced

### Objectives

1. Identify unsupported assumptions in a finished product.
2. Name the cognitive biases operating and the specific control for each.
3. Rewrite conclusions using only verified evidence.
4. Document how the revised methodology changed the findings.

### Prerequisites

- Sections 1.6 and 1.1.5, and comparison tables 2.1, 2.2, and 2.8.
- The provided report and evidence list below. This is a different case from Labs 2 through 4.

### The Provided Investigation

*Fictional. All entities and persons are invented.*

**Background.** The Coastal Basin Regional Council administers a small grants programme. An internal reviewer produced the report below concerning a grantee, Tidewater Youth Collective. Your task is to audit it.

### THE REPORT AS SUBMITTED

#### Findings: Tidewater Youth Collective grant review

(1) Tidewater Youth Collective received \$340,000 across three grant cycles. (2) The organization's reported participant numbers rose from 180 to 640 over that period, a 256% increase that is implausible for an organization of this size. (3) Site visits were conducted in cycles one and two but not cycle three, indicating that oversight lapsed. (4) The executive director, K. Ballard, previously worked at Harbor Lane Services, another grantee, which shows a pattern of individuals moving between funded organizations to maintain access to Council money. (5) Tidewater's registered address is a shared office suite that also houses two other grantees, so the three organizations are clearly connected. (6) Tidewater's annual filings show program expenses of \$290,000 against the \$340,000 received; the remaining \$50,000 is unaccounted for. (7) Three former staff members declined to be interviewed,

which suggests they were instructed not to speak. **(8)** In cycle three, Tidewater submitted its report eleven days late, and the late submission conveniently followed the Council's announcement of an audit. **(9)** No complaints have been received from participants, but this is expected in a programme serving a vulnerable population who would not know how to complain. **(10)** Taken together, the evidence establishes that Tidewater Youth Collective inflated its participant numbers to obtain funding it did not earn, and that Council oversight failures allowed this to continue. **(11)** We recommend referral for prosecution and immediate suspension of all three organizations at the shared address.

#### THE EVIDENCE LIST THE REVIEWER RELIED ON

| ID | Item                                                                                                                                                           | Notes                          |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| F1 | Grant award records: \$120,000, \$110,000, \$110,000 across three cycles                                                                                       | Council records                |
| F2 | Tidewater's submitted participant counts: 180, 410, 640                                                                                                        | Self-reported in grant reports |
| F3 | Council site visit records: visits logged for cycles one and two; cycle three visit scheduled and cancelled by the Council, with a note "reviewer unavailable" | Council records                |
| F4 | Professional profile for K. Ballard listing prior employment at Harbor Lane Services, ending four years ago                                                    | Public profile                 |
| F5 | Registry filings: Tidewater, Harbor Lane Services, and Anchor Point Learning all list the same street address, Suite 200, 44 Marina Way                        | Corporate filings              |
| F6 | Building directory for 44 Marina Way listing a coworking operator on the second floor with 60-plus member organizations                                        | Public listing                 |
| F7 | Tidewater's most recent annual filing: total expenses \$312,000, of which program expenses \$290,000, administration \$22,000                                  | Public filing                  |
| F8 | Council grant terms permitting up to 12% administrative allocation                                                                                             | Council records                |
| F9 | The three grant cycles span 26 months; Tidewater's fiscal year                                                                                                 | Council records                |

| ID  | Item                                                                                                                                                                  | Notes              |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|     | does not align with the grant cycles                                                                                                                                  |                    |
| F10 | Email records: three former staff were contacted once each; two did not reply, one replied declining, citing a new employer's policy on outside interviews            | Reviewer's file    |
| F11 | Cycle three report submitted 11 days after the deadline; Council's audit announcement was published two weeks before the report deadline                              | Council records    |
| F12 | Council complaint log: no complaints regarding Tidewater; also no complaints regarding any of the 22 grantees in the programme over four years                        | Council records    |
| F13 | Tidewater's cycle two and three reports describe a change in counting methodology, from "enrolled participants" to "participants served including drop-in attendance" | Grant reports      |
| F14 | Comparable regional programmes report participant growth of 40% to 300% following methodology changes of this kind                                                    | Sector publication |
| F15 | No independent verification of participant counts was performed in any cycle                                                                                          | Council records    |

## Steps

**Step 1 (30 min).** Claim inventory on Worksheet 5A. Break the report into its individual claims, one per row, using the numbering provided. For each: is it a fact claim, an evidence citation, or an inference? Which evidence items does it actually cite or rest on?

**Step 2 (35 min).** Assumption audit on Worksheet 5B. Perform a therefore-audit on the whole report, then the negation test on claims 2, 4, 5, 6, 7, 9, and 10. Record every assumption, its type, what rests on it, and the consequence if wrong. Minimum ten assumptions.

**Step 3 (25 min).** Bias identification on Worksheet 5C. For each claim, name any cognitive bias

operating and the specific control that would have prevented it. Minimum six biases identified, with at least four distinct types.

**Step 4 (20 min).** Evidence-to-claim mapping on Worksheet 5D. For each claim, state whether the cited evidence supports it as written, supports a weaker version, or does not support it. Identify every claim where evidence in the list contradicts the claim.

**Step 5 (35 min).** Rewrite. Produce a revised findings document using only what the evidence supports. Every claim cites item IDs. Claims that cannot be supported are either removed or restated as open questions. Include confidence levels and ranked gaps.

**Step 6 (20 min).** Change documentation on Worksheet 5E. For each original claim: what changed, why, and what the change does to the overall conclusion. Then write two paragraphs on how the revised methodology altered the outcome and what the original report's failure signature was.

**Step 7 (15 min).** Recommendations. Rewrite claim 11. State what recommendation the revised evidence actually supports, and explain what would be required before any referral could be considered.

### ***Worksheet 5A: Claim Inventory***

| Claim # | Text | Fact / evidence / inference | Items cited or relied on | Actually supports it? |
|---------|------|-----------------------------|--------------------------|-----------------------|
|---------|------|-----------------------------|--------------------------|-----------------------|

### ***Worksheet 5B: Assumption Register***

| A# | Assumption | Type | Claim it supports | Consequence if wrong | Testable? |
|----|------------|------|-------------------|----------------------|-----------|
|----|------------|------|-------------------|----------------------|-----------|

### ***Worksheet 5C: Bias Identification***

| Claim # | Bias | How it operates here | Control that would have prevented it |
|---------|------|----------------------|--------------------------------------|
|---------|------|----------------------|--------------------------------------|

### ***Worksheet 5D: Evidence Mapping***

| Claim # | Supported as written / weaker version / not supported | Contradicting items | Notes |
|---------|-------------------------------------------------------|---------------------|-------|
|---------|-------------------------------------------------------|---------------------|-------|

### ***Worksheet 5E: Change Log***

| Original claim | Revised claim or removed | Why | Effect on the conclusion |
|----------------|--------------------------|-----|--------------------------|
|----------------|--------------------------|-----|--------------------------|

## ***Expected Outcomes***

The audit should find at minimum the following. Students who find fewer than eight of these have not audited rigorously.

- **Claim 2** is contradicted by F13 and F14 in the reviewer's own evidence list. The counting methodology changed, and comparable programmes show growth in this range after such changes. "Implausible" is an unsupported judgment, and the base rate was available and ignored.
- **Claim 3** is contradicted by F3: the Council cancelled its own visit. The report attributes the lapse to the grantee's circumstances by implication. This is attribution error and a misreading of the reviewer's own record.
- **Claim 4** is a bare association plus an inference of intent. F4 shows employment ending four years ago. "Shows a pattern of individuals moving between organizations to maintain access" is unsupported by anything in the list.
- **Claim 5** fails the discrimination test decisively. F6 establishes that the address is a coworking space with 60-plus member organizations. "Clearly connected" is manufactured from a non-discriminating shared attribute, and it is the single worst error in the report because claim 11 rests on it.
- **Claim 6** is arithmetic on incomparable periods. F7 reports one fiscal year; F9 establishes that the grant cycles span 26 months and do not align with the fiscal year. F8 permits an administrative allocation that F7 shows was within the limit. The "\$50,000 unaccounted for" does not survive contact with F7, F8, and F9 together.
- **Claim 7** is contradicted by F10: three staff were contacted once; two did not reply and one declined citing an employer policy. "Declined to be interviewed" misstates the record for two of the three, and "suggests they were instructed not to speak" is an inference of coordination with no basis whatsoever.
- **Claim 9** is unfalsifiable as written. It converts the absence of complaints into support for the conclusion, and F12 shows no complaints about any of 22 grantees over four years, which indicates a complaint-channel problem rather than anything about Tidewater.
- **Claim 8** treats sequence as motive without checking it: F11 shows the audit announcement preceded the deadline by two weeks, so the ordering is real, but "conveniently" asserts intent, and no evidence addresses why the report was late.
- **Claim 10** rests on claims 2, 5, 6, and 9, three of which are contradicted by the reviewer's own evidence. It also asserts the central fact (inflated numbers) when F15 establishes that no independent verification was ever performed, which means the counts are unverified in both directions.
- **Claim 11** recommends suspending two organizations solely on the basis of a coworking

address.

- **Failure signature:** the premature narrative, plus load-bearing assumptions. Every ambiguous item is read against the grantee, and the evidence list contains the refutation of several claims, meaning the reviewer collected the disconfirming evidence and did not use it.

**The revised findings should conclude approximately:** participant counts are unverified in both directions and the reported increase coincides with a documented methodology change that is consistent with sector norms; the cycle three site visit lapse is a Council failure; the expense analysis is invalid as constructed and must be redone on aligned periods; no relationship between the three organizations is established by the shared coworking address; and the substantive open question is whether independent verification of participant counts can be performed retrospectively. Confidence in any finding of wrongdoing: none, on this evidence.

### ***Discussion Questions***

1. The reviewer's own evidence list contained the refutation of at least four claims. What does that tell you about where the failure occurred?
2. Claim 5 supports claim 11, which would affect two organizations that were never investigated. Describe the harm the original report would have caused.
3. Which of the eleven claims survives the audit essentially intact?
4. The revised conclusion is much weaker and much shorter. Explain to a stakeholder why that is an improvement rather than a failure.
5. Name the one investigative step that would do most to resolve the substantive question, and explain why it was never taken.

### ***Grading Rubric: Lab 5 (72 points)***

| <b>Criterion</b>               | <b>Weight</b> | <b>4 points looks like</b>                                                                    |
|--------------------------------|---------------|-----------------------------------------------------------------------------------------------|
| Claim inventory and typing     | x2            | Every claim classified; reliance correctly traced                                             |
| Assumption register            | x3            | Ten-plus assumptions, typed, with consequences                                                |
| Bias identification            | x3            | Six-plus, four-plus distinct types, with specific controls                                    |
| Evidence mapping               | x3            | Contradicting items in the reviewer's own list identified                                     |
| Claim 5 discrimination failure | x4            | Coworking address correctly identified as non-discriminating and as load-bearing for claim 11 |
| Claim 6 period incomparability | x3            | F7, F8, F9 correctly combined to invalidate the arithmetic                                    |

| Criterion                | Weight | 4 points looks like                                                                        |
|--------------------------|--------|--------------------------------------------------------------------------------------------|
| Claim 7 misstatement     | x3     | Record correctly read; the inference of coordination identified as baseless                |
| Claim 9 unfalsifiability | x2     | Identified, with F12's programme-wide implication noted                                    |
| Rewritten findings       | x4     | Every claim cited; unsupported claims removed or converted to questions; confidence stated |
| Change documentation     | x3     | Per-claim, with the effect on the conclusion                                               |
| Failure signature named  | x2     | Correctly diagnosed, with reasoning                                                        |
| Revised recommendations  | x2     | Proportionate; the two uninvestigated organizations not swept in                           |

Total possible: 136 weighted. Passing: 93.

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## Part 9: Capstone Assignment

### Operation Quarry Light: An End-to-End Investigation

**Weight:** 35% of chapter grade **Time required:** 6 to 9 hours **Format:** Individual **Deliverables:** Nine items, listed in Section 9.4

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#### 9.1 The Scenario

*Everything below is fictional.*

You are the analyst supporting a joint review team at the **Cedar Bluff Regional Airport Authority**, a public body governed by a seven-member appointed commission.

**The trigger.** During a routine reconciliation, the Authority's controller identified that the ground handling services contract, held by **Quarry Light Aviation Services LLC**, has been invoiced 41% above the contracted schedule over 22 months, through a series of 63 change orders. The Authority's director of operations approved all 63. The commission has asked for an assessment before its 14 June meeting, six weeks away.

**The commission must choose among:**

1. Treat this as a contract administration failure, correct the process, and continue with the vendor.
2. Terminate the contract and re-tender, which would cost an estimated \$1.4M in transition and carries operational risk.
3. Refer the matter to the state inspector general and the county prosecutor, and suspend the contract pending review.

### Complications.

- The director of operations, **M. Halloran**, is a 19-year employee with an unblemished record who has stated that every change order responded to a genuine operational need.
- The Authority is subject to public records law; your product may become public.
- A commissioner has already told a local reporter that "this looks like exactly what it looks like."
- Quarry Light employs 140 people locally, and termination has employment consequences the commission will weigh.

## 9.2 The Evidence Packet

Forty items. Some are decisive, some are noise, at least three conflict with one another, at least one is probably unreliable, and at least one raises an ethical question you must handle explicitly.

| ID | Item                                                                                                                               | Category           |
|----|------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| G1 | Original contract, awarded 26 months ago after competitive tender; base value \$3.2M annually; three bidders                       | Government records |
| G2 | Change order log: 63 orders over 22 months, cumulative value \$2.9M against a base of \$7.0M for the period                        | Government records |
| G3 | Change order distribution by month: 4 in months 1 to 6, 17 in months 7 to 14, 42 in months 15 to 22                                | Government records |
| G4 | Authority procurement policy: change orders below \$75,000 may be approved by the director of operations without commission action | Government records |
| G5 | Change order values: 58 of 63 fall between \$61,000 and \$74,500                                                                   | Government records |

| <b>ID</b> | <b>Item</b>                                                                                                                            | <b>Category</b>               |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| G6        | Five change orders exceed \$75,000; all five carry commission minutes recording approval                                               | Government records            |
| G7        | Aviation industry publication: regional ground handling costs rose 19% over the period due to labour and equipment costs               | Trade press                   |
| G8        | Two peer airports' published contract amendments show cumulative increases of 22% and 27% over comparable periods                      | Government records            |
| G9        | Change order narratives: 44 of 63 cite "additional aircraft movements"; 11 cite "equipment substitution"; 8 cite "extended hours"      | Government records            |
| G10       | Authority's published monthly aircraft movement statistics: movements rose 8% over the 22 months                                       | Government records            |
| G11       | Authority's flight schedule records show extended-hours operations on 61 dates; 8 change orders cite extended hours                    | Government records            |
| G12       | Quarry Light Aviation Services LLC: state registry, formed 6 years ago; managers J. Reyes and D. Coleman                               | Corporate filings             |
| G13       | Cedar Bluff Equipment Leasing LLC: state registry, formed 5 years ago; manager D. Coleman; registered agent Statewide Agents Inc.      | Corporate filings             |
| G14       | Eleven "equipment substitution" change orders name Cedar Bluff Equipment Leasing as the equipment provider                             | Government records            |
| G15       | No disclosure of a relationship between Quarry Light and Cedar Bluff Equipment Leasing appears in the Authority's vendor conflict file | Government records (negative) |

| <b>ID</b> | <b>Item</b>                                                                                                                                                                   | <b>Category</b>    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| G16       | Authority vendor disclosure form for Quarry Light, signed at award, listing no affiliated entities                                                                            | Government records |
| G17       | Statewide Agents Inc. serves approximately 31,000 entities                                                                                                                    | Corporate filings  |
| G18       | M. Halloran's annual conflict disclosure forms for three years: no disclosures                                                                                                | Government records |
| G19       | Property records: M. Halloran owns a residence in the county; no other property                                                                                               | Government records |
| G20       | Professional profile for M. Halloran's adult child listing employment at a regional aviation services firm; the firm is not Quarry Light and is not named in any change order | Public profile     |
| G21       | Email from Halloran to the controller, month 9: "the change order threshold is the only tool I have; the contract scope was written too narrowly"                             | Internal record    |
| G22       | Authority board minutes, month 4: the commission declined a request to increase the contract's base scope, citing budget                                                      | Government records |
| G23       | Internal memo from the previous controller, month 12, flagging change order volume; no recorded response                                                                      | Internal record    |
| G24       | The previous controller left the Authority in month 14                                                                                                                        | Internal record    |
| G25       | Exit interview record for the previous controller: cites "relocation for family reasons"                                                                                      | Internal record    |
| G26       | Change order approval timestamps: 39 of 63 were approved within 24 hours of submission                                                                                        | Government records |
| G27       | Change order supporting documentation: 21 of 63 files                                                                                                                         | Government records |

| ID  | Item                                                                                                                                                           | Category                 |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|     | contain no attachment beyond the order form itself                                                                                                             |                          |
| G28 | Authority policy requires "documentation of operational necessity" for every change order                                                                      | Government records       |
| G29 | Quarry Light's invoices reconcile exactly to the change order values in all 63 cases                                                                           | Government records       |
| G30 | Bank of the Authority's payment records: all payments made to Quarry Light's single account of record                                                          | Government records       |
| G31 | Court records: one civil case involving Quarry Light, a wage dispute, resolved by settlement 3 years ago                                                       | Court records            |
| G32 | Court records: no cases involving Cedar Bluff Equipment Leasing                                                                                                | Court records (negative) |
| G33 | Local news article, two weeks ago, quoting an unnamed commissioner describing the situation as "a scandal"                                                     | Newspapers               |
| G34 | Social media: four accounts created within the past month posting identical text alleging kickbacks at the Authority                                           | Social media             |
| G35 | Archived capture of Quarry Light's website, 18 months ago, listing "equipment partner: Cedar Bluff Equipment Leasing"; the current site does not name it       | Archived websites        |
| G36 | Satellite imagery of the Authority's ground service equipment apron, comparing month 2 and month 20: visible increase in the number of ground service vehicles | Satellite imagery        |
| G37 | Equipment substitution change orders total \$748,000 across 11 orders                                                                                          | Government records       |

| ID  | Item                                                                                                                                                        | Category        |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| G38 | A regional equipment leasing rate publication indicates market rates that would price comparable equipment at roughly \$520,000 to \$610,000 for the period | Trade press     |
| G39 | Authority's own fleet records: three Authority-owned ground service vehicles were out of service for extended periods during the relevant months            | Internal record |
| G40 | Regional fuel price index: fuel costs rose 11% over the period                                                                                              | Trade press     |

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### 9.3 Guidance on the Substance

A weak submission concludes that this looks like what the commissioner said it looks like. A strong submission separates the questions that the evidence can answer from those it cannot, distinguishes the change order volume question from the equipment relationship question from the personal benefit question, and notices that these three have different evidentiary states and different appropriate responses.

The comparative question is available and is decisive for part of the analysis. So is the negative question.

G20 concerns an identifiable private individual who is not a party to anything. G34 concerns accounts. G33 quotes an unnamed source. Each requires an explicit handling decision, and the Authority's exposure to public records law makes those decisions consequential.

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### 9.4 What You Must Produce

#### ***Deliverable 1: Question Set (10 points)***

- The decision statement: which of the three courses, who decides, by when.
- Minimum eighteen questions across all six types, at least two of each.
- Every question assessed against the seven quality properties, with repairs visible.
- Prioritization on discriminating power, answerability, and consequence, with a top six selected.
- At least two questions this evidence cannot answer, named, with what would answer them.

### ***Deliverable 2: Hypothesis Set and Matrix (14 points)***

- Minimum five hypotheses, including the null or an innocent explanation and at least one of compound, incompetence, or third-party. Each with a disproof condition and a fair one-sentence proponent summary.
- A full scoring matrix over all 40 items, scored for consistency.
- A disconfirmation pass answered per hypothesis.
- Discriminating evidence identified, with a statement of how many items the judgment rests on.
- Eliminations recorded with bases, marked provisional where appropriate.

### ***Deliverable 3: Timeline (12 points)***

- A master timeline with precision markers, entry types, source item IDs, and flags.
- Minimum three inferred entries, reasoned.
- An event-based analysis in addition: treat the change order process as a repeating episode and map its internal stages, then compare across the three volume periods from G3.
- Minimum eight inconsistency flags, typed, with a "does not establish" column.
- Derived investigative steps for each flag.

### ***Deliverable 4: Link Chart and Edge Register (12 points)***

- Minimum twenty typed nodes and twenty-five attributed edges.
- Every edge with type, direction, basis, and strength, rendered distinguishably.
- Association matrix completed before drawing.
- Structural analysis: degree, betweenness, cut-points, clusters, negative space.

### ***Deliverable 5: Hidden Relationship Analysis (12 points)***

- A shared attribute table across all entities.
- Every candidate indirect relationship put through the four tests.
- Temporal correlation analysis, including the change order volume pattern against any other dated series in the packet.
- Every finding worded per Section 1.7.4.
- An explicit statement of which candidate relationships failed the discrimination test and why.

### ***Deliverable 6: Assumption Register (10 points)***

- Minimum twelve assumptions, typed, each with what rests on it, its test status, and the consequence if wrong.

- At least two coverage assumptions and one behavioral assumption.
- A therefore-audit performed on your own draft findings, with the results recorded.

### ***Deliverable 7: Investigative Report (20 points)***

**Maximum three pages** plus annexes. Addressed to the commission, written in the knowledge that it may become public.

Required: bottom line (four sentences, judgment, confidence, and which course of action the evidence supports); findings organized by question, with facts separated from assessments and every fact citing item IDs; **separate treatment of the three distinct issues** (change order volume and process, the equipment relationship, and any question of personal benefit), each with its own confidence; the entity relationship findings worded without asserting intent; assumptions; ranked gaps each with a resolving source; at least eight recommendations that are specific, assignable, and time-bound, including at least two executable this week at no cost and at least two that are information requests or referrals rather than conclusions; what would change this assessment; evidence annex; ethics annex.

### ***Deliverable 8: Confidence and Alternatives Statement (5 points)***

One page: for each of your three issue findings, why you chose that confidence level and not the level above or below; the alternative explanation you consider most likely to be correct if you are wrong; and the single observation that would most change your assessment.

### ***Deliverable 9: Decision Justification Log (5 points)***

Minimum fifteen rows: each significant investigative decision, the alternatives, why you chose as you did, and what would have changed it. Absence of this log caps the capstone at 70.

## **9.5 Capstone Grading Rubric (100 points)**

| <b>Deliverable</b>   | <b>Criterion</b>                                                      | <b>Points</b> |
|----------------------|-----------------------------------------------------------------------|---------------|
| <b>1. Questions</b>  | Eighteen-plus, all six types, quality properties applied with repairs | 4             |
|                      | Comparative and negative questions present and substantive            | 3             |
|                      | Prioritization with reasoning; unanswerable questions named           | 3             |
| <b>2. Hypotheses</b> | Five-plus, required types, fairly stated with disproof conditions     | 5             |
|                      | Matrix scored for consistency across all 40 items                     | 4             |
|                      | Disconfirmation pass per hypothesis                                   | 3             |

| <b>Deliverable</b>             | <b>Criterion</b>                                                          | <b>Points</b> |
|--------------------------------|---------------------------------------------------------------------------|---------------|
|                                | Discriminating evidence isolated; reliance count stated                   | 2             |
| <b>3. Timeline</b>             | Precision, type, and source on every row; inferred entries reasoned       | 4             |
|                                | Event-based analysis of the change order process, compared across periods | 4             |
|                                | Eight-plus flags with "does not establish" completed                      | 3             |
|                                | Derived steps                                                             | 1             |
| <b>4. Link analysis</b>        | Node and edge volume, full attribution, distinguishable strengths         | 5             |
|                                | Association matrix before drawing                                         | 2             |
|                                | Structural analysis complete                                              | 5             |
| <b>5. Hidden relationships</b> | Attribute table systematic                                                | 3             |
|                                | Four tests applied to every candidate                                     | 4             |
|                                | Non-discriminating candidates correctly rejected                          | 3             |
|                                | Wording discipline per Section 1.7.4                                      | 2             |
| <b>6. Assumptions</b>          | Twelve-plus, typed, with consequences                                     | 4             |
|                                | Coverage and behavioral assumptions present                               | 3             |
|                                | Therefore-audit performed on own draft                                    | 3             |
| <b>7. Report</b>               | Bottom line: four sentences, judgment, confidence, course of action       | 4             |
|                                | Three issues treated separately with distinct confidence levels           | 5             |
|                                | Facts separated from assessments; every fact cited                        | 3             |
|                                | Relationship findings without asserted intent                             | 3             |

| Deliverable            | Criterion                                                                  | Points     |
|------------------------|----------------------------------------------------------------------------|------------|
|                        | Recommendations: specific, assignable, time-bound, proportionate           | 3          |
|                        | Length discipline, annexes complete                                        | 2          |
| <b>8. Confidence</b>   | Defended against adjacent levels for each issue                            | 2          |
|                        | Most likely alternative if wrong, and the observation that would change it | 3          |
| <b>9. Decision log</b> | Fifteen-plus rows, substantive                                             | 5          |
| <b>Total</b>           |                                                                            | <b>100</b> |

**Grade bands:** 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

**Automatic deductions:**

- Any judgment without a confidence level: minus 5 per instance.
- Any claim not traceable to item IDs: minus 3 per instance.
- Treating the three issues as one undifferentiated finding: minus 10.
- Asserting that any individual acted with improper intent, on this evidence: minus 10 per instance.
- Drawing a meaningful relationship on the shared registered agent (G17): minus 8.
- Including M. Halloran's adult child (G20) anywhere in the analysis or product other than to record that the item was encountered and excluded: minus 20. The correct handling is to note in the ethics annex that the item concerns a private individual with no evidenced connection to any transaction, that no research on that person was conducted, and that the item was excluded.
- Treating G33 or G34 as evidence bearing on the facts: minus 5 each. Both are commentary; G34 additionally shows indicators of coordinated inauthentic posting.
- Missing decision justification log: capped at 70.

**Guidance on the expected shape of a strong answer.** The comparative evidence (G7, G8, G40) establishes that a cumulative increase in this range is within regional norms, which substantially weakens any inference drawn from the 41% figure alone. The distribution evidence (G3, G5, G26, G27) points to a process and control problem: 58 of 63 orders sitting just below a delegation threshold, 39 approved within 24 hours, and 21 with no supporting documentation contrary to G28. The equipment relationship (G12, G13, G14, G15, G16, G35) is a genuine undisclosed affiliation with a pricing question attached (G37, G38), and it is discriminating because the intermediary is a named

individual, unlike G17. G21, G22, and G39 supply a substantive innocent explanation for the volume that a fair analysis must carry. Nothing in the packet evidences personal benefit to any Authority employee, and G18, G19, and the absence of any financial trail should be reported as negative findings with their coverage boundaries. The three issues therefore resolve at three different confidence levels and support three different responses, which is the analytic result the capstone is built to produce.

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## Part 10: Answer Keys

*Instructors may wish to remove this section from student copies.*

### 10.1 Short Answer Key

1. Collection methodology governs the sequence of work; investigation methodology governs the quality of reasoning inside it. Without it, an investigator produces fully documented, entirely traceable, wrong conclusions, which is more dangerous than sloppy work because the documentation makes the conclusion persuasive.
2. A fact is a state of the world, true independently of observation. Evidence is an item bearing on whether a proposed fact is true, always carrying a source and ratings. An inference is a reasoning step from evidence to a claim, adding something the evidence does not state. Correct wording: "Item 12 records a load loss at 02:14 (A1)"; "We assess the fire had reached the panel by that time, moderate confidence."
3. A topic is a subject area, which is converted into a concern stated as a proposition that could be true or false, and then into questions. Questions must be generated from both directions, asking what would have to be true for the concern to be justified and what would have to be true for it to be unjustified, because questions generated only from the concern produce an investigation that can only confirm.
4. Descriptive (without it you investigate a situation you have not described), temporal (you cannot distinguish cause from coincidence), relational (actors appear isolated and the network is missed), causal (correlation is mistaken for explanation), comparative (ordinary variation is investigated as anomaly), negative (the most informative evidence in many cases is missed).
5. Because absence is evidence and it is invisible to anyone who never asks for it. A permit never filed, an inspection that never occurred, a log gap, or a person absent from where their role required them each carry information, and none appears in a search for what exists.
6. Any four of: answerable, bounded, neutral, single-barrelled, falsifiable, sourced, consequential. The rewrite should convert an unbounded or loaded question into one naming entities, a period, and a source that would settle it.
7. Discriminating power, answerability, and consequence. An interesting question with low discriminating power should be deferred because every possible answer would be consistent with all hypotheses, so answering it cannot change the analysis regardless of how much effort it consumes.
8. The method holds several explanations simultaneously and deliberately so that no single one

accumulates the attachment that turns an investigator into an advocate. Single-hypothesis investigation fails because evidence consistent with almost any plausible explanation exists; the search for supporting evidence always succeeds, contradicting items are explained away, and the result is a confident wrong conclusion.

**9.** Any six of: null (guards against investigating noise), innocent explanation (presumption of wrongdoing), obvious explanation (pretending you have no prior), role reversal (fixed role assignment), third-party (closed suspect pools), compound (forcing one explanation onto two events), deception (taking a convenient evidence pattern at face value), incompetence (attributing to intent what process failure explains).

**10.** Because counting plus signs measures volume rather than discrimination, and most items are consistent with several hypotheses. What should be examined is which items are inconsistent with which hypotheses, and which items distinguish between them; a single strong inconsistency does more work than five consistencies.

**11.** Elimination means evidence is inconsistent with a hypothesis in a way that cannot reasonably be explained, and it must be recorded with its basis and item IDs. Surviving means being the best-supported explanation on current evidence, at a stated confidence, with the gaps that could change it. Survivors inherit the limits of the elimination process, so incomplete collection means incomplete elimination.

**12.** Sequence (causation requires the cause to precede the effect), opportunity (who could have done what given evidenced location and time), impossibility (the strongest analytic finding available), and anomaly (clusters, gaps, and unusual timing).

**13.** The type field records whether an entry is observed, reported, or inferred. Unmarked inference is serious because it presents the analyst's reconstruction as though it were the record, so a reader cannot distinguish what happened from what the analyst concluded, and the reasoning becomes uncheckable.

**14.** Chronological analysis places events on one absolute time axis and reveals simultaneity, convergence, and impossibility; it suits single incidents and opportunity questions. Event-based analysis groups events into episodes and traces each internally, revealing stages and repeated pattern; it suits recurring conduct such as fraud schemes and campaigns. Serious investigations build both.

**15.** Contradiction (identify the more reliable production process and seek a third independent category); impossibility (re-verify timestamps and identity attribution before concluding anything about actors); gap (identify which record type covers the period and whether it exists, was never created, or was not retained); anomaly (establish the baseline before treating it as significant); convergence (test whether the intersection is coincidence given base rates).

**16.** Type, direction, basis by item ID, and strength. Rule 6 states that a relationship is a claim and every claim needs a basis, because a chart is persuasive beyond its evidence and an unsourced line is the most convincing form of unsupported assumption in the discipline.

**17.** A direct relationship is a single edge asserted by evidence; an indirect relationship is a connection through intermediates or derived from a shared attribute, which no record states. The discriminating-

intermediary test asks how many other entities share the same intermediary: a commercial agent serving thousands makes the connection worthless, while a specific named individual or an uncommon address makes it meaningful.

**18.** Degree centrality counts connections and identifies hubs; betweenness centrality measures how often a node sits on paths between others and identifies brokers. Betweenness often matters more because the most connected node is usually a well-known principal, whereas the highest-betweenness node is frequently an administrator, agent, or facilitator nobody had noticed, whose position makes the structure function.

**19.** Assumptions cannot be eliminated because reasoning requires premises: that timestamps are approximately right, that records describe what they claim, that names refer to the entities intended. The register converts hidden assumptions into one of three visible states: an evidenced finding, an explicit and tested assumption with its consequence stated, or a removed premise.

**20.** Shared attribute analysis, temporal correlation, structural inference, negative space, and cross-domain assembly. Most hidden relationships are latent rather than concealed because the information already exists in the collected material and has simply never been assembled side by side, which means systematic assembly outperforms cleverness.

## 10.2 Multiple Choice Key

| #  | Answer | Note                                                           |
|----|--------|----------------------------------------------------------------|
| 1  | b      | Documented, traceable, and wrong                               |
| 2  | b      | An item from a source bearing on a proposed fact               |
| 3  | c      | A reasoning step beyond what the item states                   |
| 4  | c      | Asks what is absent that should be present                     |
| 5  | b      | Establishes whether anything anomalous occurred                |
| 6  | b      | At least one of third-party, compound, deception, incompetence |
| 7  | c      | Non-discriminating; contributes almost nothing                 |
| 8  | b      | Recorded and marked inferred                                   |
| 9  | c      | Recurring conduct                                              |
| 10 | c      | Impossibility                                                  |
| 11 | c      | Hypothesized, visually distinct, recorded                      |

| #  | Answer | Note                                                      |
|----|--------|-----------------------------------------------------------|
| 12 | c      | Non-discriminating intermediary                           |
| 13 | b      | Betweenness centrality                                    |
| 14 | b      | Coverage assumption                                       |
| 15 | b      | Concealment asserts an intentional act requiring evidence |

### 10.3 True or False Key

| #  | Answer | If false, why                                                                                                   |
|----|--------|-----------------------------------------------------------------------------------------------------------------|
| 1  | False  | Investigators possess evidence and make inferences; facts are rarely held directly                              |
| 2  | False  | Evidence consistent with a hypothesis exists for almost any plausible explanation and proves nothing on its own |
| 3  | True   |                                                                                                                 |
| 4  | True   |                                                                                                                 |
| 5  | False  | Connection is not participation; participation requires its own evidence                                        |
| 6  | False  | Absence reflects the record system's scope, requirements, and retention as much as the world                    |
| 7  | False  | Assumption elimination follows construction, because you cannot audit a picture you have not built              |
| 8  | True   |                                                                                                                 |
| 9  | False  | The highest-betweenness node is frequently more important than the highest-degree node                          |
| 10 | True   |                                                                                                                 |

### 10.4 Critical Thinking Guidance

**7.1** The concern should be stated as a proposition, for example: "maintenance contracts have been awarded through a process that departs from policy." Questions must cover all six types, with the

comparative ("how does the award pattern compare to peer authorities and to prior periods") and negative ("which required solicitation or justification documents are absent") ranked highest, because both are answerable from records and both discriminate.

**7.2** (a) Presupposes an actor and an offence; rewrite as "which approvals in the transaction chain required the CFO's signature, and which of those bear it?" (b) Presupposes concealment and a second facility; rewrite as "does any public record identify a second facility, and does any filing disclose it?" (c) Three questions in one; split them. (d) Unbounded with no baseline; rewrite as "how does overtime per employee in this department compare to the preceding four quarters and to comparable departments?"

**7.3** The dull, discriminating questions come first; the interesting, non-discriminating ones are deferred or dropped. The consumer sentence should name the four dropped questions, state that two require records unobtainable within the deadline and name what would obtain them, and state that two would not change the analysis whatever the answer.

**7.4** Acceptable hypotheses include: structuring by a single organizer using recruited account holders; a legitimate remittance or payroll pattern for a group of workers with a common employer; an unrelated coincidence of a common transaction size; a compound case combining a small structuring operation with unrelated ordinary accounts; and a bank process artifact in how the accounts were flagged. Each needs a disproof condition, for example "account holders have no common referral source or address" for the first.

**7.5** The steel-man should be written so a proponent would recognize it as fair, using the specific facts, and it should identify the evidence that would defeat it: for a legitimate remittance explanation, the defeating evidence would be an absence of any common employer, an absence of onward transfers consistent with remittance, and account behavior inconsistent with the stated purpose.

**7.6** The ratio is close to meaningless because most items will be consistent with several hypotheses, so a high support count measures volume rather than discrimination. The correct analysis is to score every item against every hypothesis, isolate the items inconsistent with one or more, and identify which items distinguish; the judgment typically rests on two or three of the 34.

**7.7** The timeline should show: delivery scan 09:12 Tuesday (observed, minute precision); witness sighting Tuesday afternoon (reported, bracketed to a several-hour window, and inconsistent with the 09:12 scan); no dock log entry Tuesday (negative, observed); camera gap 09:00 to 09:40 Tuesday (observed, and it covers the scan time); email sending the invoice Tuesday 16:20 (observed); invoice dated Wednesday (observed, and inconsistent with the Tuesday email). Inconsistencies: contradiction between the witness and the scan; gap in the dock log; gap in camera coverage precisely spanning the scan; impossibility, or near it, between an invoice dated Wednesday and an email transmitting it on Tuesday.

**7.8** Steps: for the witness contradiction, re-interview with the scan time as a prompt and treat the reported time as low-precision rather than discarding it; for the dock log gap, establish whether the log is continuous by policy and who is responsible for entries; for the camera gap, obtain the system's event log to establish whether the gap was a fault, a power event, or a manual action; for the invoice date,

determine whether the accounting system dates invoices at creation or at posting, which frequently resolves such conflicts benignly.

**7.9** Both. The event-based analysis maps each episode's internal stages and enables cross-episode comparison, revealing whether the conduct followed a consistent pattern. The chronological view then reveals whether episodes overlapped, which would indicate parallel operation rather than sequential repetition, and it exposes convergence with external dates such as audits, reporting deadlines, or personnel changes.

**7.10** Direct edges: Adeyemi to Company A and to Company B (both confirmed, from filings); Company B to Company D (lease, confirmed); Company C and Company D as co-defendants (confirmed, from the court exhibit). Indirect: Company A and Company C via shared suite address (strength depends on the discrimination test); Company A and Company C via a shared bank account reference (potentially strong, but requires establishing that the reference identifies the same account); Company A and Company D via a two-step path through Adeyemi and Company B.

**7.11** Adeyemi and Company B are the candidates. Company B may have the highest degree; Adeyemi is likely the highest betweenness and, in most renderings of this chart, a cut-point between the A and B clusters. Adeyemi should be investigated first, because the position on the paths between otherwise separate clusters is what makes the structure function, and because a natural person is a more discriminating intermediary than an entity.

**7.12** The test is to determine how many entities are registered at that suite, using the corporate registry's address search and the building directory. At 400 entities the edge should be removed or marked possible with an explicit note that the intermediary is non-discriminating; at three entities it becomes a probable relationship worth pursuing, though still not evidence of coordination.

**7.13** Assumptions: that the badge log captures all entry (coverage); that badge-out means the person left and did not return (behavioral and coverage); that no entry means no presence (coverage); that absence of a person implies an unattended ignition source (causal). Rewrite: "The access log records a final badge-out at 19:40 (E3). The log records only badged events at monitored doors, and one door sensor was in a fault condition (E4), so the log does not establish whether anyone was present after 19:40. No evidence in the current material establishes presence or absence after that time."

**7.14** A coverage assumption. Test it by establishing whether the regulator receives complaints of this type at all, whether complaints are public, what the base rate of complaints is for comparable suppliers, and whether the complaint channel is known to affected parties. If untestable, the finding cannot support any inference about capacity, and confidence in any conclusion resting on it should be low with the assumption stated explicitly.

**7.15** A complete answer reports what each test surfaced. Typical results: the negation test reveals a source assumption ("this is wrong only if the record is inaccurate") or a coverage assumption; the stranger test reveals that the uninvolved reader reaches a weaker conclusion, and the difference between the two conclusions is composed of context the analyst supplied rather than evidence.

**7.16** Build the table, then sort by each column in turn and record repeats. Registered agent and officers should be sorted first, but with the discrimination test applied immediately to the agent column,

because commercial agents produce large numbers of meaningless repeats. Domain and name server columns are usually the most discriminating in practice. Every repeat is then subjected to the four tests before any edge is drawn.

**7.17** Estimate the base rate: how many permits does each entity file per year, over what period, and how many pairs would fall within eight days by chance given those rates? Six occurrences at a consistent short lag over three years is unlikely to be chance if filing rates are low, and unremarkable if both file monthly. Converting it to a supported relationship requires an independent connector: a shared officer, agent, address, contractor, or a document referencing both.

**7.18** It suggests a missing intermediary: something or someone connects the clusters and has not been identified. Three steps: examine the events where members co-appear and identify every other party present, including organizers and venues; run shared attribute analysis across the two clusters looking for a common vendor, counsel, auditor, or agent; and apply negative space analysis to ask which role would be required for the two clusters to interact as they do, then look for who fills it.

**7.19** The response should explain that the access log's value depends on completeness; a disabled sensor means entry could have occurred without being recorded, so the log can no longer exclude anyone. It therefore removes the analytic basis for saying that a person who was not logged was absent, which weakens the case against every candidate equally, including the one the log previously appeared to implicate.

**7.20** The three defeating sentences should be specific and observable. The collection description should name the source category and system for each, and should state whether it is available to the investigator.

**7.21** The response should state what the evidence supports and at what confidence; distinguish the findings that are established from those that are open; name the alternative explanations and why they were downgraded; state the gaps and what would close them; and connect all of this to the decision the stakeholder faces, noting that decisions are routinely made under uncertainty and that the product's purpose is to make the uncertainty explicit rather than to eliminate it. It should not overstate to satisfy the demand.

---

## Appendix A: Reusable Templates

### A.1 Question Set

| Q# | Question | Type | Concern served | What would answer it | Discriminating (H/M/L) | Answerable (H/M/L) | Consequence (H/M/L) | Rank | Status |
|----|----------|------|----------------|----------------------|------------------------|--------------------|---------------------|------|--------|
|----|----------|------|----------------|----------------------|------------------------|--------------------|---------------------|------|--------|

### A.2 Hypothesis Record

H#: \_\_\_\_\_  
STATEMENT (declarative): \_\_\_\_\_

TYPE (null / innocent / obvious / third-party / compound / deception / incompetence / other): \_\_\_\_\_

DISPROOF CONDITION: \_\_\_\_\_

PROPONENT'S ONE-SENTENCE CASE: \_\_\_\_\_

DISCONFIRMATION: what would disprove it? \_\_\_\_\_

Did I look? \_\_\_\_\_ What did I find? \_\_\_\_\_

STATUS: active / eliminated / provisionally eliminated

BASIS (item IDs): \_\_\_\_\_

### A.3 Scoring Matrix

| Item | H1 | H2 | H3 | H4 | H5 | Discriminating? | Reason for ++ or -- |
|------|----|----|----|----|----|-----------------|---------------------|
|------|----|----|----|----|----|-----------------|---------------------|

Notation: ++ strongly consistent, + consistent, 0 neutral, - inconsistent, -- strongly inconsistent.

### A.4 Timeline

| Date/time | Precision | Type (obs/rep/inf) | Event as observed | Actors | Location | Item IDs | Categories | Confidence | Flags |
|-----------|-----------|--------------------|-------------------|--------|----------|----------|------------|------------|-------|
|-----------|-----------|--------------------|-------------------|--------|----------|----------|------------|------------|-------|

### A.5 Inconsistency Register

| Flag # | Type (contradiction / impossibility / gap / anomaly / convergence) | Items | What it establishes | What it does NOT establish | Investigative step |
|--------|--------------------------------------------------------------------|-------|---------------------|----------------------------|--------------------|
|--------|--------------------------------------------------------------------|-------|---------------------|----------------------------|--------------------|

### A.6 Edge Register

| Edge # | From | To | Type | Direction | Basis (item IDs) | Strength | Direct or indirect | Discrimination tested? | Added by |
|--------|------|----|------|-----------|------------------|----------|--------------------|------------------------|----------|
|--------|------|----|------|-----------|------------------|----------|--------------------|------------------------|----------|

### A.7 Attribute Table

| Entity | Address | Registered agent | Officers | Domains | Bank refs | Counsel / vendors | Dates | Repeats |
|--------|---------|------------------|----------|---------|-----------|-------------------|-------|---------|
|--------|---------|------------------|----------|---------|-----------|-------------------|-------|---------|

## A.8 Assumption Register

| A# | Assumption | Type | Where it entered | What rests on it | Status | Test performed | Result | Consequence if wrong | Residual |
|----|------------|------|------------------|------------------|--------|----------------|--------|----------------------|----------|
|----|------------|------|------------------|------------------|--------|----------------|--------|----------------------|----------|

## A.9 Findings Document

|                                                                    |                                                                |
|--------------------------------------------------------------------|----------------------------------------------------------------|
| BOTTOM LINE                                                        | Judgment + confidence + reason + so-what.<br>3 to 4 sentences. |
| FINDINGS BY ISSUE                                                  | Each issue separately, with its own confidence.                |
| FACTS                                                              | Observed only. Item IDs and ratings on each.                   |
| ASSESSMENT                                                         | Inference, with alternatives considered.                       |
| KEY ASSUMPTIONS                                                    | Each with the consequence if wrong.                            |
| INTELLIGENCE GAPS                                                  | Ranked; each with the source that would resolve it.            |
| RECOMMENDATIONS                                                    | Specific, assignable, time-bound, proportionate.               |
| WHAT WOULD CHANGE THIS ASSESSMENT                                  |                                                                |
| --- annexes ---                                                    |                                                                |
| EVIDENCE LIST · EDGE REGISTER · ASSUMPTION REGISTER · ETHICS ANNEX |                                                                |

## A.10 Wording Reference

| Do write                                                                                                             | Do not write                                     |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| "Item E9 records that the last inbound delivery was 1 February"                                                      | "There was no inventory"                         |
| "We assess X, at moderate confidence, based on E2, E9, E19"                                                          | "X is the case"                                  |
| "Filings list the same individual as manager of both entities (E31, E33)"                                            | "The companies are secretly connected"           |
| "No document in the file discloses a relationship"                                                                   | "The relationship was concealed"                 |
| "The access log does not record entry after 19:40; one door sensor was in fault"                                     | "No one entered after 19:40"                     |
| "A vehicle of similar description passed at 22:10; the plate is partially legible and the vehicle is not identified" | "The vehicle returned to the site"               |
| "Two entities share a registered agent serving approximately 31,000 entities; we assess this as non-discriminating"  | "The entities are linked through a shared agent" |

## Appendix B: The Investigation Checklist

Run this at the end of every investigation, and again before any product leaves your hands.

### Questions

1. Was the topic converted into a concern stated as a falsifiable proposition?
2. Are all six question types represented?
3. Is any question loaded or presupposing an actor?
4. Were questions prioritized by discriminating power, not interest?
5. Is any question unanswerable, and was that stated to the consumer early?

**Hypotheses** 6. Were at least four generated, including the null or innocent explanation? 7. Is the least likely stated as its proponent would state it? 8. Does each carry a disproof condition? 9. Was scoring done for consistency rather than support? 10. Was the disconfirmation question answered per hypothesis, honestly? 11. Are the discriminating items identified, and is the reliance count stated? 12. Is every elimination recorded with its basis?

**Timeline** 13. Does every row carry a precision marker? 14. Is every row typed as observed, reported, or inferred? 15. Does every row cite item IDs? 16. Are all five inconsistency types searched for? 17. Does each flag have a derived investigative step? 18. Was anything smoothed away to make the narrative read better?

**Link analysis** 19. Does every edge carry type, direction, basis, and strength? 20. Are strengths rendered distinguishably? 21. Was the discrimination test applied to every indirect relationship? 22. Was degree distinguished from betweenness? 23. Was negative space examined? 24. Can I defend every line on this chart, right now, by item ID?

**Assumptions** 25. Was a therefore-audit performed on the written findings? 26. Does the register include coverage, identity, continuity, and behavioral assumptions? 27. Does every conclusion survive if each untested assumption is negated? 28. Was every relevant bias's specific control applied?

**Hidden relationships** 29. Was a shared attribute table built and sorted by every column? 30. Was temporal correlation tested against base rates? 31. Did every candidate relationship pass all four tests? 32. Is any finding worded as concealment without evidence of a concealing act? 33. After each discovery, were the matrix, the timeline, and the register all revisited?

**Conclusion** 34. Does every claim cite item IDs? 35. Is every inference labeled as an assessment with a confidence level? 36. Is the confidence justified by a stated reason? 37. Are alternatives named and their downgrading explained? 38. Are gaps ranked, each with a resolving source? 39. Are recommendations proportionate to the strength of the findings? 40. Would this product survive being read by the party it disfavours, with their lawyer?

## Appendix C: Where to Go Next

**Skills to build after this chapter, in priority order:**

1. **Hypothesis generation under pressure.** The hardest habit to build, because the first explanation arrives instantly and the others must be summoned deliberately. Practise on news stories: read a headline, then write four explanations before reading the article.

2. **Timeline construction from messy sources.** Take any documented public event and build a precision-marked, typed timeline from published material. The discipline only becomes real when the sources conflict.
3. **Link charting with a register.** Build charts for anything, and make yourself defend every edge by citation before you look at the chart as a whole.
4. **Assumption auditing on your own work.** The therefore-audit takes fifteen minutes and finds something every time. Run it on everything you write for a month.
5. **Base rate discipline.** Before calling anything unusual, find out how usual it is. This single habit prevents more errors than any other in this chapter.
6. **Steel-manning.** Write the strongest version of the explanation you find least plausible, weekly, on any subject. It is the clearest marker of an analyst rather than an advocate.
7. **Reviewing others' work against the failure signatures.** Read published investigations and diagnose them: premature narrative, load-bearing assumption, confident chart, or smoothed timeline. Most contain at least one.

#### **Habits to start now:**

- Write down your first explanation before you analyse anything, and compare it to your conclusion afterward. Keep the record.
  - Never draw a line you cannot source.
  - For every conclusion, write the sentence that would defeat it.
  - When you find a hidden relationship, re-score everything, actually.
  - Ask what is missing before you ask what it means.
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## **End of Chapter 5**